

Functional Analytic Methods and Partial Differential Equations

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Chapter 1

Calculus in Banach Spaces

In this chapter we will focus on the generalization of *infinitesimal calculus* to Banach spaces. This will be particularly useful when formalizing the method of characteristics and proving the implicit/inverse function theorems. Of course, as this book will be greatly concerned with the study of solutions to various differential equations, this chapter is easy to motivate. As the reader shall soon see, the concepts explored within this chapter will be applicable when we study applications of the so-called *Sobolev spaces*.

Our approach will be different than that of a standard calculus course; we will begin by developing an integration theory for functions taking values in a Banach space $\mathcal{X}$ over the real or complex numbers. This integral will be a generalization of the *Riemann integral*. One could instead generalize the *Lebesgue integral*, giving rise to the *Bochner integral*, although for our purposes the generalized Riemann integral will suffice. In fact, the Riemann integral will be easier to construct and will be more “to the point”. However, when working with weak solutions over $\mathbb{R}^n$, we will almost always be discussing the Lebesgue integral. For this reason, the reader should have a strong background in the measure and integration theories.

After developing the generalized Riemann integral, we will introduce two notions of differentiations for functions between Banach spaces and study applications of our generalized calculus. For instance, we will apply these developments to prove generalized versions of the inverse and implicit function theorems. As it is often quite difficult to find complete notes regarding the Riemann integral on Banach spaces, we hope this first chapter will be particularly useful to curious students.

Let us now get some notation out of the way. Throughout this chapter, $\mathbb{K}$ will denote one of the fields $\mathbb{R}$ or $\mathbb{C}$, and $\mathcal{X}$ will represent a *Banach space* over the given field $\mathbb{K}$. Let us recall that a Banach space $\mathcal{X}$ is a vector space over $\mathbb{K}$ equipped with a norm $\|\cdot\|$ such that $\mathcal{X}$ is a complete metric space with respect to the metric

$$d(x_1, x_2) := \|x_1 - x_2\|, \quad x_1, x_2 \in \mathcal{X}.$$

Henceforth, we will assume that the reader has taken courses in basic topology and functional analysis. We now move towards the development of the generalized Riemann integral.

1.1 Constructing the Generalized Riemann Integral

Fix a compact interval $[a, b] \subset \mathbb{R}$ and let $\mathcal{X}$ be a Banach space over $\mathbb{K}$. A function $f : [a, b] \rightarrow \mathcal{X}$ is called *bounded* if there exists $M > 0$ such that

$$\|f\|_\infty := \sup_{x \in [a, b]} \|f(x)\| \leq M.$$

Here, $\|\cdot\|$ is the given norm on $\mathcal{X}$. The set of all bounded maps $f : [a, b] \rightarrow \mathcal{X}$ will be denoted by $B([a, b], \mathcal{X})$. In practice, one often takes $\mathcal{X}$ to be one of the vector spaces $\mathbb{R}$ or $\mathbb{C}$.

Definition 1. Let $a < b$ be real numbers. A partition of a compact interval $[a, b]$ is a finite family $\mathcal{P}$ consisting of closed sub-intervals of $[a, b]$ whose union is $[a, b]$. More precisely,

$$\mathcal{P} = \{ [x_{i-1}, x_i] : x_j \in [a, b], i = 1, \dots, n \} \quad (1.1)$$

and $\bigcup_1^n [x_{i-1}, x_i] = [a, b]$. By convention, we always require that

$$a = x_0 < x_1 < \dots < x_n = b.$$

Partitions themselves are not fundamental to the Riemann integral. However, a *tagged partition* is.

Definition 2. Let $\mathcal{P} = \{[x_{i-1}, x_i]\}_{i=1}^n$ be a partition of a compact interval $[a, b]$. A set of *tags* for the partition $\mathcal{P}$ is a finite collection $\{t_1, \dots, t_n\}$ of points in $[a, b]$ satisfying $t_i \in [x_{i-1}, x_i]$ for all $i = 1, \dots, n$. A *tagged partition* is then defined to be a partition together with a set of tags, i.e.

$$\dot{\mathcal{P}} := \{\mathcal{P}, \{t_i\}_{i=1}^n\}. \quad (1.2)$$

By convention, we will often write $\dot{\mathcal{P}}$ instead of $\mathcal{P}$ to indicate that the partition $\mathcal{P}$ comes equipped with a set of tags. Given a partition $\mathcal{P}$ (possibly tagged), we define the *mesh* of $\mathcal{P}$ as

$$\|\mathcal{P}\| := \max_{1 \leq i \leq n} (x_i - x_{i-1}) > 0. \quad (1.3)$$

With this out of the way, we come to the definition of the Riemann integral. As in standard calculus, this begins with the introduction of a *Riemann sum*.

Definition 3. Let $f : [a, b] \rightarrow \mathcal{X}$ be a function and let $\dot{\mathcal{P}}$ be a tagged partition of $[a, b]$. The *Riemann sum* of f over the partition $\dot{\mathcal{P}}$ is defined via the equation

$$S(f; \dot{\mathcal{P}}) := \sum_{i=1}^n (x_i - x_{i-1}) \cdot f(t_i) = \sum_{i=1}^n (x_i - x_{i-1}) f(t_i). \quad (1.4)$$

This brings us to another convention: for $u \in \mathcal{X}$ and $\alpha \in \mathbb{K}$, we will typically write αu instead of the scalar product $\alpha \cdot u$.

We have now come to the definition of Riemann integrability for functions taking values in a Banach space $\mathcal{X}$.

Definition 4. A function $f : [a, b] \rightarrow \mathcal{X}$ is called *Riemann integrable*¹ (on $[a, b]$) if there is a vector $\Lambda \in \mathcal{X}$ with the property that, for each $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|S(f; \dot{\mathcal{P}}) - \Lambda\| < \varepsilon$$

for **all** tagged partitions $\dot{\mathcal{P}}$ of $[a, b]$ having $\|\dot{\mathcal{P}}\| < \delta$. If f is Riemann integrable, we say that Λ is the *Riemann integral* of f over $[a, b]$ and denote this quantity by

$$\int_a^b f.$$

Finally, we will denote by $\mathcal{R}([a, b], \mathcal{X})$ the collection of all Riemann integrable functions $[a, b] \rightarrow \mathcal{X}$. We will also agree to the following convention:

$$\int_b^a f := - \int_a^b f.$$

¹Or, sometimes *Extended Riemann integrable*.

Remark 1.1. Our construction technically only allows us to define the quantity $\int_a^b f$ for $a < b$ and suitable f . However, there is nothing that prevents us from using an analogous definition with $a = b$. Certainly, let $f : [a, b] \rightarrow \mathcal{X}$; there exists but a single tagged partition of $[a, a] = \{a\}$, given by

$$\dot{\mathcal{P}} = \{[a, a], a\}.$$

Then,

$$S(f; \dot{\mathcal{P}}) = f(a)(a - a) = 0.$$

Hence, it would make sense to define $\int_a^a f = 0$ for *all* functions $[a, b] \rightarrow \mathcal{X}$. In general, we will only allow this extreme case for functions that are Riemann integrable on non-trivial compact intervals $[c, d]$ containing the point a . Unless stated otherwise, one should assume that $[a, b]$ is a non-trivial interval.

1.1.1 Immediate Consequences and Trivial Examples

We would like to check that the Riemann integral is well defined whenever it exists. This is accomplished by way of the following proposition.

Proposition 1.1. *Let $f : [a, b] \rightarrow \mathcal{X}$ be Riemann integrable. Then its Riemann integral is unique.*

Proof. Suppose that L and L' are two candidates for the “Riemann integral of f ”. Let $\varepsilon > 0$ be given and choose $\delta > 0$ such that

$$\|S(f; \dot{\mathcal{P}}) - L\| < \varepsilon \quad \text{and} \quad \|S(f; \dot{\mathcal{P}}) - L'\| < \varepsilon$$

for all tagged partitions $\dot{\mathcal{P}}$ of $[a, b]$ whose mesh is strictly less than δ . Clearly, such a partition of $[a, b]$ will always exist and as such we get that

$$\|L - L'\| \leq \|S(f; \dot{\mathcal{P}}) - L\| + \|S(f; \dot{\mathcal{P}}) - L'\| < 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we conclude that $L = L'$. □

In our development of the integral, it will be useful to have a class of “prototype functions” that we know are Riemann integrable. This begins with a quick proof that constant functions are integrable.

Proposition 1.2. Fix a vector $u \in \mathcal{X}$ and define a function $f : [a, b] \rightarrow X$ via the equation $f(x) \equiv u$. Then, $f(x)$ is Riemann integrable on $[a, b]$. Moreover,

$$\int_a^b f = (b - a)u.$$

Proof. First, let $\dot{\mathcal{P}}$ be any tagged partition of the interval $[a, b]$, following of course the notation in Definition 2. It is easy to see that

$$S(f; \dot{\mathcal{P}}) = \sum_{i=1}^n (x_i - x_{i-1})f(t_i) = \sum_{i=1}^n (x_i - x_{i-1})u = (b - a)u.$$

Thus, given $\varepsilon > 0$, take $\delta := 1$. Then, if $\dot{\mathcal{P}}$ is a tagged partition of $[a, b]$ whose mesh is strictly less than δ , the above yields

$$\|S(f; \dot{\mathcal{P}}) - (b - a)u\| < \varepsilon.$$

This concludes the proof. $\square$

1.1.2 Basic Properties of the Extended Riemann Integral

We now turn towards some basic regularity properties that our integral enjoys. Again, these all hold true for the standard Riemann and Lebesgue integrals and are to be expected of any integral. As I like to say, an integral that does not satisfy the following theorem is no integral at all!

Theorem 1.3. Let $[a, b] \subset \mathbb{R}$ and $\mathcal{X}$ a Banach space over $\mathbb{K}$. Suppose that $f, g : [a, b] \rightarrow \mathcal{X}$ are Riemann integrable. The following hold true.

(1) If $\alpha \in \mathbb{K}$, then αf is also Riemann integrable on $[a, b]$. Moreover,

$$\int_a^b \alpha f = \alpha \int_a^b f.$$

(2) The function $f + g$ is Riemann integrable on $[a, b]$ and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

Proof. We first prove (1). Let $\varepsilon > 0$ be given and choose $\delta > 0$ such that

$$\left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| < \varepsilon$$

whenever $\|\dot{\mathcal{P}}\| < \delta$. For any such tagged partition $\dot{\mathcal{P}}$, one also has

$$\begin{aligned} \left\| S(\alpha f; \dot{\mathcal{P}}) - \alpha \int_a^b f \right\| &= \left\| \sum_{i=1}^n (x_i - x_{i-1}) \cdot \alpha f(t_i) - \alpha \int_a^b f \right\| \\ &= |\alpha| \left\| \sum_{i=1}^n (x_i - x_{i-1}) f(t_i) - \int_a^b f \right\| \\ &= |\alpha| \left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| \\ &\leq |\alpha| \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was taken arbitrarily, we see from the above that the first point holds true. For the second, we simply make use of the triangle inequality. Let $\varepsilon > 0$ and fix $\delta > 0$ such that both

$$\left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| < \frac{\varepsilon}{2} \quad \text{and} \quad \left\| S(g; \dot{\mathcal{P}}) - \int_a^b g \right\| < \frac{\varepsilon}{2}$$

whenever $\|\dot{\mathcal{P}}\| < \delta$. Of course, this can be done since both f and g are Riemann integrable on $[a, b]$. If $\dot{\mathcal{P}}$ is a tagged partition of $[a, b]$ whose mesh is strictly less than δ , the triangle inequality gives

$$\begin{aligned} &\left\| S(f + g; \dot{\mathcal{P}}) - \left(\int_a^b f + \int_a^b g \right) \right\| \\ &= \left\| \sum_{i=1}^n (x_i - x_{i-1}) (f(t_i) + g(t_i)) - \int_a^b f - \int_a^b g \right\| \\ &\leq \left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| + \left\| S(g; \dot{\mathcal{P}}) - \int_a^b g \right\| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

This proves (2) and the proof is complete. $\square$

The following bound is intuitively obvious, but incredibly useful.

Theorem 1.4. *Let $f : [a, b] \rightarrow \mathcal{X}$ be Riemann integrable and bounded. There holds the following:*

$$\left\| \int_a^b f \right\| \leq (b - a) \|f\|_\infty, \quad (1.5)$$

Proof. Let $\varepsilon > 0$ be given and choose $\delta > 0$ such that

$$\left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| < \varepsilon$$

for all tagged partitions $\dot{\mathcal{P}}$ of $[a, b]$ whose mesh is strictly less than δ . Now, fix any such partition $\dot{\mathcal{P}}$ and write

$$\left\| \int_a^b f \right\| \leq \left\| \int_a^b f - S(f; \dot{\mathcal{P}}) \right\| + \|S(f; \dot{\mathcal{P}})\| < \|S(f; \dot{\mathcal{P}})\| + \varepsilon.$$

Following the notation used in Definition 2, we subsequently see from the above that

$$\begin{aligned} \left\| \int_a^b f \right\| &\leq \varepsilon + \sum_{i=1}^n |(x_i - x_{i-1})| \|f(t_j)\| \\ &\leq \varepsilon + \left(\sum_{i=1}^n (x_i - x_{i-1}) \right) \|f\|_\infty \\ &= \varepsilon + (b - a) \|f\|_\infty. \end{aligned}$$

Passing to the limit as $\varepsilon \searrow 0$ yields the desired inequality. $\square$

Before we move on towards the next unrelated subsection, we prove a “boundedness theorem” which states that all Riemann integrable functions are necessarily bounded. In most theories of Riemann integration, one assumes a priori that the function f is bounded. In our case, however, this is something that can be deduced from integrability.

Theorem 1.5. *Let $f : [a, b] \rightarrow \mathcal{X}$ be Riemann integrable. Then, f is bounded on $[a, b]$.*

Proof. We proceed by way of contradiction. Suppose there exists a Riemann integrable function f on $[a, b]$ that is unbounded. Let L denote the value $\int_a^b f$. Taking $\varepsilon := 1$, there exists $\delta > 0$ such that

$$\|S(f; \dot{\mathcal{P}}) - L\| < 1$$

for all tagged partitions $\dot{\mathcal{P}}$ of $[a, b]$ having mesh strictly less than δ . The reverse triangle inequality then yields

$$\|S(f; \dot{\mathcal{P}})\| < 1 + \|L\| \quad (1.6)$$

for all such $\dot{\mathcal{P}}$. We now construct a special tagged partition $\dot{\mathcal{Q}}$ which will contradict the above. To this end, let $\mathcal{Q} = \{[x_{i-1}, x_i]\}_{i=1}^n$ be any partition of $[a, b]$ whose mesh is strictly less than δ . Since f is unbounded on $[a, b]$, it is unbounded one at least one interval in $\mathcal{Q}$. Hence, there is an index $k = 1, \dots, n$ and $t_k \in [x_{k-1}, x_k]$ such that

$$\|(x_k - x_{k-1})f(t_k)\| > 1 + \|L\| + \left\| \sum_{j \neq k} (x_j - x_{j-1})f(t_k) \right\|.$$

Here, we have chosen $t_j := x_j$ for all $j \neq k$. But then, if we give $\mathcal{Q}$ these tags,

$$\begin{aligned} \|S(f; \dot{\mathcal{Q}})\| &= \left\| (x_k - x_{k-1})f(t_k) + \sum_{j \neq k} (x_j - x_{j-1})f(t_k) \right\| \\ &\geq \|(x_k - x_{k-1})f(t_k)\| - \left\| \sum_{j \neq k} (x_j - x_{j-1})f(t_k) \right\| \\ &> 1 + \|L\|. \end{aligned}$$

This of course contradicts (1.6). Our proof is now complete. $\square$

Remark 1.2. We conclude from the theorem above that the estimate in Theorem 1.4 holds for all Riemann integrable functions $[a, b] \rightarrow \mathcal{X}$.

1.2 Classes of Extended Riemann Integrable Functions

The ultimate goal of this section is to prove that continuous functions $[a, b] \rightarrow \mathcal{X}$ are Riemann integrable in the extended sense. This result is to be expected, but it nonetheless fundamental to the calculus we are developing. Certainly, we will mostly be dealing with functions that are *at least* continuous.

We begin with a useful criterion for Riemann integrability.

Theorem 1.6. *A function $f : [a, b] \rightarrow \mathcal{X}$ is Riemann integrable if and only if for each $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})\| < \varepsilon$$

for all tagged partitions $\dot{\mathcal{P}}, \dot{\mathcal{Q}}$ of $[a, b]$ whose meshes are strictly less than δ .

Proof. First, suppose that $f \in \mathcal{R}([a, b], \mathcal{X})$. Given $\varepsilon > 0$, we choose $\delta > 0$ with the property that

$$\left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| < \frac{\varepsilon}{2}$$

for all tagged partitions $\dot{\mathcal{P}}$ of $[a, b]$ whose mesh is strictly less than δ . Then, if both $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ have meshes strictly less than δ , the triangle inequality gives

$$\|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})\| \leq \left\| S(f; \dot{\mathcal{P}}) - \int_a^b f \right\| + \left\| \int_a^b f - S(f; \dot{\mathcal{Q}}) \right\| < \varepsilon,$$

as was required. Conversely, we must show that f is Riemann integrable. Given $n \in \mathbb{N}$, let $\delta_n > 0$ be such that

$$\|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})\| < \frac{1}{n}$$

for all tagged partitions $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ having meshes strictly less than δ_n . Of course, we may assume that $\delta_{n+1} \leq \delta_n$ for all $n \in \mathbb{N}$. For each index $n \in \mathbb{N}$, let $\dot{\mathcal{P}}_n$ be a tagged partition of $[a, b]$ whose mesh is strictly less than δ_n . For any two $n \geq m$ we then have

$$\|S(f; \dot{\mathcal{P}}_n) - S(f; \dot{\mathcal{P}}_m)\| < \frac{1}{m}.$$

The above shows that the sequence $(S(f; \dot{\mathcal{P}}_n))_{n \in \mathbb{N}}$ is Cauchy in $\mathcal{X}$. Since $\mathcal{X}$ is a Banach space, there exists a point $\Lambda \in \mathcal{X}$ such that

$$\lim_{n \rightarrow \infty} S(f; \dot{\mathcal{P}}_n) = \Lambda.$$

We now prove that f is Riemann integrable on $[a, b]$, and that $\int_a^b f$ is equal to Λ . Let $\varepsilon > 0$ be given and select $N_1 \in \mathbb{N}$ such that

$$\|S(f; \dot{\mathcal{P}}_n) - \Lambda\| < \frac{\varepsilon}{2}$$

for all $n \geq N_1$. Let $N_2 \in \mathbb{N}$ be such that

$$\frac{1}{N_2} < \frac{\varepsilon}{2}.$$

Thus, if $n, m \geq N_2$ there holds

$$\|S(f; \dot{\mathcal{P}}_n) - S(f; \dot{\mathcal{P}}_m)\| < \frac{1}{N_2} < \frac{\varepsilon}{2}.$$

Set now $N := \max(N_1, N_2)$ and let $\dot{\mathcal{P}}$ be any tagged partition of $[a, b]$ whose mesh is strictly less than δ_N . The triangle inequality subsequently yields

$$\begin{aligned} \|S(f; \dot{\mathcal{P}}) - \Lambda\| &\leq \|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{P}}_N)\| + \|S(f; \dot{\mathcal{P}}_N) - \Lambda\| \\ &< \frac{1}{N} + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

This yields the desired conclusion. $\square$

Using the above criterion, we will show that all continuous functions from $[a, b]$ into $\mathcal{X}$ are Riemann integrable. But first, we must quickly introduce some terminology. Suppose that $\mathcal{P}$ and $\mathcal{Q}$ are partitions of an interval $[a, b]$. Let

$$a = x_0 < x_1 < \cdots < x_n = b$$

denote the endpoints of those intervals in $\mathcal{P}$; let

$$a = y_0 < y_1 < \cdots < y_m = b$$

be those of intervals in $\mathcal{Q}$. The partition $\mathcal{Q}$ of $[a, b]$ is called a **refinement** of $\mathcal{P}$ if every x_i is equal to some y_j (in the lists above). In this case, we will agree to write $\mathcal{P} \subseteq \mathcal{Q}$.

The **common refinement** of the partitions $\mathcal{P}$ and $\mathcal{Q}$ consists of all those intervals formed by rearranging the x_i 's and y_j 's into a strictly increasing list of real numbers. This common refinement is often denoted by $\mathcal{P} \vee \mathcal{Q}$.

Lemma 1.7. *Let $f : [a, b] \rightarrow \mathcal{X}$ be continuous. For each $\varepsilon > 0$ there exists $\delta > 0$ such that for any tagged partition $\dot{\mathcal{P}}$ of $[a, b]$ whose mesh is strictly less than δ , and any refinement $\dot{\mathcal{Q}}$ of $\dot{\mathcal{P}}$, there holds*

$$\|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})\| < \varepsilon.$$

Proof. Since f is continuous and $[a, b]$ is compact, the function f is uniformly continuous. Therefore, given $\varepsilon > 0$, we may find $\delta > 0$ such that

$$\|f(x) - f(y)\| < \frac{\varepsilon}{b - a}$$

for all $x, y \in [a, b]$ satisfying $|x - y| < \delta$. Let $\dot{\mathcal{P}}$ be a tagged partition of $[a, b]$ with $\|\dot{\mathcal{P}}\| < \delta$. Suppose that $\dot{\mathcal{Q}}$ is any refinement of $\dot{\mathcal{P}}$ with its own set of tags. Any subinterval $[x_{i-1}, x_i]$ of $\dot{\mathcal{P}}$ may then be decomposed into a finite union of subintervals from $\dot{\mathcal{Q}}$

$$[x_{i-1}, x_i] = \bigcup_{j=1}^{n_i} [y_{i,j-1}, y_{i,j}].$$

Therefore,

$$\begin{aligned} \|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})\| &= \left\| \sum_{i=1}^n (x_i - x_{i-1}) f(x_i^*) - \sum_{i=1}^n \sum_{j=1}^{n_i} (y_{i,j} - y_{i,j-1}) f(y_{i,j}^*) \right\| \\ &= \left\| \sum_{i=1}^n \sum_{j=1}^{n_i} (y_{i,j} - y_{i,j-1}) f(x_i^*) - \sum_{i=1}^n \sum_{j=1}^{n_i} (y_{i,j} - y_{i,j-1}) f(y_{i,j}^*) \right\| \\ &\leq \sum_{i=1}^n \sum_{j=1}^{n_i} (y_{i,j} - y_{i,j-1}) \|f(x_i^*) - f(y_{i,j}^*)\| \end{aligned}$$

where, of course, the x_i^* are tags from $\dot{\mathcal{P}}$ and the $y_{i,j}^*$ are those from $\dot{\mathcal{Q}}$. However, since the mesh of $\dot{\mathcal{P}}$ is strictly less than δ , the above implies that

$$\|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})\| < \frac{\varepsilon}{b - a} \sum_{i=1}^n \sum_{j=1}^{n_i} (y_{i,j} - y_{i,j-1}) = \varepsilon.$$

This proves the lemma. $\square$

We can now prove our cornerstone theorem.

Theorem 1.8. *Let $f : [a, b] \rightarrow \mathcal{X}$ be a continuous function. Then, f is extended Riemann integrable on $[a, b]$.*

Proof. This proof relies upon the previous lemma. Let $\varepsilon > 0$ be given and fix $\delta > 0$ as in the statement of our lemma. Let $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ be two tagged partitions of

$[a, b]$, both having mesh strictly less than δ . Letting $\dot{\mathcal{P}} \vee \dot{\mathcal{Q}}$ denote their common refinement, we get

$$\begin{aligned} \|s(f; \dot{\mathcal{P}}) - s(f; \dot{\mathcal{Q}})\| &\leq \|s(f; \dot{\mathcal{P}}) - s(f; \dot{\mathcal{P}} \vee \dot{\mathcal{Q}})\| \\ &\quad + \|s(f; \dot{\mathcal{P}} \vee \dot{\mathcal{Q}}) - s(f; \dot{\mathcal{Q}})\| \end{aligned}$$

whence

$$\|s(f; \dot{\mathcal{P}}) - s(f; \dot{\mathcal{Q}})\| < 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, Theorem 1.6 gives that f is integrable. This completes the proof. $\square$

We finally know that continuous functions $[a, b] \rightarrow \mathcal{X}$ are (extended) Riemann integrable. Although this was a significant amount of work, it also serves as a reminder that our definition is “sane”.

Lemma 1.9. *Let $f : [a, b] \rightarrow \mathcal{X}$ be integrable. If $a \leq c \leq b$ then*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

As the proof is tedious and technical, we choose to omit the proof. Of course, the proof of this identity for the classical Riemann integral can easily be adapted for the lemma above. We leave the details to the reader.

Let us now conclude with a theorem which sharpens the estimate proven in Theorem 1.4. Here, we will use the fact that continuous functions are Riemann integrable. We also recall that continuous functions $[a, b] \rightarrow \mathbb{R}$ are Riemann integrable in the standard sense. Alternatively, one can note that our definition with $\mathcal{X} = \mathbb{R}$ corresponds with the classical Riemann integral.

Theorem 1.10. *Let $f : [a, b] \rightarrow \mathcal{X}$ be continuous. Then,*

$$\left\| \int_a^b f \right\| \leq \int_a^b \|f\| \, dx.$$

Proof. Other than the Riemann integrability of f on $[a, b]$, we only need the continuity of f to ensure that $\|f\|$ is Riemann integrable (in the classical sense) on $[a, b]$. Indeed, by composition, the map $\|f\| : [a, b] \rightarrow \mathbb{R}$ is continuous and hence *classically Riemann integrable* on $[a, b]$.

Let $\{\dot{\mathcal{P}}_n\}_{n=1}^{\infty}$ be a sequence of partitions of $[a, b]$ whose mesh tends to zero as $n \rightarrow \infty$. Then, since $f \in \mathcal{R}([a, b], \mathcal{X})$, we have

$$\int_a^b f = \lim_{n \rightarrow \infty} S(f; \dot{\mathcal{P}}_n).$$

By continuity of the norm, this extends to

$$\left\| \int_a^b f \right\| = \lim_{n \rightarrow \infty} \|S(f; \dot{\mathcal{P}}_n)\|.$$

Fix an index $n \in \mathbb{N}$ and let

$$a = x_0 < \cdots < x_m = b$$

denote the end points of the sub-intervals in $\dot{\mathcal{P}}_n$, and let x_j^* be the tag from the j^{th} interval. Clearly,

$$\|S(f; \dot{\mathcal{P}}_n)\| \leq \sum_{j=1}^m (x_j - x_{j-1}) \|f(x_j^*)\|$$

which is itself a Riemann sum for the function $\|f(\cdot)\|$ on $[a, b]$. Since $\|f\|$ is Riemann integrable on $[a, b]$, putting all the pieces together gives

$$\left\| \int_a^b f \right\| = \lim_{n \rightarrow \infty} \|S(f; \dot{\mathcal{P}}_n)\| \leq \lim_{n \rightarrow \infty} S(\|f\|; \dot{\mathcal{P}}_n) = \int_a^b \|f\|.$$

This completes the proof. $\square$

1.3 Transformations of Riemann Integrable Functions

In this subsection, we fix Banach spaces $\mathcal{X}$ and $\mathcal{Y}$ over the same ground field $\mathbb{K}$. Before giving a useful proposition, we recall some definitions from functional analysis. We say a linear operator $T : \mathcal{X} \rightarrow \mathcal{Y}$ is *bounded* if there exists $M > 0$ such that

$$\|T(x)\| \leq M \|x\|, \quad \forall x \in \mathcal{X}.$$

One can show that a linear operator $T : \mathcal{X} \rightarrow \mathcal{Y}$ is bounded if and only if it is continuous. Commonly, one writes $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ to denote the set of all bounded

linear operators $\mathcal{X} \rightarrow \mathcal{Y}$. Obviously, $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is a vector space over $\mathbb{K}$. One can make $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ into a normed space by defining for each $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ the norm

$$\|T\| := \inf \{M \geq 0 : \|T(x)\| \leq M \|x\|, \forall x \in \mathcal{X}\}.$$

It can be shown that the infimum above is achieved since the set

$$\{M \geq 0 : \|T(x)\| \leq M \|x\|, \forall x \in \mathcal{X}\}$$

is non-empty, closed, and bounded from below. Thus, $\|Tx\| \leq \|T\| \|x\|$ for all $x \in \mathcal{X}$. Actually, since $\mathcal{Y}$ is a Banach space, the space $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is also a Banach space.

With these formalities out of the way, we state the following.

Proposition 1.11. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces and $A : [a, b] \rightarrow \mathcal{L}(\mathcal{X}, \mathcal{Y})$ continuous. Fix $x \in \mathcal{X}$. The function*

$$A(\cdot)x : [a, b] \longrightarrow \mathcal{Y}, \quad t \mapsto A(t)(x)$$

is continuous, and hence Riemann integrable, on $[a, b]$. Furthermore,

$$\int_a^b A(t)x = \left(\int_a^b A(t) \right) x.$$

Proof. First, fix $t, s \in [a, b]$. Then, $A(t) - A(s)$ is itself a bounded linear operator $\mathcal{X} \rightarrow \mathcal{Y}$. As a result, we get that

$$\|A(t)x - A(s)x\| \leq \|A(t) - A(s)\| \|x\|.$$

Since A is continuous, we see from the above that $A(\cdot)x$ is a continuous function $[a, b] \rightarrow \mathcal{Y}$. In particular, $A(\cdot)x$ is Riemann integrable. Next, we prove the identity. Let $\{\dot{\mathcal{P}}_n\}_{n=1}^\infty$ be a sequence of tagged partitions of $[a, b]$ whose mesh tends to zero as $n \rightarrow \infty$. Since $A(\cdot)x$ is Riemann integrable on $[a, b]$, we have that

$$\int_a^b A(t)x = \lim_{n \rightarrow \infty} S(A(\cdot)x; \dot{\mathcal{P}}_n).$$

However, it is not difficult to see that

$$S(A(\cdot)x; \dot{\mathcal{P}}_n) = \left(S(A(\cdot); \dot{\mathcal{P}}_n) \right) (x)$$

for each $n \in \mathbb{N}$. Moreover, the function $A(\cdot)$ is continuous and takes values in a Banach space $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. Therefore, $A(\cdot)$ is integrable and $\int_a^b A(t)$ will be a bounded linear operator $\mathcal{X} \rightarrow \mathcal{Y}$. In particular, we get that

$$\lim_{n \rightarrow \infty} S(A(\cdot); \dot{\mathcal{P}}_n) = \int_a^b A(t) \quad (1.7)$$

in the space $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. But, if a sequence $\{T_n\}_{n=1}^\infty$ of bounded linear operators in $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ converges to an element $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, then $T_n(x)$ converges to $T(x)$ in $\mathcal{Y}$ for each fixed $x \in \mathcal{X}$. Indeed, this follows from the inequality

$$\|T_n(x) - T(x)\| \leq \|T_n - T\| \|x\|.$$

Finally, putting all the pieces together, we infer that

$$\int_a^b A(t)x = \lim_{n \rightarrow \infty} S(A(\cdot); \dot{\mathcal{P}}_n)(x) = \left(\int_a^b A(t) \right)(x).$$

□

The following proposition admits a similar method of proof.

Proposition 1.12. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces over a field $\mathbb{K}$. Assume that $f : [a, b] \rightarrow \mathcal{X}$ is continuous and let $A : \mathcal{X} \rightarrow \mathcal{Y}$ be a bounded linear operator. There holds the following,*

$$\int_a^b A(f) = A\left(\int_a^b f\right).$$

Proof. Clearly, the composite

$$[a, b] \xrightarrow{f} \mathcal{X} \xrightarrow{A} \mathcal{Y}$$

is continuous, and hence $A(f)$ is Riemann integrable on $[a, b]$. Let $\{\dot{\mathcal{P}}_n\}_{n=1}^\infty$ be a sequence of tagged partitions for the interval $[a, b]$ whose mesh tends to 0 as $n \rightarrow \infty$. Using the integrability of $A(f)$, one has

$$\int_a^b A(f) = \lim_{n \rightarrow \infty} S(A(f); \dot{\mathcal{P}}_n).$$

By the linearity of the operator A , it is easy to see that for each n

$$S(A(f); \dot{\mathcal{P}}_n) = A(S(f; \dot{\mathcal{P}}_n)).$$

Invoking the integrability of f , we simultaneously obtain

$$\lim_{n \rightarrow \infty} S(f; \dot{\mathcal{P}}_n) = \int_a^b f.$$

Finally, the continuity of A yields

$$\int_a^b A(f) = \lim_{n \rightarrow \infty} S(A(f); \dot{\mathcal{P}}_n) = \lim_{n \rightarrow \infty} A(S(f; \dot{\mathcal{P}}_n)) = A\left(\int_a^b f\right).$$

This concludes the proof. $\square$

1.3.1 The Continuity of the Riemann Integral

Let $f : [a, b] \rightarrow \mathcal{X}$ be a continuous map. Here, $[a, b] \subset \mathbb{R}$ is a non-trivial compact interval. For each $c \in [a, b]$, we will adopt the convention that $\int_c^c f = 0$. As pointed out at the start of this chapter, we can certainly make sense of this when viewing $\{c\}$ as the trivial interval $[c, c]$. What we show here, is that this “special case” does not affect the continuity of the Riemann integral.

Theorem 1.13. *Let $f : [a, b] \rightarrow \mathcal{X}$ be Riemann integrable. Consider the function*

$$F : [a, b] \rightarrow \mathcal{X}, \quad F(t) := \int_a^t f(t).$$

Then, $F(t)$ is well defined, continuous, and satisfies $F(a) = 0$.

Proof. Since f is Riemann integrable, it is bounded on $[a, b]$. By definition, we also have $F(a) = 0$. Now, to prove the continuity of the integral. We first check continuity at a . Let $t \geq a$; if $t = a$ then

$$\left\| \int_a^t f \right\| = 0$$

by definition. If instead $t \in (a, b]$, then

$$\left\| \int_a^t f(t) \right\| \leq (t - a) \sup_{t \in [a, b]} \|f(x)\| =: M|t - a|,$$

where $M > 0$ is defined in the obvious way. Notice that the above actually holds even when $t = a$. This shows that $\int_a^t f$ is continuous at $t = a$. Similarly, one can show that $\int_a^t f$ is a continuous function at $t = b$. If $c \in (a, b)$ the argument is a little different. Let $\varepsilon > 0$ be given and choose M such that

$$\sup_{t \in [a, b]} \|f(t)\| < M < \infty.$$

Pick $\delta > 0$ satisfying the inequality

$$\delta < \frac{\varepsilon}{M}.$$

If $0 < t - c < \delta$, then

$$\left\| \int_a^t f - \int_a^c f \right\| = \left\| \int_c^t f \right\| \leq M|t - c| < \varepsilon.$$

If $-\delta < t - c < 0$ one similarly has

$$\left\| \int_a^t f - \int_a^c f \right\| = \left\| \int_t^c f \right\| \leq M|c - t| < \varepsilon.$$

Since $\left\| \int_a^t f - \int_a^c f \right\| = 0$ when $t = c$, the continuity of $F(t)$ on $[a, b]$ follows. $\square$

Actually, one can partially strengthen the above.

Corollary 1.14. *Let $f : [a, b] \rightarrow \mathcal{X}$ be Riemann integrable and fix $c \in [a, b]$. The function defined by*

$$F : [a, b] \rightarrow \mathcal{X}, \quad F(t) := \int_c^t f$$

is continuous.

Proof. By the previous theorem, the function

$$F_1(t) := \begin{cases} 0, & t \leq c, \\ \int_c^t f, & t > c \end{cases}$$

is continuous on $[a, b]$. By the same token, so is the map

$$F_2(t) := \begin{cases} 0, & t \geq c, \\ \int_t^c f, & t < c. \end{cases}$$

The result then follows from the identity

$$F(t) = F_1(t) - F_2(t), \quad t \in [a, b].$$

$\square$

1.4 The Fundamental Theorem of Calculus

So far, our main goal has been to develop a generalized calculus for functions taking values in Banach spaces. So far, we have only discussed the integration portion of such a theory. In this section, we discuss a notion of differentiation and prove a generalization of the fundamental theorem of calculus. In later sections, we will discuss alternative derivatives/differentials and it is important that the reader distinguish these various definitions.

Throughout this section, $\mathcal{X}$ denotes a Banach space and I is a bounded interval in $\mathbb{R}$. We recall some asymptotic notation.

Definition 5. A function $f : I \rightarrow \mathcal{X}$ is called *differentiable* at a point $c \in I$ if there exists $\lambda \in \mathcal{X}$ such that

$$f(t) - f(c) = \lambda(t - c) + o(|t - c|)$$

as $t \rightarrow c$, with $t \in I$. That is, if for each $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|f(t) - f(c) - \lambda(t - c)\| \leq \varepsilon|t - c|, \quad \text{for all } t \in I \text{ with } |t - c| < \delta.$$

In this case, we write $f'(c) = \lambda$ and call this the *derivative* of f at c . The map f is called differentiable on I if f is differentiable at all points $c \in I$.

Our definition of differentiability appears to be substantially different from the “standard definition” one encounters in calculus. Below, we show that this is not at all the case – we have simply chosen to give the definition in a form that is more common in functional analysis. After all, one will not always be able to work with functions defined on an interval. For such functions, the formulation above is easier to extend.

Lemma 1.15. Let $I \subseteq \mathbb{R}$ be an interval and fix a point $c \in I$. A function $f : I \rightarrow \mathcal{X}$ is differentiable at c , with derivative $f'(c) \in \mathcal{X}$, if and only if

$$\lim_{\substack{t \rightarrow c \\ t \in I}} \frac{f(t) - f(c)}{t - c}$$

exists and is equal to $f'(c)$.

Proof. Suppose first that $f'(c)$ exists according to the definition. Given $\varepsilon > 0$, we choose $\delta > 0$ such that

$$\|f(t) - f(c) - (t - c)f'(c)\| \leq \frac{\varepsilon}{2}|t - c|$$

for all $t \in I$ with $|t - c| < \delta$. If $0 < |t - c| < \delta$, we then have

$$\|f(t) - f(c) - (t - c)f'(c)\| \leq \frac{\varepsilon}{2}|t - c| < \varepsilon|t - c|.$$

For such $t > c$, we divide through by $|t - c| = (t - c) > 0$ to obtain

$$\left\| \frac{f(t) - f(c)}{t - c} - f'(c) \right\| < \varepsilon.$$

If instead $-\delta < t - c < 0$, we divide both sides by $|t - c| = (c - t)$ yielding

$$\begin{aligned} \left\| \frac{f(t) - f(c)}{c - t} - \frac{t - c}{c - t}f'(c) \right\| &= \left\| -\frac{f(t) - f(c)}{t - c} + f'(c) \right\| \\ &= \left\| \frac{f(t) - f(c)}{t - c} - f'(c) \right\| < \varepsilon. \end{aligned}$$

Hence,

$$\lim_{\substack{t \rightarrow c \\ t \in I}} \frac{f(t) - f(c)}{t - c} = f'(c) \quad \text{in } \mathcal{X}.$$

Conversely, suppose this last equation holds true for f . Fixing $\varepsilon > 0$, we find $\delta > 0$ such that

$$\left\| \frac{f(t) - f(c)}{t - c} - f'(c) \right\| < \varepsilon$$

for all $t \in I$ with $0 < |t - c| < \delta$. Clearly, for these t :

$$\left\| \frac{f(t) - f(c)}{t - c} - f'(c) \right\| = \frac{1}{|t - c|} \|f(t) - f(c) - (t - c)f'(c)\| < \varepsilon.$$

But this implies that

$$\|f(t) - f(c) - (t - c)f'(c)\| < \varepsilon|t - c|.$$

Since equality holds for $t = c$, it follows that

$$\|f(t) - f(c) - (t - c)f'(c)\| \leq \varepsilon|t - c|$$

for all $t \in I$ with $|t - c| < \delta$. □

It is not difficult to see that the usual addition rule for derivatives continues to hold in this abstract structure. Indeed, this is a simple application of the triangle inequality. Similarly, scalar multiplication commutes with differentiation. We summarize these with the following proposition.

Proposition 1.16. *Let $f, g : I \rightarrow \mathcal{X}$ be differentiable at a point $c \in I$. Then,*

1. $f + g$ is differentiable at c with derivative $(f + g)'(c) = f'(c) + g'(c)$.
2. If $\gamma \in \mathbb{K}$, the function $\gamma \cdot f(t)$ is differentiable at c with derivative equal to $c \cdot f'(c)$.
3. Suppose in addition that $\mathcal{X}$ is a Banach algebra over $\mathbb{K}$ with multiplication operation $*$. Then $f * g$ is differentiable at $c \in I$ with derivative equal to:

$$(f * g)'(c) = f'(c) * g(c) + f(c) * g'(c).$$

Proof. The proof is left as a routine exercise to the reader. □

As we should expect, differentiable functions are at least continuous.

Proposition 1.17. *Let $f : I \rightarrow \mathcal{X}$ be differentiable at a point $c \in I$. Then, f is continuous at c .*

Proof. First, let $\delta > 0$ be such that

$$\|f(t) - f(c) - \lambda(t - c)\| \leq |t - c|, \quad \text{for all } t \in I \text{ with } |t - c| < \delta.$$

If $t \in I$ satisfies $|t - c| < \delta$, the triangle inequality gives

$$\|f(t) - f(c)\| \leq |t - c| + \|\lambda(t - c)\| = |t - c| + \|\lambda\| |t - c|.$$

Let $\varepsilon > 0$ be given; since the right hand side of the above equation tends to 0 as $t \rightarrow c$, we can choose $\delta' > 0$ such that

$$|t - c| + \|\lambda\| |t - c| < \varepsilon, \quad \text{whenever } |t - c| < \delta'.$$

Take δ'' to be the minimum of δ and δ' . For each $t \in I$ satisfying $|t - c| < \delta''$, all of this gives

$$\|f(t) - f(c)\| \leq |t - c| + \|\lambda\| |t - c| < \varepsilon.$$

We infer that f is continuous at the point c . □

1.4.1 The Fundamental Theorem of Calculus: Form 1

In this subsection, we prove a version of the fundamental theorem of calculus. The proof is standard, but the result will be used to deduce the second (and more applicable) form of the fundamental theorem. This will be accomplished in the subsequent subsection. For now, we handle the following.

Theorem 1.18. *Suppose that $\mathcal{X}$ is a Banach space over $\mathbb{K}$ and let $f : [a, b] \rightarrow \mathcal{X}$ be continuous. Define*

$$F : [a, b] \rightarrow \mathcal{X}, \quad F(x) := \int_a^x f.$$

Then, F is continuously differentiable on $[a, b]$ and satisfies the rule $F' \equiv f$ on $[a, b]$.

Proof. Let us fix x, y with $a \leq x < y \leq b$. Then, using the additivity of the integral, we get that

$$F(y) - F(x) = \int_a^y f - \int_a^x f = \int_x^y f.$$

Moreover, if the integral is taken with respect to a variable t ,

$$\int_x^y f(x) dt = (y - x)f(x).$$

Therefore,

$$\|F(y) - F(x) - (y - x)f(x)\| = \left\| \int_x^y (f(t) - f(x)) dt \right\| \quad (1.8)$$

$$\leq \|x - y\| \sup_{x \leq t \leq y} \|f(t) - f(x)\|. \quad (1.9)$$

Being continuous on $[a, b]$, the function f is also uniformly continuous on this interval. The above therefore implies that

$$\lim_{y \searrow x} \frac{F(y) - F(x)}{y - x}$$

exists and is equal to $f(x)$. Let us now treat the “left hand derivative”. Of course, the above already covers the case $x = a$ and as such we may as well assume that $x > a$. Let y be such that $a \leq y < x$. As above, we get

$$F(y) = \int_a^y f \quad \text{and} \quad F(x) = \int_a^x f$$

so that $F(x) - F(y) = \int_y^x f$. Or, rather,

$$F(y) - F(x) = - \int_y^x f.$$

Therefore,

$$\begin{aligned} \|F(y) - F(x) - (y - x)f(x)\| &= \left\| - \int_y^x f - (y - x)f(x) \right\| \\ &= \left\| \int_y^x f - (x - y)f(x) \right\| \\ &\leq (x - y) \sup_{y \leq t \leq x} \|f(t) - f(x)\|. \end{aligned}$$

Thus, the same arguments used in (1.8)-(1.9) apply to give

$$\lim_{y \nearrow x} \frac{F(y) - F(x)}{y - x} = f(x).$$

This proves the claim. $\square$

By way of a similar argument, one can establish the following. We leave the details of the proof as an exercise to the reader.

Proposition 1.19. *Suppose that $\mathcal{X}$ is a Banach space over $\mathbb{K}$ and let $f : [a, b] \rightarrow \mathcal{X}$ be continuous. Define*

$$F : [a, b] \rightarrow \mathcal{X}, \quad F(x) := \int_a^x f.$$

Then, F is continuously differentiable on $[a, b]$ and satisfies $F' \equiv f$ on $[a, b]$.

1.4.2 The Fundamental Theorem of Calculus: Form 2

This section is devoted to a proof of the following identity:

$$u(b) - u(a) = \int_a^b u'.$$

Of course, we will need to place additional regularity conditions on this function u . At the very least, we need u to be differentiable with a derivative that is extended Riemann integrable.

Let $f : [a, b] \rightarrow \mathcal{X}$ be differentiable. Then, $f'(t)$ exists for all $t \in [a, b]$. Thus, we can view $f'(t)$ as a function $[a, b] \rightarrow \mathcal{X}$. We will say that f is *continuously differentiable* if f' exists and is continuous on $[a, b]$. Notice then that f' must also be Riemann integrable on $[a, b]$. More generally, for $k \geq 1$, we define

$$C^k([a, b], \mathcal{X}) := \{f : [a, b] \rightarrow \mathcal{X} : f^{(k)} \text{ exists and is continuous}\}.$$

Here, we will mostly be interested in the space $C^1([a, b], \mathcal{X})$. Our main goal here is to establish the following:

Theorem 1.20. *Let $\mathcal{X}$ be a Banach space over $\mathbb{R}$ or $\mathbb{C}$ and suppose $f : [a, b] \rightarrow \mathcal{X}$ is continuously differentiable on $[a, b]$. Then, there holds the following*

$$\int_a^b f' = f(b) - f(a). \quad (1.10)$$

The proof will require a “lemma” that is of significant interest in its own right. Of course, the statement will not be the least bit surprising. However, the fact that the claim holds true is certainly reassuring.

Proposition 1.21. *Let $f : [a, b] \rightarrow \mathcal{X}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f'(t) = 0$ for all $t \in (a, b)$, then, f is constant.*

Proof. Let $a < c < d < b$; we will show that $f(c) = f(d)$. It would then follow by continuity of f is constant. Let $\varepsilon > 0$ be given and define

$$E := \{t \in [c, d] : \|f(t) - f(c)\| \leq \varepsilon(t - c)\}.$$

The set E is non-empty since $c \in E$ by definition. Let s denote $\sup E$. Since $s = \sup E$, there exists a sequence of elements (t_n) in E converging to s as $n \rightarrow \infty$. Invoking the continuity of f and the norm, it is easy to see that

$$\|f(s) - f(c)\| = \lim_{n \rightarrow \infty} \|f(t_n) - f(c)\| \leq \lim_{n \rightarrow \infty} \varepsilon(t_n - c) = \varepsilon(s - c).$$

Hence, $s \in E$ as well. Suppose, by way of contradiction, that $s < d$. Using that $f'(s) = 0$, we can find $t \in (s, d)$ such that

$$\|f(t) - f(s)\| = \|f(t) - f(s) - f'(s)(t - s)\| \leq \varepsilon(t - s)$$

implying that

$$\begin{aligned} \|f(t) - f(c)\| &\leq \|f(t) - f(s)\| + \|f(s) - f(c)\| \leq \varepsilon(t - s) + \varepsilon(s - c) \\ &= \varepsilon(t - c). \end{aligned}$$

That is, $t \in E$ which contradicts the upper-bound property of s . As a result, we conclude that $d = s$ and that

$$\|f(d) - f(c)\| \leq \varepsilon(d - c).$$

Since $\varepsilon > 0$ was taken arbitrarily, the proof is complete. $\square$

Having dispensed with this technicality, we can prove (1.10).

Proof of Theorem 1.20. Consider the function $G : [a, b] \rightarrow \mathcal{X}$ defined by the quantity

$$G(x) := \left(\int_a^x f' \right) - f(x).$$

By Theorem 1.18, $G(x)$ is continuously differentiable on $[a, b]$ and satisfies

$$G'(x) = f'(x) - f'(x) = 0$$

for each $x \in [a, b]$. Invoking the previous proposition, we see that $G(x)$ must be constant. That is,

$$-f(a) = G(a) = G(b) = -f(b) + \int_a^b f'.$$

The proof is now complete. $\square$

From this one can easily prove the so called “mean value inequality”. We leave its proof as an exercise to the reader.

Corollary 1.22. *Let $f : [a, b] \rightarrow \mathcal{X}$ be continuously differentiable. Then,*

$$\|f(b) - f(a)\| \leq (b - a) \cdot \|f'\|_\infty.$$

1.5 The Inverse and Implicit Function Theorems

In this section, we establish the infamous inverse and implicit function theorems, well known to us from calculus, within the context of Banach spaces. In fact, we will prove these theorems for functions between *arbitrary* Banach spaces. To do so, we must first introduce the *Fréchet* and *Gateaux* derivatives.

As before, we remind the reader that $\mathbb{K}$ denotes one of the ground fields $\mathbb{R}$ or $\mathbb{C}$. Moreover, $\mathcal{X}$ and $\mathcal{Y}$ are always Banach spaces over this field $\mathbb{K}$.

Definition 6. Let $U \subseteq \mathcal{X}$ be an open set. A mapping $f : U \rightarrow \mathcal{Y}$ is called *Fréchet differentiable* at a point $x \in U$ if

$$f(x + h) = f(x) + \Lambda(h) + o(\|h\|) \quad \text{as } h \rightarrow 0, h \in \mathcal{X},$$

for some bounded linear operator $\Lambda : \mathcal{X} \rightarrow \mathcal{Y}$. We then define $Df(x) := \Lambda$. The function f is called *Fréchet differentiable* if $Df(x)$ exists at all points $x \in U$.

If f is indeed Fréchet differentiable on U , then there is a natural map

$$Df : U \rightarrow \mathcal{L}(\mathcal{X}, \mathcal{Y}), \quad x \mapsto Df(x).$$

This resulting function Df shall henceforth be called the *Fréchet derivative* of the function f . We say that f is continuously Fréchet differentiable if Df is continuous. The set of all such maps will be denoted by $C^1(U, \mathcal{Y})$ or $C_F^1(U, \mathcal{Y})$.

It is not difficult to see that Fréchet differentiability at a point implies continuity at that point. The verification of this fact is left as an exercise to the reader.

Lemma 1.23. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces over a the ground field $\mathbb{K}$. Suppose that $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is a bijective (and hence an isomorphism). If $H \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is a bounded linear operator such that $\|H\|$ is sufficiently small, then $T + H$ is also bijective (and hence an isomorphism).*

Proof. Consider the map $(T + H)T^{-1} = I + HT^{-1}$, where I is the identity map $\mathcal{Y} \rightarrow \mathcal{Y}$. Clearly, this is a bounded linear operator. On the other hand, it is clear that

$$\|(T + H)T^{-1} - I\| = \|HT^{-1}\| \leq \|H\| \|T^{-1}\|.$$

For $\|H\|$ very small, the above can be made to be strictly less than 1. Consequently, we get that $(T + H)T^{-1}$ is invertible. It follows that $T + H$ is invertible, as was required. $\square$

The following corollary is simply a restated version of the above.

Corollary 1.24. *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. Let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ be a bijection. There exists $\delta > 0$ such that $\|T - S\| < \delta$, with $S \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, implies that S is bijective.*

Proof. Follows from the previous lemma with $S = (S - T) + T$. $\square$

We are now prepared to give the rather messy proof of the inverse function theorem.

Theorem 1.25 (Implicit Function Theorem). *Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces over $\mathbb{K}$, and let $U \subseteq \mathcal{X}$ be open. Suppose that $f : U \rightarrow \mathcal{Y}$ is continuously Fréchet differentiable. Fix $x \in U$ and assume that $Df(x)$ is bijective. There exists a neighbourhood $V \subseteq U$, of x , such that $f : V \rightarrow \mathcal{Y}$ is injective and $f^{-1} : f(V) \rightarrow \mathcal{X}$ is continuously Fréchet differentiable.*

Proof. If $z \in f(\mathcal{X}) \subseteq \mathcal{Y}$, we consider the map

$$\phi : U \rightarrow \mathcal{X}, \quad \phi(y) := y + [Df(x)]^{-1} (z - f(y)).$$

Since $Df(x)$ is injective, so is $[Df(x)]^{-1}$. Therefore, we see that $\phi(y) = y$ if and only if $f(y) = z$. Let $y, y' \in U$ be sufficiently close to the point $x \in U$. Then,

$$\phi(y) - \phi(y') = y - y' + [Df(x)]^{-1} (f(y') - f(y)) \quad (1.11)$$

$$= [Df(x)]^{-1} \{f(y') - f(y) + Df(x)(y - y')\}. \quad (1.12)$$

Now, since y and y' are “close” to the point x , they are close to themselves. Using that f is Fréchet differentiable over $U \subseteq \mathcal{X}$, we get that

$$f(y') - f(y) = Df(y)(y' - y) + o(\|y' - y\|).$$

Using this together with

$$\begin{aligned} \phi(y) - \phi(y') &= [Df(x)]^{-1} \{(Df(x) - Df(y))(y - y')\} \\ &\quad + [Df(x)]^{-1} (o(\|y' - y\|)). \end{aligned}$$

Since $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces, the inverse $[Df(x)]^{-1}$ is a bounded linear operator $\mathcal{Y} \rightarrow \mathcal{X}$. Therefore, the above can be refined to obtain

$$\phi(y) - \phi(y') = [Df(x)]^{-1} \{(Df(x) - Df(y))(y - y')\} + o(\|y' - y\|). \quad (1.13)$$

By assumption, the map $y \mapsto Df(y)$ is continuous and so, (1.13), we can choose a closed ball $B \subseteq \mathcal{X}$ centered at x , whose interior contains y and y' , having the property that the restriction

$$\phi : B \rightarrow \mathcal{Y}$$

is a contraction. Furthermore, for each $y \in B$ one would have

$$\begin{aligned} \phi(y) - x &= [Df(x)]^{-1} [z - f(y) + Df(x)(y - x)] \\ &= [Df(x)]^{-1} [z - f(x) + f(x) + Df(x)(y - x) - f(y)] \\ &= [Df(x)]^{-1} (z - f(x)) + o(\|y - x\|); \end{aligned}$$

where this last step is a consequence of the Fréchet differentiability of f at the point $x \in B$. By the above, we can choose a closed ball $B' \subseteq B$, centered at the point x , and an open ball $E \subseteq \mathcal{Y}$ centered at $f(x)$ such that

$$\phi(B') \subseteq B', \quad \forall z \in E \subseteq \mathcal{Y}.$$

To see that this can be done, let $r > 0$ be such that

$$\|\phi(y) - x\| \leq M \|z - f(x)\| + \frac{1}{4} \|y - x\|, \quad \forall y \in B(x, r) \subseteq B.$$

Then, by taking a sufficiently small ball E about $f(x)$ in $\mathcal{Y}$, we that

$$\|\phi(y) - x\| \leq \frac{r}{4} + \frac{1}{4} \|y - x\| \leq \frac{r}{2} < r, \quad \forall z \in E, y \in B(x, r) \subseteq B.$$

Hence, the restriction $\phi : B' \rightarrow B'$ is also a contraction. By the Banach fixed point theorem, there exists a *unique point* $y \in B'$ such that

$$\phi(y) = y.$$

By our observations at the start of this proof, we get that $f(y) = z$. Since $z \in E$ was arbitrary, we conclude that $f : B' \rightarrow E$ is a bijection.

Choose $y \in U$ so close to x that $Df(y)$ is invertible. This is made possible by the continuity of Df , with the help of Corollary 1.24. If $y' \in U$ is sufficiently close to y , then differentiability gives

$$f(y') = f(y) + Df(y)(y' - y) + o(\|y' - y\|) \quad (1.14)$$

whence

$$[Df(y)]^{-1} \{f(y') - f(y)\} = y' - y + o(\|y' - y\|). \quad (1.15)$$

From the above, we may find $C > 0$ such that

$$\|y' - y\| \leq C \|f(y') - f(y)\|$$

for all y' sufficiently close to y . By our work above, we might as well take $y, y' \in B'$ where our function $f : B' \rightarrow E$ is a bijection. Put then

$$z := f(y) \quad \text{and} \quad z' := f(y');$$

using (1.15) we see that

$$f^{-1}(z') = f^{-1}(z) + [Df(y)]^{-1} \{z' - z\} + o(\|z' - z\|).$$

Hence, $Df^{-1}(z)$ exists and is equal to $[Df(y)]^{-1}$. In particular, f^{-1} is continuous on $f(B')$. This completes the proof. $\square$

Corollary 1.26 (Implicit Function Theorem). *Let $\mathcal{X}, \mathcal{Y}$ and $\mathcal{Z}$ be Banach spaces over the ground field $\mathbb{K}$. Suppose $A \subseteq \mathcal{X} \times \mathcal{Y}$ is open and let $g \in C^1(A, \mathcal{Z})$. Assume that $(a, b) \in A$ has the property that $g(a, b) = 0$ and that*

$$D_y g(a, b) : \mathcal{X} \times \mathcal{Y} \longrightarrow \mathcal{Z}$$

is invertible. Here, $D_y g(x, y)$ denotes the Fréchet derivative of the map $y \mapsto g(x, y)$ with $x \in \mathcal{X}$ fixed. There exists an open set $U \subseteq \mathcal{X}$ and a function $h \in C^1(U, \mathcal{Y})$ such that

$$h(a) = b \quad \text{and} \quad g(x, h(x)) \equiv 0, \quad \forall x \in U.$$

Proof. Let $f(x, y) := (x, g(x, y))$. Apply the previous theorem to this function. We leave the details as an exercise to the reader. $\square$

1.6 Applications to Ordinary Differential Equations

We study applications of our generalized calculus to ordinary differential equations. Let $\mathcal{X}$ be a Banach space and suppose that $I \subseteq \mathbb{R}$ is an open interval containing the point 0. Let $\Omega \subseteq \mathcal{X}$ be an open set and suppose $f : \Omega \times I \rightarrow \mathcal{X}$ is continuous. We are interested in the *initial value problem*

$$\begin{cases} u'(t) = f(u(t), t), & t \in I, \\ u(0) = x, \end{cases} \quad (1.16)$$

where $x \in \Omega$ is given. If this function f does not depend on the parameter t , then we say the system in (1.16) is *autonomous*. Let us give some crucial definitions regarding the classes of functions satisfying (1.16).

Definition 7. A *classical solution* to (1.16) is a function continuously differentiable function $u : I \rightarrow \mathcal{X}$ satisfying (1.16). A *strong solution* to (1.16) is a continuous function $u : I \rightarrow \mathcal{X}$ such that

$$u(t) = x + \int_0^t f(u(s), s) \, ds, \quad t \in I. \quad (1.17)$$

As we should intuitively expect, these two notions are equivalent. The reader should also note that we implicitly require $u(t) \in \Omega$, for all $t \in I$.

Proposition 1.27. *Let $u : I \rightarrow \mathcal{X}$ be a continuous function. Then, u is a strong solution of (1.16) if and only if it is a classical solution to (1.16).*

Proof. Suppose that u is a strong solution to (1.16). If $t \in I$ is given, then either $[t, 0]$ or $[0, t]$ is contained within I . By the continuity of u and f , the composite $f(u(s), s)$ will be continuous on $[t, 0]$ or $[0, t]$. Clearly, $u(0) = x$. Furthermore, it is obvious from the definitions that the constant map $t \mapsto x$ is differentiable on the whole of $\mathbb{R}$ with everywhere vanishing derivative. Actually, this shows that $t \mapsto x$ is continuously differentiable on I . On the other hand, invoking Theorem 1.18 and Proposition 1.19 shows that

$$\int_0^t f(u(s), s) \, ds$$

is continuously differentiable on I with derivative equal to $f(u(t), t)$ at each t . By the linearity of differentiation, it follows that u satisfies (1.16).

Conversely, suppose that u satisfies (1.16), i.e. u is a classical solution to the initial value problem. We distinguish two cases.

1. First, consider $t \geq 0$. By Theorem 1.20, it is obvious that

$$u(t) - x = u(t) - u(0) = \int_0^t u'(s) \, ds = \int_0^t f(u(s), s) \, ds.$$

Thus, $u(t) = x + \int_0^t f(u(s), s) \, ds$.

2. Suppose that $t < 0$. Then, by this same result,

$$x - u(t) = u(0) - u(t) = \int_t^0 u'(s) \, ds = \int_t^0 f(u(s), s) \, ds.$$

This certainly implies that

$$u(t) = x - \int_t^0 f(u(s), s) \, ds = x + \int_0^t f(u(s), s) \, ds.$$

□

The existence of a solution to a particular problem (e.g. a boundary, or initial value problem) is an fundamental question in analysis. Unfortunately, such results are often quite difficult to establish. It is also possible that no solutions to a given problem exist! Luckily for us, there exists a very nice existence result for “continuous enough” initial value problems.

Definition 8. A continuous map $f : \Omega \times I \rightarrow \mathcal{X}$ is called *locally Lipschitz* in the first variable if for each $r > 0$, and every compact interval $J \subset I$, there exists $C_{r,J} > 0$ such that

$$\|f(x, t) - f(y, t)\| \leq C_{r,J} \|x - y\|, \quad \forall x, y \in Q_r, t \in J.$$

Here, we define

$$Q_r := \left\{ x \in \Omega : \|x\| \leq r, \text{dist}\left(x, \Omega^c\right) \geq \frac{1}{r} \right\}. \quad (\text{Q})$$

We are now prepared to present our fundamental local existence result in full generality.

Theorem 1.28 (Picard-Lindelöf). *Let $f : \Omega \times I \rightarrow \mathcal{X}$ be locally Lipschitz in the first variable (and hence continuous). For each $r > 0$, there exists an open interval $J \ni 0$, contained in I , such that for any initial data $x \in Q_r$, the initial value problem (1.16) (with J) has a strong solution $u \in C(J, \mathcal{X})$ that depends continuously (in a particular space of functions) upon x .*

Proof. Fix arbitrarily $T > 0$ such that $J := [-T, T] \subset I$. For $R > r$, let us also define the space $U_R := C(J, Q_R)$. When endowed with the sup-norm, it is easy to see that U_R is a Banach space. For $u \in U_R$, consider the map

$$\phi_x(u)(t) := x + \int_0^t f(u(s), s) \, ds, \quad t \in J \subset I.$$

Clearly, $\phi_x(u)$ returns a continuous function $J \rightarrow \mathcal{X}$. We will prove that, for a suitable choice of $T, R > 0$, the map ϕ_x is a contraction on the space U_R .

Let $u, v \in U_R$ be given. It is not difficult to see that $\phi_x(u)$ will be a continuous function of t on J . Certainly, this follows by continuity of f on $Q_R \times J \subseteq \Omega \times I$. Moreover, the following uniform estimate holds:

$$\|\phi_x(u) - \phi_x(v)\|_\infty = \sup_{t \in J} \left| \int_0^t (f(u(s), s) - f(v(s), s)) \, ds \right| \quad (1.18)$$

$$\leq \sup_{t \in J} |t| \|f(u(s), s) - f(v(s), s)\|_\infty \quad (1.19)$$

$$\leq C_{R,J} \cdot T \|u - v\|_\infty. \quad (1.20)$$

In a similar vein, for a fixed $z \in Q_R$, we obtain

$$\begin{aligned}
\|\phi_x(u) - x\|_\infty &= \sup_{t \in J} \left\| \int_0^t f(u(s), s) \, ds \right\| \\
&\leq \int_0^{|t|} \|f(u(s), s)\|_\infty \, ds \\
&\leq \int_0^{|t|} \|f(u(s), s) - f(z, s)\|_\infty \, ds + \int_0^{|t|} \|f(z, s)\|_\infty \, ds \\
&\leq TC_{R,J} \|u - z\|_\infty + T \|f(z, \cdot)\|_\infty.
\end{aligned}$$

Consequently yielding

$$\|\phi_x(u) - x\|_\infty \leq 2TRC_{R,J} + T \|f(z, \cdot)\|_\infty. \quad (1.21)$$

Notice that “shrinking” the interval J need not affect the constant $C_{R,J}$. Therefore,

$$\begin{aligned}
\|\phi_x(u)\|_\infty &\leq 2TRC_{R,J} + T \|f(z, \cdot)\|_\infty + \|x\| \\
&\leq 2TRC_{R,J} + T \|f(z, \cdot)\|_\infty + r.
\end{aligned}$$

Furthermore, by the reverse triangle inequality,

$$\begin{aligned}
\text{dist}(\phi_x(u(t)), \Omega^C) &= \inf_{y \in \Omega^C} \|\phi_x(u(t)) - y\| \\
&\geq \inf_{y \in \Omega^C} [\|x - y\| - \|\phi_x(u)(t) - x\|] \\
&\geq \frac{1}{r} - \|\phi_x(u) - x\|_\infty \\
&\geq \frac{1}{r} - (TC_{R,J} \|u - z\|_\infty + T \|f(z, \cdot)\|_\infty).
\end{aligned}$$

Hence, by choosing $T > 0$ sufficiently small, we can ensure that

$$\phi_x : U_R \rightarrow U_R$$

is a contraction. In fact, this choice of T can be taken independently of x . By the Banach fixed point theorem, there exists a unique function $u \in U_R$ such that $\phi_x(u) = u$. Of course, this is to say that

$$u(t) = x + \int_0^t f(u(s), s) \, ds, \quad \forall t \in (-T, T)$$

for some $T > 0$ sufficiently small. This certainly means that u is a strong solution to the initial value problem (1.16) with J instead of I .

We now show that this u depends continuously on the initial data $x \in Q_r$. Suppose that $v \in U_R$ is a solution to the initial value problem with initial data $y \in Q_r$ and the interval J as above. Then, v must satisfy

$$v(t) = y + \int_0^t f(v(s), s) \, ds = \phi_y(v)(t)$$

for all $t \in J$. Thus, using that ϕ is a contraction on U_R for any initial data in Q_r ,

$$\|u - v\|_\infty \leq \|x - y\| + \|\phi_0(u) - \phi_0(v)\|_\infty \leq \|x - y\| + \gamma \|u - v\|_\infty$$

for some $0 < \gamma < 1$. It follows that

$$\|u - v\|_\infty \leq \frac{1}{1 - \gamma} \|x - y\|$$

which proves the claim. $\square$

1.6.1 Gronwall's Inequality and Local Uniqueness

We would like to complement our local existence criterion with a “local uniqueness” corollary. This can certainly be achieved, and the proof begins with a classical lemma due to Gronwall.

Proposition 1.29 (Gronwall's Inequality). *Suppose that $T > 0$ and the functions $y, b : [0, T] \rightarrow \mathbb{R}$ are continuous with $b(t) \geq 0$. Assume further that*

$$y(t) \leq A + \int_0^t b(s)y(s) \, ds, \quad \forall t \in [0, T]$$

for some $A \in \mathbb{R}$. Then,

$$y(t) \leq A \exp \left(\int_0^t b(s) \, ds \right), \quad \forall t \in [0, T].$$

Proof. Define

$$g(t) := A + \int_0^t b(s)y(s) \, ds \quad \text{and} \quad z(t) := g(t)e^{-\int_0^t b(s) \, ds}, \quad \forall t \in [0, T].$$

Elementary calculus then gives

$$g'(t) = b(t)y(t) \leq b(t) \left(A + \int_0^t b(s)y(s) \, ds \right) = b(t)g(t).$$

Thus,

$$\begin{aligned} z'(t) &= g'(t)e^{-\int_0^t b(s) \, ds} - g(t)b(t)e^{-\int_0^t b(s) \, ds} \\ &\leq b(t)g(t)e^{-\int_0^t b(s) \, ds} - g(t)b(t)e^{-\int_0^t b(s) \, ds} = 0. \end{aligned}$$

It follows that $z(t) \leq g(0) = A$ for all $t \in [0, T]$. Of course, this is equivalent to having

$$g(t) \leq A \exp \left(\int_0^t b(s) \, ds \right).$$

Finally, putting all of the pieces together,

$$y(t) \leq A + \int_0^t b(s)y(s) \, ds = g(t) \leq A \exp \left(\int_0^t b(s) \, ds \right).$$

This completes the proof. $\square$

Corollary 1.30 (Gronwall's Inequality – Form 2). *Suppose that $T < 0$ and let $y, b : [T, 0] \rightarrow \mathbb{R}$ be continuous with $b(t) \geq 0$. Assume there exists $A \in \mathbb{R}$ such that*

$$y(t) \leq A + \int_t^0 b(s)y(s) \, ds, \quad \forall t \in [T, 0].$$

Then,

$$y(t) \leq A \exp \left(\int_t^0 b(s) \, ds \right), \quad \forall t \in [T, 0].$$

Proof. Naturally, we will invoke the classical form of Gronwall's inequality. Define a function

$$z : [0, -T] \xrightarrow{t \mapsto -t} [T, 0] \xrightarrow{y} \mathbb{R}, \quad z(t) := y(-t).$$

By composition, this function z is continuous. Given $t \in [0, -T]$, we have

$$z(t) = y(-t) \leq A + \int_{-t}^0 b(s)y(s) \, ds.$$

Making the substitution $\sigma := -s$, the above becomes

$$\begin{aligned} z(t) &\leq A + \int_{-t}^0 b(s)y(s) \, ds = A + \int_0^t b(-\sigma)y(-\sigma) \, d\sigma \\ &= A + \int_0^t b(-\sigma)z(\sigma) \, d\sigma \end{aligned}$$

By the usual Gronwall inequality, we discover that

$$z(t) \leq A \exp \left(\int_0^t b(-\sigma) \, d\sigma \right), \quad \forall t \in [0, -T].$$

Finally, make the substitution $\eta := -\sigma$ to obtain

$$z(t) \leq A \exp \left(- \int_0^{-t} b(\eta) \, d\eta \right) = A \exp \left(\int_{-t}^0 b(\eta) \, d\eta \right).$$

Or, rather,

$$y(-t) \leq A \exp \left(\int_{-t}^0 b(\eta) \, d\eta \right).$$

□

Using this, we will deduce the subsequent uniqueness theorem.

Theorem 1.31 (Local Uniqueness Criterion). *Let $I_1, I_2 \ni 0$ be two open intervals and I an open interval containing $I_1 \cup I_2$. Suppose that*

$$f : \Omega \times I \rightarrow \mathcal{X}$$

is locally Lipschitz in the first variable. Let $u_1 \in C(I_1, \mathcal{X})$ and $u_2 \in C(I_2, \mathcal{X})$ be two strong solutions to their respective initial value problems (with the same initial datum). Then $u_1 \equiv u_2$ on $I_1 \cap I_2$.

Proof. First, let $T > 0$ be such that $T \in I_1 \cap I_2$. Then, $[0, T] \subset I_1 \cap I_2$. For $t \in [0, T]$ we will apply Gronwall's inequality with

$$y(t) := \|u_1(t) - u_2(t)\|,$$

and $A = 0$. For the moment however, we notice that

$$\begin{aligned} \|u_1(t) - u_2(t)\| &\leq \left\| \int_0^t (f(u_1(s), s) - f(u_2(s), s)) \, ds \right\| \\ &\leq C_{R, [0, T]} \int_0^t \|u_1(s) - u_2(s)\| \, ds; \end{aligned}$$

here we have chosen $R > 0$ such that $u_1, u_2 \in C([0, T], Q_R)$ (which may be achieved by the continuity of u_1, u_2). Invoking Gronwell's inequality (where we take $b(t) \equiv C_{R,[0,T]}$) implies that

$$\|u_1(t) - u_2(t)\| \leq 0 \cdot e^{\int_0^t C_{R,[0,T]} ds} = 0,$$

as was asserted. A similar applies for $T < 0$ belonging to $I_1 \cap I_2$. $\square$

1.6.2 Extensions and Maximal Solutions

Let I_1 and I_2 be two open intervals containing the point 0. Let $u_1 \in C(I_1, \mathcal{X})$ and $u_2 \in C(I_2, \mathcal{X})$ be strong solutions to the (respective) initial value problem (1.16). If $I_1 \supseteq I_2$, we will say that u_1 is an *extension* of u_2 .

Remark 1.3. In light of our “local uniqueness”, we have a simple extension argument. Suppose that $u_1 \in C(I_1, \mathcal{X})$ and $u_2 \in C(I_2, \mathcal{X})$ are solutions to respective initial value problems. Then, the function

$$u(t) := \begin{cases} u_1(t), & t \in I_1, \\ u_2(t), & t \in I_2 \end{cases}$$

is an extension of both u_1 and u_2 .

Naturally, we have the following definition of maximality.

Definition 9. A solution $u \in C(I, \mathcal{X})$ to (1.16) is called *maximal* if it does not admit a proper extension.

Theorem 1.32 (Maximality Theorem). *There exists a unique maximal solution to any initial value problem of the form (1.16).*

Proof. Existence of a single solution is evident from the Picard-Lindelöf theorem (after all, any initial data $x \in \Omega$ will belong to Q_r for some $r > 0$). Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be the set of all intervals in $\mathbb{R}$, containing the origin, over which there exists a solution to (1.16). Put

$$I := \bigcup_{\alpha \in \mathcal{A}} I_\alpha$$

and notice that I is non-empty. Since $\bigcap_{\alpha \in \mathcal{A}} I_\alpha$ is non-empty, I must be an open interval. For each $t_\alpha \in I_\alpha$, with $\alpha \in \mathcal{A}$, let $u_\alpha(\cdot)$ be a solution to the corresponding initial value problem. Put then

$$u(t) := u_\alpha(t), \quad t \in I_\alpha \text{ for some } \alpha \in \mathcal{A}.$$

By our local uniqueness theorem, the function u is well defined and will satisfy (1.16) on I .

It is clear that this map u is maximal. Now, to prove uniqueness. Again invoking the local uniqueness theorem, if $I_1 = I_2$ then any two given solutions $u_1 \in C(I_1, \mathcal{X})$ and $u_2 \in C(I_2, \mathcal{X})$ to (1.16) will be identical. However, by maximality, we must have $I_1 = I_2$ for otherwise one could construct an extension as in Remark 1.3. $\square$

Chapter 2

Prototypical Linear Differential Equations

In this chapter we will explore three fundamental differential operators, and their induced homogeneous equations. Despite being simple and easy to formulate, these problems are *prototypical* in the sense that they give rise to large classes of well understood differential equations. The theorems we will prove here give us a reasonable idea of what to expect when studying more general equations and systems. We will begin with the study of the *Laplace equation*.

2.1 Harmonic Functions

Throughout this section, $n \geq 2$ will be an integer. As before, Ω will always denote an open subset of $\mathbb{R}^n$. Also, for $k \in \mathbb{N}_0$, we denote by $C^k(\Omega)$ the $\mathbb{R}$ -vector space of all continuous and k -times differentiable maps $u : \Omega \rightarrow \mathbb{R}$. Finally, we will write $C(\Omega)$ instead of $C^0(\Omega)$.

Definition 10. The *Laplacian*, or Laplace operator, is the linear functional

$$\Delta : C^2(\Omega) \longrightarrow C(\Omega), \quad u \mapsto \sum_{j=1}^n \frac{\partial^2 u}{\partial x_j^2}.$$

In other terms, Δ is the differential operator $\sum_{j=1}^n \frac{\partial^2}{\partial x_j^2}$. A function $u \in C^2(\Omega)$ is then said to be *harmonic* in Ω if it satisfies

$$\Delta u(x) = 0, \quad \forall x \in \Omega.$$

At this point, we would simply write “ $\Delta u \equiv 0$ in Ω ”.

As a first step, let us attempt to explicitly solve the problem

$$\Delta u \equiv 0 \quad \text{in } \Omega \quad (2.1)$$

by looking for radically symmetric $C^2(\Omega)$ -solutions. Of course, this is not done naively; I know that this approach will be successful. For the time being however, let's focus on seeing it in action. Let $r \in \mathbb{R}$ denote the norm

$$r := |x| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad x = (x_1, \dots, x_n) \in \mathbb{R}^n$$

and suppose a solution u to (2.1) can be written in terms of r . That is, assume that $u(x) = v(r)$ for some twice continuously differentiable function v . Now, for each index $j = 1, \dots, n$ there holds the obvious equality:

$$\frac{\partial r}{\partial x_j} = x_j \cdot (x_1^2 + \cdots + x_n^2)^{-1/2} = \frac{x_j}{r}, \quad \Omega \ni x \neq 0.$$

For these x and any index $j = 1, \dots, n$ one then has

$$\frac{\partial u(x)}{\partial x_j} = v'(r) \cdot \frac{x_j}{r}$$

and

$$\frac{\partial^2 u(x)}{\partial x_j^2} = v''(r) \frac{x_j^2}{r^2} + v'(r) \left[\frac{1}{r} - \frac{x_j^2}{r^3} \right].$$

Thus, using that u is harmonic in Ω , we get for $x \in \Omega \setminus \{0\}$ that

$$\begin{aligned} 0 &= \sum_{j=1}^n \frac{\partial^2 u(x)}{\partial x_j^2} = \sum_{j=1}^n \left(v''(r) \frac{x_j^2}{r^2} + v'(r) \left[\frac{1}{r} - \frac{x_j^2}{r^3} \right] \right) \\ &= v''(r) + \frac{v'(r)}{r} \sum_{j=1}^n \left(1 - \frac{x_j^2}{r^2} \right) \\ &= v''(r) + \frac{n-1}{r} v'(r). \end{aligned}$$

This shows that $\Delta u = 0$ if and only if v satisfies the above *ordinary* differential equation. Of course, $r \neq 0$ in all cases. Since we are only looking for a single solution, we might as well make the assumption that $v'(r)$ is **never** zero. Hence,

by the intermediate value theorem, $v'(r)$ cannot ever “change sign”. We thus get that

$$\frac{v''(r)}{v'(r)} = \frac{1-n}{r}$$

whence

$$\ln |v'(r)| = (1-n) \ln r + C = \ln r^{1-n} + C.$$

Of course, after a relabeling of C , we obtain

$$|v'(r)| = \frac{a}{r^{n-1}}$$

for some constant $a \in \mathbb{R}$. Recalling our previous observation (that v' is constant in sign), it follows that

$$v'(r) = \frac{c}{r^{n-1}}$$

for some constant $c \in \mathbb{R} \setminus \{0\}$. Finally, we conclude that

$$u(x) = v(r) = \begin{cases} c \ln r + d, & n = 2, \\ \frac{c}{r^{n-2}} + d, & n \geq 3. \end{cases}$$

This gives rise to the following definition.

Definition 11 (Fundamental Solution). The *fundamental solution* to the Laplace equation is the $L^1_{\text{loc}}(\mathbb{R}^n)$ function given by

$$\Phi(x) := \begin{cases} -\frac{1}{2\pi} \ln |x|, & n = 2, \\ \frac{1}{n(n-2)\omega_n \cdot |x|^{n-2}}, & n \geq 3. \end{cases} \quad (2.2)$$

Here, ω_n denotes the measure of the unit ball in $\mathbb{R}^n$. Notice that $\Phi(x)$ is radially symmetric and thus may be considered as a function solely in $|x|$. One can easily verify by direct calculation that Φ is harmonic in the region $\mathbb{R}^n \setminus \{0\}$. By setting $\Phi(0) = \infty$, we can assume that Φ belongs to $L^1_{\text{loc}}(\mathbb{R}^n)$.

The above function is particularly useful in determining solutions to the fundamental, and prototypical, *Poisson equation* which we describe below.

Definition 12 (Poisson Equation). Let $f \in C(\Omega)$ be given. The Poisson equation for f in Ω is the differential equation

$$\Delta u \equiv f, \quad \text{in } \Omega. \quad (2.3)$$

2.1.1 Solving the Poisson Equation for f

Our goal here is to determine a function $u \in C^2(\Omega)$ which satisfies (2.3), where f is given. Fortunately for us, the function Φ described in (2.2) always gives us precisely such a solution.

Theorem 2.1. *Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by*

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y) f(y) \, dy,$$

where $f \in C^2(\mathbb{R}^n)$ has compact support. Then, $u \in C^2(\mathbb{R}^n)$ and satisfies the equation

$$-\Delta u = f \quad \text{in } \mathbb{R}^n.$$

Proof. First let us observe that the function u is well defined. Certainly, for each $x \in \mathbb{R}^n$, the map $y \mapsto \Phi(x - y)$ belongs to $L^1_{\text{loc}}(\mathbb{R}^n)$ and the function f has compact support. Thus, u is a real valued function defined on the entirety of $\mathbb{R}^n$. Now, commutativity of the convolution gives

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y) f(y) \, dy = \int_{\mathbb{R}^n} \Phi(y) f(x - y) \, dy, \quad \forall x \in \mathbb{R}^n. \quad (2.4)$$

Let $K \subset \mathbb{R}^n$ be a compact set outside of which $f(y)$ vanishes. Then, we can write

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y) \mathbb{1}_K(y) f(y) \, dy$$

where the function $\Phi(x - y) \mathbb{1}_K(y)$ belongs to $L^1(\mathbb{R}^n)$. Since f is $C^2(\mathbb{R}^n)$ with compact support, it then follows from Proposition 8.10 in [FLND] that $u \in C^2(\mathbb{R}^n)$.

For the second part, let $\varepsilon > 0$ be given. For each $x \in \mathbb{R}^n$, we may write $u(x)$ as the following sum:

$$u(x) = \int_{B(0, \varepsilon)} \Phi(y) f(x - y) \, dy + \int_{B(0, \varepsilon)^c} \Phi(y) f(x - y) \, dy. \quad (2.5)$$

Repeating the same arguments as above, it is easy to see that

$$\Delta u(x) = \int_{B(0, \varepsilon)} \Phi(y) \Delta_x f(x - y) \, dy + \int_{B(0, \varepsilon)^c} \Phi(y) \Delta_x f(x - y) \, dy \quad (2.6)$$

$$=: I_\varepsilon + J_\varepsilon. \quad (2.7)$$

Observe that Δf is continuous on $\mathbb{R}^n$ with compact support. Therefore, direct calculation easily gives

$$|I_\varepsilon| \leq \|\Delta f\|_{L^\infty(\mathbb{R}^n)} \int_{B(0,\varepsilon)} |\Phi(y)| \, dy \leq \begin{cases} C\varepsilon^2 |\ln \varepsilon|, & n = 2, \\ C\varepsilon^2, & n \geq 3. \end{cases}$$

Thus, we see that I_ε tends to 0 as $\varepsilon \rightarrow 0$. With J_ε , we must instead integrate by parts in each coordinate. Doing so gives

$$\begin{aligned} J_\varepsilon &= \int_{B(0,\varepsilon)^c} \Phi(y) \Delta_x f(x-y) \, dy = - \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \nabla \Phi(y) \cdot \nabla f(x-y) \, dy \\ &\quad + \int_{\partial B(0,\varepsilon)} \Phi(y) \frac{\partial f}{\partial \nu}(x-y) \, dS(y) \\ &= K_\varepsilon + L_\varepsilon. \end{aligned}$$

Here, ν denotes the inward pointing unit normal vector to $\partial B(0, \varepsilon)$. Proceeding as above, we get

$$|L_\varepsilon| \leq \|\nabla f\|_{L^\infty(\mathbb{R}^n)} \int_{\partial B(0,\varepsilon)} |\Phi(y)| \, dy \leq C \begin{cases} \varepsilon |\ln \varepsilon|, & n = 2, \\ \varepsilon, & n \geq 3. \end{cases}$$

As before, this tends to 0 as $\varepsilon \rightarrow 0$. We hence need only consider K_ε . Once again integrating by parts yields

$$\begin{aligned} K_\varepsilon &= \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Delta \Phi(y) f(x-y) \, dy - \int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y) \, dS(y) \\ &= - \int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y) \, dS(y). \end{aligned}$$

Here, we have used the property that Φ is harmonic in $\mathbb{R}^n \setminus \{0\}$. On $\partial B(0, \varepsilon)$, we obviously have

$$\nu = \frac{-y}{|y|} = -\frac{y}{\varepsilon}.$$

Also, direct calculation gives

$$\frac{\partial \Phi}{\partial \nu}(y) = \frac{1}{n\omega_n \varepsilon^{n-1}}.$$

It therefore follows that

$$K_\varepsilon = -\frac{1}{n\omega_n \varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y) \, dS(y) = - \oint_{\partial B(x,\varepsilon)} f(y) \, dS.$$

Since f is continuous over $\mathbb{R}^n$, this converges to $-f(x)$ as ε tends to 0. This completes the proof. $\square$

2.1.2 The Mean Value Property

The previous subsection was far more quantitative than qualitative; and this was reflected within the proof of the previous theorem. This subsection instead focuses on the averaging properties of harmonic functions. From these, we will be able to deduce powerful maximum principles giving rise to uniqueness of solutions to particular problems involving the differential operator Δ .

Theorem 2.2 (Mean Value Property of Harmonic Functions). *Let $\Omega \subseteq \mathbb{R}^n$ be an open set. Suppose that $u \in C^2(\Omega)$ is harmonic within Ω . Then, for every open ball $B(x_0, \rho)$ whose closure is contained in Ω , there holds the following*

$$u(x_0) = \int_{B(x_0, \rho)} u(y) \, dy = \int_{\partial B(x_0, \rho)} u(y) \, dS(y). \quad (2.8)$$

Equation (2.8) is called the mean value formula for harmonic functions.

Proof. Suppose that Δu vanishes in Ω . If $\overline{B(x_0, \rho)} \subset \Omega$ then

$$\begin{aligned} 0 &= \int_{B(x_0, \rho)} \Delta u(y) \, dy = \int_{\partial B(x_0, \rho)} \nabla u(y) \cdot \nu \, dS(y) \\ &= \int_{\partial B(0, 1)} \nabla u(x_0 + \rho z) \cdot \nu \rho^{n-1} \, dS(z) \\ &= \rho^{n-1} \int_{\partial B(0, 1)} \nabla u(x_0 + \rho z) \cdot z \, dS(z) \\ &= \rho^{n-1} \int_{\partial B(0, 1)} \frac{\partial}{\partial \rho} u(x_0 + \rho z) \, dS(z) \\ &= \rho^{n-1} \frac{\partial}{\partial \rho} \int_{\partial B(x_0, \rho)} u(y) \rho^{1-n} \, dS(y) \\ &= \rho^{n-1} n \omega_n \frac{\partial}{\partial \rho} \left(\frac{1}{n \omega_n \rho^{n-1}} \int_{\partial B(x_0, \rho)} u(y) \, dS(y) \right) \\ &= \rho^{n-1} n \omega_n \frac{\partial}{\partial \rho} \left(\int_{\partial B(x_0, \rho)} u(y) \, dS(y) \right). \end{aligned}$$

Thus, we see that the map

$$\rho \mapsto \int_{\partial B(x_0, \rho)} u(y) \, dS(y)$$

is constant for $\rho > 0$. Using the continuity of u in Ω , we conclude that

$$\oint_{\partial B(x_0, \rho)} u(y) \, dS(y) = \lim_{\rho \searrow 0} \oint_{\partial B(x_0, \rho)} u(y) \, dS(y) = u(x_0).$$

It remains to show that the first equality in (2.8) holds true. This is done using the so-called “coarea formula”. By the second part of (2.8), we can write

$$\begin{aligned} \oint_{B(x_0, \rho)} u(y) \, dy \frac{1}{\rho^n \omega_n} &= \int_{B(x_0, \rho)} u(y) \, dy = \frac{1}{\rho^n \omega_n} \int_0^\rho \left(\int_{\partial B(x_0, \delta)} u(y) \, dS(y) \right) d\delta \\ &= \frac{1}{\rho^n \omega_n} \int_0^\rho u(x_0) n \omega_n \delta^{n-1} \, d\delta \\ &= \frac{nu(x_0)}{\rho^{n-1}} \int_0^\rho \delta^{n-1} \, d\delta \\ &= u(x_0). \end{aligned}$$

This completes the proof. $\square$

The theorem we have just proven is certainly surprising. A priori, it is not clear that harmonic functions enjoy such nice averaging properties. Perhaps even more surprising is the fact that these averaging properties *characterize* harmonic functions.

Theorem 2.3 (Converse to the Mean Value Property). *Let $\Omega \subseteq \mathbb{R}^n$ be an open set and suppose $u \in C^2(\Omega)$ satisfies*

$$u(x_0) = \oint_{\partial B(x_0, \rho)} u(y) \, dS(y)$$

for each ball $\overline{B(x_0, \rho)} \subset \Omega$. Then u is harmonic in Ω .

Remark 2.1. We will later show that the conclusion above holds even when we only assume that u is continuous in Ω .

Proof. We argue by contradiction. Suppose that Δu does not vanish entirely within Ω . Since $u \in C^2(\Omega)$, the function Δu is continuous on Ω . Choose a point $x_0 \in \Omega$ such that $\Delta u(x_0) \neq 0$. Without loss of generality, we may assume that $\Delta u(x_0) > 0$. Choose a ball $\overline{B(x_0, \varepsilon)} \subset \Omega$ such that

$$\Delta u > 0 \quad \text{on } \overline{B(x_0, \varepsilon)}.$$

For all ρ in the interval $(0, \varepsilon)$, one has

$$u(x_0) = \oint_{\partial B(x_0, \rho)} u(y) \, dS(y)$$

whence

$$\frac{\partial}{\partial \rho} \oint_{\partial B(x_0, \rho)} u(y) \, dS(y) = 0 \quad \text{on } (0, \varepsilon).$$

On the other hand,

$$\begin{aligned} \oint_{\partial B(x_0, \rho)} u(y) \, dS(y) &= \frac{1}{n\omega_n \rho^{n-1}} \int_{\partial B(x_0, \rho)} u(y) \, dS(y) \\ &= \frac{1}{n\omega_n} \int_{\partial B(0, 1)} u(x_0 + \rho z) \, dS(z) \end{aligned}$$

so that

$$\begin{aligned} \frac{\partial}{\partial \rho} \oint_{\partial B(x_0, \rho)} u(y) \, dS(y) &= \frac{1}{n\omega_n} \int_{\partial B(0, 1)} \nabla u(x_0 + \rho z) \cdot z \, dS(z) \\ &= \frac{1}{n\omega_n \rho^{n-1}} \int_{\partial B(x_0, \rho)} \nabla u(y) \cdot \frac{y - x_0}{\rho} \, dS(y) \\ &= \int_{\partial B(x_0, \rho)} \nabla u(y) \cdot \nu \, dS(y) \\ &= \frac{1}{n\omega_n \rho^{n-1}} \int_{B(x_0, \rho)} \Delta u(y) \, dy > 0. \end{aligned}$$

This contradiction completes the proof. $\square$

2.1.3 The Maximum Principles

We shall now put the mean value property to use. Arguably, the maximum principles are the most useful properties of harmonic functions. Similar results will also arise when considering arbitrary elliptic linear operators as well as other types of differential equations (e.g. heat and wave equations).

Theorem 2.4 (Strong Maximum Principle). *Let $\Omega \subset \mathbb{R}^n$ be a connected open set and let $u \in C^2(\Omega)$ be a harmonic function in Ω . Assume further that u is continuous on $\bar{\Omega}$. If u achieves a local extremum in Ω , then u is constant on $\bar{\Omega}$.*

Proof. Suppose that u achieves a local extremum at a point $x_0 \in \Omega$. By replacing u with $-u$, we might as well assume that x_0 is a local *maximum* for u . Let $\delta > 0$ be such that $B(x_0, \delta) \Subset \Omega$ ¹ and $u(x) \leq u(x_0)$ for all $x \in \overline{B(x_0, \delta)}$. By invoking (2.8), we obtain

$$\int_{B(x_0, \delta)} u(x_0) \, dx = u(x_0) = \int_{B(x_0, \delta)} u(x) \, dx.$$

It follows that

$$\int_{B(x_0, \delta)} (u(x_0) - u(x)) \, dx = 0.$$

Since $u(x) \leq u(x_0)$ on the entirety of $B(x_0, \delta)$, we conclude that $u(x_0) = u(x)$ almost everywhere on $B(x_0, \delta)$. Because $u(x)$ is continuous on this same ball, we also conclude that $u(x) = u(x_0)$ at **all** points $x \in B(x_0, \delta)$. This argument immediately shows that

$$\Xi := \{x \in \Omega : u(x) = u(x_0)\}$$

is open in Ω . By the continuity of u , Ξ is clearly closed within Ω . Since Ω is a connected subspace of $\mathbb{R}^n$ by hypothesis and $x_0 \in \Xi$, we must have $\Omega \setminus \Xi = \emptyset$. We deduce that $u \equiv u(x_0)$ in the whole of Ω .

Let now $x \in \overline{\Omega}$ be given. There exists a sequence $(x_n)_{n=1}^\infty$ of points in Ω converging to x . By the continuity of u , one then has

$$u(x) = \lim_{n \rightarrow \infty} u(x_n) = \lim_{n \rightarrow \infty} u(x_0) = u(x_0).$$

This completes the proof. □

2.1.3.1 The Weak Maximum Principles

From the strong maximum principle described above one can deduce other so-called maximum principles. The first is often also called the “strong maximum principle”. We will instead choose to call it the “first weak maximum principle”.

Corollary 2.5 (First Weak Maximum Principle). *Suppose that $\Omega \subseteq \mathbb{R}^n$ is a connected open set. Let $u \in C^2(\Omega)$ be continuous on the closure of Ω and harmonic within Ω . If there exists a point $x_0 \in \Omega$ such that*

$$u(x_0) = \max_{x \in \overline{\Omega}} u(x),$$

¹If U and V are open subsets of $\mathbb{R}^n$, we write $U \Subset V$ to say that $\overline{U}$ is compact and $U \subseteq \overline{U} \subseteq V$.

then u is constant on $\overline{\Omega}$.

Proof. Choose an open ball $B(x_0, \delta) \subseteq \Omega$. In particular, $u(x) \leq u(x_0)$ for all x in this ball. Hence, x_0 is a local extremum. The result then follows from the strong maximum principle. $\square$

Theorem 2.6 (Second Weak Maximum Principle). *Let $\emptyset \subsetneq \Omega \subsetneq \mathbb{R}^n$ be open and bounded. Suppose that $u \in C^2(\Omega)$ is harmonic in Ω and continuous up to $\partial\Omega$. Then,*

$$\max_{x \in \overline{\Omega}} u(x) = \max_{\partial\Omega} u(x).$$

Remark 2.2. Let us first explain the requirement that $\emptyset \subsetneq \Omega \subsetneq \mathbb{R}^n$. One can show that Ω is clopen if and only if $\Omega = \mathbb{R}^n$ or $\Omega = \emptyset$. On the other hand, it is possible to prove that $\partial\Omega$ is empty if and only if Ω is clopen. Thus, $\partial\Omega$ will never be empty and the equality

$$\max_{x \in \overline{\Omega}} u(x) = \max_{\partial\Omega} u(x)$$

is well defined.

Proof. First, we assume that Ω is connected. Clearly,

$$\max_{x \in \overline{\Omega}} u(x) \geq \max_{\partial\Omega} u(x).$$

Suppose that equality does not hold. Then u achieves its maximum at a point $x_0 \in \Omega$. By the first weak maximum principle, we see that u is constant in $\overline{\Omega}$. In particular,

$$\max_{x \in \overline{\Omega}} u(x) = \max_{\partial\Omega} u(x).$$

We now relax the requirement that Ω be connected. Since $\mathbb{R}^n$ is locally connected, we know that the connected components of Ω are open in $\mathbb{R}^n$. Once again, it is clear that

$$\max_{x \in \overline{\Omega}} u(x) \geq \max_{\partial\Omega} u(x).$$

Denote by $\mathcal{C}$ any connected component of Ω . Clearly, u is harmonic within $\mathcal{C}$ and so the first part gives

$$\max_{x \in \overline{\mathcal{C}}} u(x) = \max_{x \in \partial\mathcal{C}} u(x).$$

Suppose for a contradiction that

$$\max_{x \in \overline{\Omega}} u(x) > \max_{\partial\Omega} u(x)$$

and choose a point $x_0 \in \Omega$ at which u achieves the global maximum. Let $\mathcal{C}$ be the connected component of x_0 . Since u is harmonic within $\mathcal{C}$, the first part of the proof yields

$$u(x_0) \leq \max_{x \in \overline{\mathcal{C}}} u(x) = \max_{x \in \partial\mathcal{C}} u(x) \leq \max_{x \in \partial\Omega} u(x) < \max_{x \in \overline{\Omega}} u(x) = u(x_0).$$

Here, we have used that $\partial\mathcal{C} \subseteq \partial\Omega$. This contradiction completes the proof. $\square$

Corollary 2.7. *Let $\emptyset \subsetneq \Omega \subsetneq \mathbb{R}^n$ be open and bounded. Suppose $u \in C^2(\Omega)$ is harmonic in Ω and continuous up to $\partial\Omega$. Then,*

$$\min_{x \in \overline{\Omega}} u(x) = \min_{\partial\Omega} u(x).$$

Proof. Apply the previous theorem to $v := -u$. $\square$

2.1.3.2 Uniqueness and Positivity of Solutions

Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. We are largely interested in those functions $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfying the following problem

$$\begin{cases} \Delta u \equiv 0, & \text{in } \Omega, \\ u \equiv g, & \text{on } \partial\Omega, \end{cases} \quad (2.9)$$

where $g : \partial\Omega \rightarrow [0, \infty)$ is continuous and given. Suppose that u is a solution to (2.9) satisfying the desired properties. The function $v := -u$ will also be harmonic in Ω . In fact, it will satisfy the analogous system

$$\begin{cases} \Delta v \equiv 0, & \text{in } \Omega, \\ v \equiv -g, & \text{on } \partial\Omega. \end{cases}$$

Suppose for the moment that there exists $x \in \overline{\Omega}$ with $u(x) < 0$. By the boundary condition in (2.9), we must have $x \in \Omega$. Thus, $v(x) > 0$. However, the second weak maximum principle tells us that

$$\max_{x \in \overline{\Omega}} v(x) = \max_{x \in \partial\Omega} v(x) = \max_{x \in \partial\Omega} -g(x) \leq 0.$$

This contradiction shows that $u(x) \geq 0$ in the whole of $\overline{\Omega}$. This phenomenon is known as the *positivity of harmonic functions*.

Theorem 2.8 (Uniqueness of Solutions–Poisson Equation). *Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. There exists at most one $C^2(\Omega) \cap C(\overline{\Omega})$ solution to the problem*

$$\begin{cases} \Delta u \equiv f, & \text{in } \Omega, \\ u \equiv g, & \text{on } \partial\Omega, \end{cases}$$

where $f : \Omega \rightarrow \mathbb{R}$ and $g : \partial\Omega \rightarrow \mathbb{R}$ are continuous.

Proof. Let u and v be two solutions to the problem above, satisfying the desired criteria. Then, $w := (u - v)$ is harmonic in Ω . Moreover, the map w is of class $C^2(\Omega) \cap C(\overline{\Omega})$ and solves

$$\begin{cases} \Delta w \equiv 0, & \text{in } \Omega, \\ w \equiv 0, & \text{on } \partial\Omega. \end{cases}$$

Then, it follows that $w \equiv 0$ on the whole of $\overline{\Omega}$. □

2.1.4 Smoothness Properties and Local Estimates

In this part of the notes, we investigate the differentiability of harmonic functions. We will show that harmonic functions are in fact smooth, and analytic in their domains. This surprising property is highly reminiscent of holomorphic functions on $\mathbb{C}$. For this subsection, Ω denotes a non-empty open subset of $\mathbb{R}^n$. We will also give local bounds for derivatives of arbitrary order.

Theorem 2.9 (Smoothness of Harmonic Functions). *Let $\Omega \subseteq \mathbb{R}^n$ be an open set and suppose $u : \Omega \rightarrow \mathbb{R}$ is a continuous function satisfying the mean value property in Ω (i.e. (2.8)). Then, $u \in C^\infty(\Omega)$.*

Proof. Let $x \in \Omega$ be given, and choose $r > 0$ such that $\overline{B(x, r)} \subseteq \Omega$. Consider the mollifier

$$\phi_r(s) := \begin{cases} e^{\frac{s^2}{s^2-r^2}}, & \text{if } 0 \leq |s| < r, \\ 0, & \text{otherwise.} \end{cases}$$

Subsequently, we define $\Phi_r(x) := \phi_r(|x|)$. Then, this function Φ_r is smooth and compactly supported on $\mathbb{R}^n$. Now, if $0 \leq s < r$, one always has $B(x, s) \subseteq \Omega$. Integration by parts with the “polar coordinates differentiation formula” gives

$$u(x) = \frac{1}{\int_0^r \omega_n s^n \phi_r'(s) ds} \int_0^r \left(\int_{B(x, s)} u(y) dy \right) \phi_r'(s) ds. \quad (2.10)$$

Indeed, using that $\phi_r(r) = 0$, one gets

$$\begin{aligned} \int_0^r \int_{B(x,s)} (u(y) \, dy) \, \phi'_r(s) \, ds &= \phi_r(r) \int_{B(x,r)} u(y) \, dy - \int_0^r \int_{\partial B(x,s)} u(y) \, dS(y) \, \phi_r(s) \, ds \\ &= - \int_0^r n \omega_n s^{n-1} u(x) \phi_r(s) \, ds \\ &= -u(x) \int_0^r n \omega_n s^{n-1} \phi_r(s) \, ds. \end{aligned}$$

Similarly,

$$\int_0^r \omega_n s^n \phi'_r(s) \, ds = - \int_0^r n \omega_n s^{n-1} \phi_r(s) \, ds.$$

Returning to equation (2.10), integration by parts in both the numerator and denominator yield²

$$\begin{aligned} u(x) &= \frac{1}{\int_0^r \omega_n s^n \phi'_r(s) \, ds} \int_0^r \left(\int_{B(x,s)} u(y) \, dy \right) \phi'_r(s) \, ds \\ &= \frac{1}{\int_0^r n \omega_n s^{n-1} \phi_r(s) \, ds} \int_0^r \phi_r(s) \int_{\partial B(x,s)} u(y) \, dS(y) \, ds \\ &= \frac{1}{\|\Phi_r\|_{L^1(\mathbb{R}^n)}} \int_0^\infty \int_{\partial B(x,s)} \phi_r(|x-y|) u(y) \, dS(y) \, ds \\ &= \frac{1}{\|\Phi_r\|_{L^1(\mathbb{R}^n)}} \int_0^\infty \int_{\partial B(x,s)} \Phi_r(x-y) u(y) \, dS(y) \, ds \\ &= \frac{1}{\|\Phi_r\|_{L^1(\mathbb{R}^n)}} \int_{\mathbb{R}^n} \Phi_r(x-y) u(y) \, dy; \end{aligned}$$

in this last step we have used Theorem 4 from Appendix C in [EVNS]. Of course, it follows from the above that, for some constant c independent of $x \in \Omega$,

$$u(x) = c \cdot (u * \Phi)(x).$$

Since Φ_r is $C_c^\infty(\mathbb{R}^n)$ and u is locally integrable, it follows from Proposition 8.10 in [FLND] that u is smooth in Ω . $\square$

²In the numerator take $v := \int_{B(x,s)} u(y) \, dy$ and $dw := \phi'_r(s)$. In the denominator, instead use $v := s^n$ and $dw := \phi'_r(s)$.

Corollary 2.10. *Let $u \in C^2(\Omega)$ be harmonic within Ω , where Ω is an open set in $\mathbb{R}^n$. Then, u is smooth in Ω .*

Corollary 2.11. *Let $\Omega \subseteq \mathbb{R}^n$ be an open set and suppose $u : \Omega \rightarrow \mathbb{R}$ is a continuous function satisfying the mean value property in Ω (i.e. (2.8)). Then, u is harmonic within Ω .*

Suppose that $\Omega \subseteq \mathbb{R}^n$ is open and let $u \in C^2(\Omega)$ be harmonic within Ω . We have shown above that u admits derivatives of *all* orders, and that these are continuous. Now, Laplace's equation tells us that

$$\Delta u \equiv 0, \quad \text{in } \Omega.$$

Let $(x_1, \dots, x_n)$ be the coordinates in $\mathbb{R}^n$. By the commutativity of partial derivatives (everything is smooth), we find that

$$\frac{\partial}{\partial x_i} \Delta u \equiv \Delta (u_{x_i}) \equiv 0, \quad \text{in } \Omega,$$

for all $i \in \{1, \dots, n\}$. Hence, all partial derivatives of u will also be harmonic in Ω . This also extends to higher order partial derivatives. For instance, $u_{x_i x_j}$ will be harmonic in Ω as well. More generally, $\partial^\alpha u$ will be harmonic in Ω for all multiindices α .

Theorem 2.12 (Harmonic Derivative Estimates). *Let $\Omega \subseteq \mathbb{R}^n$ be open and suppose that $u \in C^2(\Omega)$ is harmonic in Ω . For each ball $B(x_0, \rho) \subset \Omega$ and every multiindex α of index k , one has*

$$|\partial^\alpha u(x_0)| \leq \frac{C_k}{\rho^{n+|\alpha|}} \|u\|_{L^1(B(x_0, \rho))}, \quad (2.11)$$

where

$$C_0 := \frac{1}{\omega_n} \quad \text{and} \quad C_k := \frac{(2^{n+1}nk)^k}{\omega_n}, \quad \forall k \geq 1. \quad (2.12)$$

Proof. Our proof goes by way of induction and follows that from [EVNS]. First, suppose that $k := |\alpha| = 0$. Then, the mean value property in (2.8) gives

$$|\partial^\alpha u(x_0)| = |u(x_0)| \leq \frac{1}{\omega_n \rho^n} \int_{B(x_0, \rho)} |u(y)| \, dy = \frac{C_0}{\rho^n} \|u\|_{L^1(B(x_0, \rho))},$$

as was required. Now, let $k := |\alpha| = 1$. Then,

$$\partial^\alpha = \frac{\partial}{\partial x_i}$$

for some index $i = 1, \dots, n$. Thus,

$$|\partial^\alpha u(x_0)| = |u_{x_i}(x_0)|. \quad (2.13)$$

By our earlier remarks, u_{x_i} is harmonic in Ω and also satisfies (2.8) there. Observing that $B(x_0, \rho/2) \subseteq \Omega$, it follows from this same equation that

$$\begin{aligned} |u_{x_i}(x_0)| &= \left| \int_{B(x_0, \rho/2)} u_{x_i}(y) \, dy \right| = \left| \frac{2^n}{\omega_n \rho^n} \int_{B(x_0, \rho/2)} u(y) \, dy \right| \\ &= \frac{2^n}{\omega_n \rho^n} \left| \int_{\partial B(x_0, \rho/2)} u(y) v^i \, dS(y) \right|. \end{aligned}$$

In this last step, we have invoked the Gauss-Green theorem. Therefore,

$$|u_{x_i}(x_0)| \leq \frac{2^n}{\omega_n \rho^n} \cdot \frac{\rho^{n-1} n \omega_n}{2^{n-1}} \|u\|_{L^\infty(\partial B(x_0, \rho/2))} = \frac{2n}{\rho} \|u\|_{L^\infty(\partial B(x_0, \rho/2))}. \quad (2.14)$$

Suppose that $x \in \partial B(x_0, \rho/2)$. The triangle inequality gives

$$B(x, \rho/2) \subseteq B(x_0, \rho) \subset \Omega$$

whence our inequality for $|\alpha| = 0$ yields

$$|u(x)| \leq \frac{1}{\omega_n} \left(\frac{2}{\rho} \right)^n \|u\|_{L^1(B(x, \rho/2))} \leq \frac{1}{\omega_n} \left(\frac{2}{\rho} \right)^n \|u\|_{L^1(B(x_0, \rho))}.$$

Since $x \in \partial B(x_0, \rho/2)$ was arbitrary, we conclude that

$$\|u\|_{L^\infty(\partial B(x_0, \rho/2))} \leq \frac{1}{\omega_n} \left(\frac{2}{\rho} \right)^n \|u\|_{L^1(B(x_0, \rho))}.$$

Combining this with (2.13)-(2.14) gives

$$|\partial^\alpha u(x_0)| \leq \frac{2n}{\rho} \cdot \frac{1}{\omega_n} \left(\frac{2}{\rho} \right)^n \|u\|_{L^1(B(x_0, \rho))} = \frac{2^{n+1} n}{\rho^{n+1} \omega_n} \|u\|_{L^1(B(x_0, \rho))}.$$

Hence, the statement of the theorem holds true for $|\alpha| = 0, 1$.

We now handle the general α with $|\alpha| \geq 2$. Thus, we put $k := |\alpha| \geq 2$ and assume that the statement of the theorem holds for multiindices of order no larger than $k - 1$. There exists a multiindex β with $|\beta| = k - 1$ and an index $i = 1, \dots, n$ such that

$$\partial^\alpha u \equiv \left(\partial^\beta \right)_{x_i} \quad \text{in } \Omega,$$

Repeating the calculations in the first part of this proof, we obtain

$$|\partial^\alpha u(x_0)| \leq \frac{nk}{\rho} \left\| \partial^\beta u \right\|_{L^\infty(\partial B(x_0, \rho/k))}. \quad (2.15)$$

Let now $x \in \partial B(x_0, \rho/k)$; a basic estimate gives

$$B\left(x, \frac{k-1}{k}\rho\right) \subseteq B(x_0, \rho) \subseteq \Omega.$$

Hence, the induction hypothesis allows us to obtain

$$\begin{aligned} |\partial^\beta u(x)| &\leq \frac{\left(2^{n+1}n(k-1)\right)^{k-1}}{\omega_n \left(\frac{k-1}{k}\rho\right)^{n+k-1}} \|u\|_{L^1(B(x, \frac{k-1}{k}\rho))} \\ &\leq \frac{\left(2^{n+1}n(k-1)\right)^{k-1}}{\omega_n \left(\frac{k-1}{k}\rho\right)^{n+k-1}} \|u\|_{L^1(B(x_0, \rho))}. \end{aligned}$$

Combining this estimate with (2.15) yields the desired inequality, thereby concluding the proof. $\square$

2.1.5 Representation of Solutions and Analyticity

The estimates we obtained in the previous subsection have a surprising application. As we shall soon see, a quick application of these estimates shows that there cannot exist a non-constant harmonic function $u : \mathbb{R}^n \rightarrow \mathbb{R}$. This result is known as *Liouville's theorem*.

Theorem 2.13 (Liouville). *Let $u \in C^2(\mathbb{R}^n)$ be harmonic in the whole of $\mathbb{R}^n$. Then, u is constant.*

Proof. We will show that ∇u vanishes at each point in $\mathbb{R}^n$. Consider an index $i = 1, \dots, n$, it suffices to show that u_{x_i} vanishes on $\mathbb{R}^n$. To this end, fix a point

$x_0 \in \mathbb{R}^n$ and let $\rho > 0$. The estimates in (2.11)-(2.12) show that for a suitable constant $M > 0$

$$|u_{x_i}(x_0)| \leq \frac{2^{n+1}n}{\omega_n \rho^{n+1}} \int_{B(x_0, \rho)} |u(x)| \, dm \leq \frac{M 2^{n+1}n}{\omega_n \rho^{n+1}} \cdot \omega_n \rho^n = \frac{C}{\rho}.$$

Letting $\rho \rightarrow \infty$ in the above yields $u_{x_i}(x_0) = 0$. Since x_0 was taken arbitrarily, we get $u_{x_i} \equiv 0$. This completes the proof. $\square$

Liouville's theorem allows us to somewhat characterize solutions to entire Poisson problems. This is illustrated within the following proposition.

Proposition 2.14. *Let $f \in C_c^2(\mathbb{R}^n)$ with $n \geq 3$. Then, any bounded $C^2(\mathbb{R}^n)$ solution to the equation*

$$-\Delta u = f \quad \text{in } \mathbb{R}^n,$$

will have the form

$$u(x) \equiv \int_{\mathbb{R}^n} \Phi(x - y) f(y) \, dy + C,$$

for some $C \in \mathbb{R}$.

Proof. Since f has compact support and $\lim_{|x| \rightarrow \infty} \Phi(x) = 0$ for $n \geq 3$, we see that the function

$$v(x) := \int_{\mathbb{R}^n} \Phi(x - y) f(y) \, dy$$

is a *bounded* solution to the equation in question. Let $u(x)$ be another bounded solution to this problem. Then, $w := u - v$ is a bounded harmonic function in $\mathbb{R}^n$. By Liouville's theorem, w is constant. $\square$

We follow up with the infamous *Harnack inequality*. A more general version of this result still holds for elliptic linear operators, which we have not yet touched upon. Again, this shows how prototypical the Laplace equation is.

Let (X, d) be a metric space and fix subsets $U, V \subseteq X$. We will write $U \Subset V$ to say that

1. $U \subseteq V$ and $\overline{U} \subseteq V$,
2. $\overline{U}$ is compact in X .

Theorem 2.15 (Harnack's Inequality). *Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. Let $u \in C^2(\Omega)$ be non-negative and harmonic in Ω . For each connected open set $V \Subset \Omega$ there exists a constant $C > 0$, depending only on V (hence not on u), such that*

$$\sup_{x \in V} u(x) \leq C \inf_{x \in V} u(x).$$

Remark 2.3. Clearly, we must then have $u(x) \leq Cu(y)$ for all $x, y \in V$. Likewise, for each $x, y \in V$ there holds

$$\frac{1}{C}u(y) \leq \frac{1}{C} \sup_{x \in V} u(x) \leq \inf_{x \in V} u(x) \leq u(x).$$

Hence, Harnack's inequality tells us that

$$\frac{1}{C}u(y) \leq u(x) \leq u(y), \quad \forall x, y \in V.$$

Proof of Theorem 2.15. Define

$$r := \frac{\text{dist}(V, \partial U)}{4}.$$

Since U is bounded, it is precompact. As a consequence, ∂U is compact. If $\text{dist}(V, \partial U) = 0$, there exists sequences $(x_n)_{n=1}^\infty \subset \partial U$, $(y_n)_{n=1}^\infty \subset V$, such that

$$\lim_{n \rightarrow \infty} |x_n - y_n| = 0.$$

After passing to a subsequence, we might as well assume that x_n converges to some $x \in \partial U$. Then,

$$|y_n - x| \leq |y_n - x_n| + |x_n - x| \xrightarrow{n \rightarrow \infty} 0$$

shows that $y_n \rightarrow x$ as well. However, this implies that $x \in \overline{V} \subset U$. Since $U \cap \partial U = \emptyset$, this is a contradiction. Hence, $r > 0$.

Now, given a point $x \in V$, we have $B(x, 2r) \subseteq U$.³ Let $x, y \in V$ be such that

³Clearly, $B(x, 2r)$ does not intersect ∂U , as this would contradict the distance requirement. Hence, the only possibility remaining is that $B(x, 2r)$ intersects the interior of U^c . By way of contradiction, suppose that there exists $y \in \text{Int}(U^c)$ with $|x - y| \leq 2r$. Let $\Gamma : [0, 1] \rightarrow \mathbb{R}^n$ be the straight line connecting x and y , beginning at x and ending at y . Clearly, the distance between the point $\Gamma(t)$ and x is *strictly* increasing in t . Hence, we would have a contradiction if the image of Γ intersects ∂U . The only other option is that

$$\Gamma([0, 1]) \subseteq U \sqcup \text{Int}(U^c).$$

But this implies that $[0, 1] \subseteq \Gamma^{-1}(U) \sqcup \Gamma^{-1}(\text{Int}(U^c))$. Of course, by the continuity of Γ , this contradicts the connectedness of $[0, 1]$.

$|x - y| \leq r$; a simple inequality gives $B(y, r) \subseteq B(x, 2r)$. Since $u \geq 0$ in U , we get

$$\begin{aligned} u(x) &= \int_{B(x, 2r)} u(z) \, dz = \frac{1}{\omega_n 2^n r^n} \int_{B(x, 2r)} u(z) \, dz \geq \frac{1}{\omega_n 2^n r^n} \int_{B(y, r)} u(z) \, dz \\ &= \frac{1}{2^n} \int_{B(y, r)} u(z) \, dz \\ &= \frac{1}{2^n} u(y). \end{aligned}$$

Hence, $u(x) \geq 2^{-n} u(y)$ for all $x, y \in V$ with $|x - y| \geq 2r$. Using that V is precompact and connected, we choose finitely many open balls $\{B_1, \dots, B_N\}$ of radius $r/2$ such that $B_i \cap B_{i+1} \neq \emptyset$, for all $i = 1, \dots, N - 1$.⁴ Finally, for any given $x, y \in V$ there holds

$$u(x) \geq \frac{1}{2^{n(N+1)}} u(y).$$

This completes the proof. $\square$

2.1.5.1 The Analyticity of Harmonic Functions

Using the knowledge that a harmonic function is smooth in its domain, we finally show that such a function is in fact *analytic*. We have saved this for the end of the subsection since, unfortunately, the procedure is messy.

Theorem 2.16. *Let $\Omega \subseteq \mathbb{R}^n$ be open and assume $u \in C^2(\Omega)$ is harmonic within Ω . Then u is real analytic in Ω . Often, this is written $u \in \text{Ann}(\Omega)$.*

The reader may find the proof (for bounded Ω) in [EVNS].

2.1.6 Green's Formulas

We continue with our analysis of harmonic functions, albeit from a different point of view. Essentially, this subsection is merely preparation for one that will

⁴Choose balls $B_1, \dots, B_m$, each of radius $r/2$, covering the compact subspace $\overline{V}$. Obviously, we only choose balls that intersect the open set V . For any index $i = 1, \dots, N - 1$, if $B_i \cap B_{i+1}$ is empty, choose points x_i and x_{i+1} from $B_i \cap V$ and $B_{i+1} \cap V$, respectively. Since V is connected, it is path connected. Choose a curve Γ , contained in V , whose endpoints are x_i and x_{i+1} . We may clearly cover this curve with finitely many balls, each having radius $r/2$ and intersecting the ball adjacent to it. By repeating this procedure one can clearly construct the desired collection of balls that cover V . Indeed, by the argument we used earlier, any such ball will be contained within U .

appear shortly within this section. We will review two useful identities from vector calculus and deduce from them the important “integration by parts” identities. Using these, we shall prove the infamous “Green’s formulas” for C^2 -functions.

These identities will be helpful when introducing the Dirichlet principle (in the next subsection). Moreover, they will be *essential* when deriving Green’s formula (this is done at the end of the section). If the reader is very comfortable with vector calculus, they may without worry skip over this subsection. I only provide this review for people (such as myself) who haven’t used calculus identities for a long time.

We begin by first recalling the Gauss-Green theorem from calculus.

Theorem 2.17 (Gauss-Green). *Let $\Omega \subset \mathbb{R}^n$ be a bounded, non-empty, open subset of $\mathbb{R}^n$. Assume further that $\partial\Omega$ is C^1 .*

1. *Let $u \in C^1(\overline{\Omega})$. Then,*

$$\int_{\Omega} u_{x_i} \, dx = \int_{\partial\Omega} u v^i \, dS, \quad i = 1, \dots, n.$$

2. *There holds*

$$\int_{\Omega} \operatorname{div}(\mathbf{u}) \, dx = \int_{\partial\Omega} \mathbf{u} \cdot \nu \, dS$$

for each vector field $\mathbf{u} : \overline{\Omega} \rightarrow \mathbb{R}^n$ with $\mathbf{u} \in C^1(\overline{\Omega})$.

An immediate consequence is the well known integration by parts identity.

Theorem 2.18 (Integration by Parts). *Let $\Omega \subset \mathbb{R}^n$ be a bounded, non-empty, open subset of $\mathbb{R}^n$. Assume further that $\partial\Omega$ is C^1 . If $u, v \in C^1(\overline{\Omega})$, then*

$$\int_{\Omega} u_{x_i} v \, dm = \int_{\partial\Omega} u v v^i \, dS - \int_{\Omega} u v_{x_i} \, dm.$$

Proof. By the Gauss-Green theorem, we have

$$\int_{\Omega} (uv)_{x_i} \, dm = \int_{\partial\Omega} u v v^i \, dS.$$

On the other hand,

$$\int_{\Omega} (uv)_{x_i} \, dm = \int_{\Omega} u_{x_i} v \, dm + \int_{\Omega} u v_{x_i} \, dm.$$

Combining these two equations gives the desired identity. $\square$

Sometimes (especially when dealing with harmonic functions!) the integration by parts identity is most useful in the following form:

Theorem 2.19. *Assume the hypotheses of the previous theorem, but also suppose that $u, v \in C^2(\overline{\Omega})$. Then,*

$$\int_{\Omega} v \Delta u \, dm = \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, dS - \int_{\Omega} \nabla u \cdot \nabla v \, dm.$$

Here,

$$\frac{\partial u}{\partial \nu} := \nabla u \cdot \nu.$$

Proof. We apply the standard integration by parts identity in the following way

$$\begin{aligned} \int_{\Omega} v \Delta u \, dm &= \sum_{i=1}^n \int_{\Omega} v u_{x_i x_i} \, dm = \sum_{i=1}^n \left[\int_{\partial\Omega} v u_{x_i} \nu^i \, dS - \int_{\Omega} u_{x_i} v_{x_i} \, dm \right] \\ &= \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, dS - \int_{\Omega} \nabla u \cdot \nabla v \, dm. \end{aligned}$$

Of course, ν is the outer normal unit surface vector to $\partial\Omega$. □

We now come to the famous 'Green formulas'.

Theorem 2.20 (Green Formulas). *Let $\emptyset \neq \Omega \subset \mathbb{R}^n$ be open and bounded with C^1 -boundary. Let $u, v \in C^2(\overline{\Omega})$. Then,*

1. $\int_{\Omega} \Delta u \, dm = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \, dS;$
2. $\int_{\Omega} \nabla u \cdot \nabla v \, dm = \int_{\partial\Omega} u \frac{\partial v}{\partial \nu} \, dS - \int_{\Omega} u \Delta v \, dm;$
3. $\int_{\Omega} (u \Delta v - v \Delta u) \, dm = \int_{\partial\Omega} \left(u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} \right) \, dS.$

Proof. Let m be the Lebesgue measure on $\mathbb{R}^n$.

1. We directly calculate

$$\int_{\Omega} \Delta u \, dm = \sum_{i=1}^n \int_{\Omega} u_{x_i x_i} \, dm.$$

On each summand we use the first part of the Gauss-Green theorem to obtain

$$\int_{\Omega} \Delta u \, dm = \sum_{i=1}^n \int_{\partial\Omega} u_{x_i} v^i \, dS = \int_{\partial\Omega} \nabla u \cdot \nu \, dS = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \, dS.$$

2. We proceed similarly:

$$\begin{aligned} \int_{\Omega} \nabla u \cdot \nabla v \, dm &= \sum_{i=1}^n \int_{\Omega} u_{x_i} v_{x_i} \, dm = \sum_{i=1}^n \left[\int_{\partial\Omega} u v_{x_i} v^i \, dS - \int_{\Omega} u v_{x_i x_i} \, dm \right] \\ &= \int_{\partial\Omega} u \nabla v \cdot \nu \, dS - \int_{\Omega} u \Delta v \, dm \\ &= \int_{\partial\Omega} u \frac{\partial v}{\partial \nu} \, dS - \int_{\Omega} u \Delta v \, dm. \end{aligned}$$

In the third step, we used the integration by parts identity.

3. We utilize 2. in order to obtain

$$\int_{\Omega} u \Delta v \, dm = \int_{\partial\Omega} u \frac{\partial v}{\partial \nu} \, dS - \int_{\Omega} \nabla u \cdot \nabla v \, dm$$

and

$$\int_{\Omega} v \nabla u \, dm = \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, dS - \int_{\Omega} \nabla u \cdot \nabla v \, dm.$$

Subtracting the second from the first, we find that

$$\int_{\Omega} (u \Delta v - v \Delta u) \, dm = \int_{\partial\Omega} \left(u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} \right) \, dS.$$

This completes the proof. □

2.1.7 Energy Methods

In this subsection, we introduce an alternative method for the analysis of harmonic functions. This “new” method is often called the “energy method” since it involves the L^2 -norm. Such a perspective will become central when discussing

the calculus of variations and Sobolev spaces. For the moment, we will be content with this primitive exposition of *energy*.

Suppose that $\Omega \subseteq \mathbb{R}^n$ is open and bounded. We now return to the classical boundary problem

$$\begin{cases} -\Delta u \equiv f, & \text{in } \Omega, \\ u \equiv g, & \text{on } \partial\Omega, \end{cases} \quad (2.16)$$

where $f \in C(\Omega)$ and $g \in C(\partial\Omega)$. Of course, by a solution to (2.16) we usually mean a function $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfying (2.16). Using the maximum principle, we have already shown that the above admits *at most* one solution. Here we offer an alternative proof which better extends to more general equations.

Theorem 2.21 (Uniqueness Theorem). *If $\partial\Omega$ is C^1 , there exists at most a single $C^2(\overline{\Omega})$ solution to (2.16).*

Proof. Let u and v be two such solutions and put $w := u - v$ in Ω . Clearly, $w \in C^2(\overline{\Omega})$ satisfies

1. $\Delta w \equiv 0$ in Ω ,
2. $w \equiv 0$ on $\partial\Omega$.

Integration by parts then yields

$$\begin{aligned} 0 &= - \int_{\Omega} w(x) \Delta w(x) \, dx = - \sum_{i=1}^n \int_{\Omega} w(x) w_{x_i x_i}(x) \, dx \\ &= - \sum_{i=1}^n \left[\int_{\partial\Omega} w(x) w_{x_i}(x) \nu^i \, dS - \int_{\Omega} w_{x_i}(x)^2 \, dx \right] \\ &= \sum_{i=1}^n \int_{\Omega} w_{x_i}(x)^2 \, dx \\ &= \int_{\Omega} |\nabla w|^2 \, dx. \end{aligned}$$

That is, $\|\nabla w\|_{L^2(\Omega)} = 0$. It follows that $\nabla w = 0$ almost everywhere on Ω , and by continuity, on the whole of Ω . We conclude that w is constant on Ω , and hence on $\overline{\Omega}$. Since w vanishes on $\partial\Omega$, we get that $w \equiv 0$ on $\overline{\Omega}$. $\square$

2.1.7.1 The Dirichlet Principle (A First Look)

For this subsection, we assume that $\Omega \subset \mathbb{R}^n$ is bounded and open. For simplicity, we also make the assumption that $\partial\Omega$ is of class C^1 . The Dirichlet principle

asserts that determining a $C^2(\overline{\Omega})$ solution to (2.16) is equivalent to minimizing the appropriate *energy functional*. This approach is very useful when considering the existence of solutions to general equations/systems.

Consider the functional

$$I : C^2(\overline{\Omega}) \rightarrow \mathbb{R}, \quad w \mapsto \int_{\Omega} \left(\frac{1}{2} |\nabla w|^2 - wf \right) dm.$$

Throughout this subsection, I will be called the *energy functional*. The entire space $C^2(\overline{\Omega})$ is too broad for our analysis, and we will instead study the restriction of $I[\cdot]$ to the family

$$\mathcal{F} := \left\{ w \in C^2(\overline{\Omega}) : w \equiv g \text{ on } \partial\Omega \right\}.$$

For the Dirichlet principle we will require an elementary inequality. We provide a proof, despite it being trivial.

Lemma 2.22. *For all $a, b \geq 0$ there holds*

$$ab \leq \frac{a^2}{2} + \frac{b^2}{2}.$$

Proof. This follows at once from the fact that $(a + b)^2 = a^2 + 2ab + b^2 \geq 0$. $\square$

Remark 2.4. Suppose that $u \in C^2(\overline{\Omega})$ is a solution to (2.16). Then, f must be bounded on the whole of Ω . Certainly, this follows from the fact that Δu will be bounded on and near $\partial\Omega$. Since $-\Delta u \equiv f$ in Ω , we see that f is bounded on Ω . Therefore, we might as well assume that f is *bounded* on Ω .

We now give the main theorem of this subsection. As per our remark above, we can assume without harm that f is bounded on Ω . In particular, f belongs to $L^p(\Omega)$ for all $1 \leq p \leq \infty$.

Theorem 2.23 (Dirichlet Principle). *Suppose that $u \in C^2(\overline{\Omega})$ satisfies (2.16). Then, u minimizes $I[\cdot]$ over $\mathcal{F}$, i.e.*

$$I[u] = \min_{w \in \mathcal{F}} I[w]. \tag{2.17}$$

Conversely, if a function $u \in \mathcal{F}$ satisfies (2.17), then u solves (2.16). That is, both these problems are equivalent. We then call $I[u]$ the energy of the solution u .

Proof. First, assume that $u \in C^2(\overline{\Omega})$ satisfies (2.16). Obviously, $u \in \mathcal{F}$. Now, for each $w \in \mathcal{F}$,

$$\int_{\Omega} (-\Delta u - f)(u - w) \, dx = 0.$$

We then integrate by parts to discover that

$$\begin{aligned} 0 &= \int_{\Omega} (-\Delta u)(u - w) \, dx - \int_{\Omega} f(u - w) \, dx \\ &= \int_{\Omega} \nabla u \cdot \nabla(u - w) \, dx - \int_{\Omega} f(u - w) \, dx; \end{aligned}$$

here we have used that $u \equiv w \equiv g$ on $\partial\Omega$ to rid ourselves of the boundary term. The above expands to give

$$0 = \int_{\Omega} |\nabla u|^2 \, dx - \int_{\Omega} \nabla u \cdot \nabla w \, dx - \int_{\Omega} f u \, dx + \int_{\Omega} f w \, dx.$$

Or, rather,

$$\begin{aligned} \int_{\Omega} (|\nabla u|^2 - f u) \, dx &= \int_{\Omega} (\nabla u \cdot \nabla v - w f) \, dx \\ &\leq \int_{\Omega} (|\nabla u| |\nabla w| - w f) \, dx \\ &\leq \int_{\Omega} \left(\frac{|\nabla u|^2}{2} + \frac{|\nabla w|^2}{2} - w f \right) \, dx. \end{aligned}$$

Here, we have used both the Cauchy-Schwarz inequality and the previous lemma. Now, we notice that the above gives

$$I[u] = \int_{\Omega} \left(\frac{|\nabla u|^2}{2} - f u \right) \, dx \leq \int_{\Omega} \left(\frac{|\nabla w|^2}{2} - w f \right) \, dx = I[w].$$

It follows that u satisfies (2.17). Conversely, assume that (2.17) holds for a function $u \in \mathcal{F}$. We must show that u solves (2.16).

Let $v \in C_c^\infty(\Omega)$ be taken arbitrarily. We define a map

$$\tau : \mathbb{R} \rightarrow \mathbb{R}, \quad \tau(s) := I[u + sv].$$

Clearly, $\tau(\cdot)$ achieves a local minimum at $s = 0$. Direct manipulation also shows that

$$\begin{aligned} \tau(s) &= \int_{\Omega} \frac{1}{2} (|\nabla u + s \nabla v|^2 - (u + sv)f) \, dx \\ &= \int_{\Omega} \frac{1}{2} (|\nabla u|^2 + 2s \nabla u \cdot \nabla v + |\nabla v|^2 - (u + sv)f) \, dx. \end{aligned}$$

The expression above is clearly differentiable with respect to $s \in \mathbb{R}$. As a result, we see that $\tau'(0) = 0$. That is,

$$0 = \int_{\Omega} (\nabla u \cdot \nabla v - v f) \, dx.$$

Applying Green's formula then implies that

$$\begin{aligned} 0 &= - \int_{\Omega} (v \Delta u + v f) \, dx + \int_{\partial\Omega} \frac{\partial u}{\partial \nu} v \, dS \\ &= - \int_{\Omega} (\Delta u + f) v \, dx. \end{aligned}$$

Here, $\Delta u + f$ belongs to $L^2(\Omega)$. Since $C_c^\infty(\Omega)$ is a dense subspace of $L^2(\Omega)$, and v was taken arbitrarily, we conclude that $-\Delta u \equiv f$ almost everywhere in Ω . By continuity, we have equality everywhere in Ω . $\square$

2.1.8 Green's Function

We return to the classical initial value problem that is determining a function $u \in C^2(\Omega)$, continuous up to $\overline{\Omega}$, satisfying the Poisson problem:

$$\begin{cases} -\Delta u \equiv f, & \text{in } \Omega, \\ u \equiv g, & \text{on } \partial\Omega. \end{cases}$$

Obviously, we will need to impose certain regularity conditions on f, g , and $\partial\Omega$ in order to conduct our analysis. These will be mentioned as we progress in the derivation.

Deriving Green's function. Let $\Omega \subset \mathbb{R}^n$ be non-empty and bounded. Suppose also that $\partial\Omega$ is of class C^1 . Let us fix a point $x \in \Omega$ and choose $\varepsilon > 0$ such that the open ball $B(x, \varepsilon)$ is contained in Ω . Subsequently, we put

$$V_\varepsilon := \Omega \setminus B(x, \varepsilon).$$

Notice that ∂V_ε is also C^1 since $\partial B(x, \varepsilon)$ is C^∞ . Assume now that $u \in C^2(\overline{\Omega})$ is some real valued function. If ν denotes the outwards pointing unit normal on ∂V_ε , the third Green formula (see §2.1.6) yields

$$\int_{V_\varepsilon} [u(y) \Delta \Phi(y - x) - \Phi(y - x) \Delta u(y)] \, dy \quad (2.18)$$

$$= \int_{\partial V_\varepsilon} \left[u(y) \frac{\partial \Phi(y - x)}{\partial \nu} - \Phi(y - x) \frac{\partial u(y)}{\partial \nu} \right] \, dS(y). \quad (2.19)$$

Now, $\Delta\Phi \equiv 0$ in $\mathbb{R}^n \setminus \{0\}$. Since $x \notin V_\varepsilon$, the above actually implies that

$$-\int_{V_\varepsilon} \Phi(y-x) \Delta u(y) \, dy = \int_{\partial V_\varepsilon} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) \, dS(y) \quad (2.20)$$

$$-\int_{\partial V_\varepsilon} \Phi(y-x) \frac{\partial u(y)}{\partial \nu} \, dS(y). \quad (2.21)$$

As shown in Theorem 2.1, direct calculation gives

$$\int_{\partial B(x,\varepsilon)} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) \, dS(y) = \int_{\partial B(x,\varepsilon)} u(y) \, dS(y) \xrightarrow{\varepsilon \rightarrow 0^+} u(x).$$

Moreover, by the continuity of ∇u on the compact set $\partial B(x, \varepsilon) \subset \Omega$:

$$\begin{aligned} \int_{\partial B(x,\varepsilon)} \Phi(y-x) \left| \frac{\partial u}{\partial \nu}(y) \right| \, dS(y) &\leq C \int_{\partial B(x,\varepsilon)} \Phi(y-x) \, dS(y) \\ &\leq \begin{cases} C\varepsilon \ln \varepsilon, & n = 2, \\ C\varepsilon, & n \geq 3. \end{cases} \end{aligned}$$

Hence, letting $\varepsilon \rightarrow 0^+$ in (2.20)-(2.21) yields, by dominated convergence (all integrands are uniformly bounded on the compact set $\bar{\Omega}$),

$$\begin{aligned} -\int_{\Omega} \Phi(y-x) \Delta u(y) \, dy &= \int_{\partial \Omega} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) \, dS(y) \\ &\quad - \int_{\partial \Omega} \Phi(y-x) \frac{\partial u}{\partial \nu}(y) \, dS(y) + u(x). \end{aligned}$$

Or, rather,

$$u(x) = \int_{\partial \Omega} \left[\Phi(y-x) \frac{\partial u}{\partial \nu}(y) - u(y) \frac{\partial \Phi}{\partial \nu}(y-x) \right] \, dS(y) \quad (2.22)$$

$$- \int_{\Omega} \Phi(y-x) \Delta u(y) \, dy, \quad (2.23)$$

for each $x \in \Omega$. We are not yet done. Certainly, the above expression for $u(x)$ is only useful if we know

- (i) the values of Δu in Ω ,
- (ii) how ∇u behaves pointwise on $\partial \Omega$.

Although we know the values of Δu in Ω (recall that u solves the Poisson problem), the behaviour of ∇u on $\partial\Omega$ is unknown to us. Hence, we must rid ourselves of this term.

Fix a point $x \in \Omega$ and assume we have found a corrector function $\phi^x = \phi^x(y)$, in $C^2(\overline{\Omega})$, satisfying the following initial value problem:

$$\begin{cases} \Delta \phi^x \equiv 0, & \text{in } \Omega, \\ \phi^x = \Phi(y - x), & \text{on } \partial\Omega \end{cases} \quad (2.24)$$

Then, an application of the Green's formula (again, see §2.1.6) offers

$$\begin{aligned} - \int_{\Omega} \phi^x(y) \Delta u(y) \, dy &= \int_{\Omega} [u(y) \Delta \phi^x(y) - \phi^x(y) \Delta u(y)] \, dy \\ &= \int_{\partial\Omega} \left[u(y) \frac{\partial \phi^x}{\partial \nu}(y) - \phi^x(y) \frac{\partial u}{\partial \nu}(y) \right] \, dS(y) \\ &= \int_{\partial\Omega} \left[u(y) \frac{\partial \phi^x}{\partial \nu}(y) - \Phi(y - x) \frac{\partial u}{\partial \nu}(y) \right] \, dS(y). \end{aligned}$$

We now add this last equation to (2.22)-(2.23), thereby obtaining

$$\begin{aligned} u(x) - \int_{\Omega} \phi^x(y) \Delta u(y) \, dy &= \int_{\partial\Omega} \left[\Phi(y - x) \frac{\partial u}{\partial \nu}(y) - u(y) \frac{\partial \Phi}{\partial \nu}(y - x) \right] \, dS(y) \\ &\quad + \int_{\partial\Omega} \left[u(y) \frac{\partial \phi^x}{\partial \nu}(y) - \Phi(y - x) \frac{\partial u}{\partial \nu}(y) \right] \, dS(y) \\ &\quad - \int_{\Omega} \Phi(y - x) \Delta u(y) \, dy. \end{aligned}$$

Hence,

$$\begin{aligned} u(x) &= - \int_{\partial\Omega} \left[u(y) \frac{\partial \Phi}{\partial \nu}(y - x) - u(y) \frac{\partial \phi^x}{\partial \nu}(y) \right] \, dS(y) \\ &\quad - \int_{\Omega} \Delta u(y) (\Phi(y - x) - \phi^x(y)) \, dy. \end{aligned}$$

This motivates the following definition.

Definition 13 (Green's Function for Ω). The Green's function for the given region Ω is defined to be

$$G(x, y) := \Phi(y - x) - \phi^x(y), \quad x, y \in \Omega,$$

with $x \neq y$.

With this notation, our last equation can be recast in the following way:

$$u(x) = - \int_{\partial\Omega} u(y) \frac{\partial G}{\partial \nu_y}(x, y) \, dS(y) - \int_{\Omega} G(x, y) \Delta u(y) \, dy.$$

Of course, by $\frac{\partial G}{\partial \nu_y}$ we *really* mean $\nabla_y G(x, y) \cdot \nu(y)$. Now, we make the natural assumption that for each $x \in \Omega$ one can find a corrector function ϕ^x as in (2.24). Then, the equation above would hold for each and every $x \in \Omega$. In light of this, we obtain the following theorem.

Theorem 2.24 (Green's Representation Theorem). *Let $\emptyset \neq \Omega \subset \mathbb{R}^n$ be open and bounded. Suppose further that $\partial\Omega$ is of class C^1 . If $u \in C^2(\overline{\Omega})$ solves the Poisson equation*

$$\begin{cases} -\Delta u \equiv f, & \text{in } \Omega, \\ u \equiv g, & \text{on } \partial\Omega, \end{cases}$$

for given functions $f \in C(\Omega)$ and $g \in C(\partial\Omega)$. Then,

$$u(x) = - \int_{\partial\Omega} u(y) \frac{\partial G}{\partial \nu_y}(x, y) \, dS(y) - \int_{\Omega} G(x, y) \Delta u(y) \, dy.$$

for every $x \in \Omega$. Notice that this allows us to describe u on the whole of $\overline{\Omega}$.

Of course, there is a caveat here. It is usually very difficult to find the Green's function $G(x, y)$ —when it exists. Luckily, if the region Ω has a nice geometry, it can be possible. We will shortly illustrate this with an example. First, however, let us prove a simple property that all Green functions must satisfy.

2.1.8.1 Symmetry of Green's Function

Theorem 2.25. *Let $\Omega \neq \emptyset$ be an open and bounded subset of $\mathbb{R}^n$ with C^1 boundary. Assume the Green's function $(x, y) \mapsto G(x, y)$ exists in Ω , for $x \neq y$. Then G is symmetric, i.e.*

$$G(x, y) = G(y, x)$$

for all $x, y \in \Omega$ with $x \neq y$.

Proof. Let $x, y \in \Omega$ be such that $x \neq y$. Define

$$v(z) := G(x, z) \quad \text{for } z \in \Omega \setminus \{x\}$$

and

$$w(z) := G(y, z) \quad \text{for } z \in \Omega \setminus \{y\}.$$

Then, $\Delta v(z) = 0$ for $z \neq x$ and $\Delta w(z) = 0$ for $z \neq y$. Moreover, we will have $v \equiv 0 \equiv w$ on $\partial\Omega$ by definition of the Green's function. Let $\varepsilon > 0$ be so small that $B(x, \varepsilon) \cup B(y, \varepsilon) \subset \Omega$. We may also assume that the closures of these balls are disjoint. Then, set

$$V_\varepsilon := \Omega \setminus (B(x, \varepsilon) \cup B(y, \varepsilon)).$$

A routine application of the Green formulas (see §2.1.6) grants us the following:

$$0 = \int_{V_\varepsilon} (v \Delta w - w \Delta v) \, dm = \int_{\partial V_\varepsilon} \left(w(z) \frac{\partial v}{\partial \nu}(z) - v(z) \frac{\partial w}{\partial \nu}(z) \right) dS(z).$$

Recalling that v and w vanish on $\partial\Omega$, the above reduces to

$$\begin{aligned} & \int_{\partial B(x, \varepsilon)} \left(w(z) \frac{\partial v}{\partial \nu}(z) - v(z) \frac{\partial w}{\partial \nu}(z) \right) dS(z) \\ &= \int_{\partial B(y, \varepsilon)} \left(v(z) \frac{\partial w}{\partial \nu}(z) - w(z) \frac{\partial v}{\partial \nu}(z) \right) dS(z). \end{aligned}$$

Here, ν is the inward pointing unit normal on each of these balls. Recall that $w(z)$ is smooth near x (since it only has a singularity near $y \neq x$). As before, it follows that

$$\lim_{\varepsilon \searrow 0} \int_{\partial B(x, \varepsilon)} v(z) \frac{\partial w}{\partial \nu}(z) dS(z) = 0.$$

Moreover, the calculations from Theorem 2.1 tell us once more that

$$\begin{aligned} \lim_{\varepsilon \searrow 0} \int_{\partial B(x, \varepsilon)} w(z) \frac{\partial v}{\partial \nu}(z) dS(z) &= \lim_{\varepsilon \searrow 0} \int_{\partial B(x, \varepsilon)} w(z) \frac{\partial \Phi}{\partial \nu}(x - z) dS(z) \\ &\xrightarrow{y:=x-z} \lim_{\varepsilon \searrow 0} \int_{\partial B(0, \varepsilon)} w(x - y) \frac{\partial \Phi}{\partial \nu}(y) dS(y) \\ &= w(x). \end{aligned}$$

This implies that

$$\lim_{\varepsilon \searrow 0} \int_{\partial B(x, \varepsilon)} \left(w(z) \frac{\partial v}{\partial \nu}(z) - v(z) \frac{\partial w}{\partial \nu}(z) \right) dS(z) = w(x).$$

Similarly, one can show that

$$\lim_{\varepsilon \searrow 0} \int_{\partial B(y, \varepsilon)} \left(v(z) \frac{\partial w}{\partial \nu}(z) - w(z) \frac{\partial v}{\partial \nu}(z) \right) dS(z) = v(y).$$

This means that $w(x) = v(y)$, which completes the proof. $\square$

2.1.9 Poisson's Formula

Typically, it is difficult to explicitly compute the Green's function for a given region Ω , unless Ω itself has a simple geometric structure. In this subsection, we derive the Green's function for two simple regions and prove the corresponding representation formulas.

Green's function for a half-space. Let $n \geq 2$ be given, we consider the open half space

$$\mathbb{R}_+^n := \{x = (x_1, \dots, x_n) \in \mathbb{R}^n : x_n > 0\}.$$

Clearly, $\partial \mathbb{R}_+^n$ consists of the plane $\{x_n = 0\}$. The determination of the Green's function for $\mathbb{R}_+^n$ rests upon finding a corrector function for $\mathbb{R}_+^n$. Also, since this region is unbounded, we cannot directly use the calculations from the previous section. Hence, we have a great deal of verification ahead of us.

Definition 14. Let $x \in \mathbb{R}_+^n$, its reflection across the plane $\partial \mathbb{R}_+^n$ is denoted by

$$\tilde{x} = (x_1, \dots, x_{n-1}, -x_n).$$

For a given $x \in \mathbb{R}_+^n$, we define the *corrector function*

$$\phi^x(y) := \Phi(y - \tilde{x}) = \Phi(y_1 - x_1, \dots, y_{n-1} - x_{n-1}, y_n + x_n), \quad y \in \mathbb{R}_+^n.$$

As required, we have $\phi^x(y) \equiv \Phi(y - x)$ for $y \in \partial \mathbb{R}_+^n$. Moreover, direct calculation shows that

$$\Delta \phi^x \equiv \Delta_y \Phi(y - \tilde{x}) \equiv 0$$

for $y \in \mathbb{R}_+^n$; this is because $y_n + x_n > 0$ for all $y_n \in \mathbb{R}_+^n$. In other words, $y - \tilde{x}$ is isolated from the singularity of Φ . To summarize,

$$\begin{cases} \Delta \phi^x \equiv 0, & \text{in } \mathbb{R}_+^n, \\ \phi^x \equiv \Phi(y - x), & \text{on } \partial \mathbb{R}_+^n. \end{cases}$$

We can now define the *Green's function for the half-space*:

$$G(x, y) := \Phi(y - x) - \phi^x(y) = \Phi(y - x) - \Phi(y - \tilde{x}), \quad x, y \in \mathbb{R}_+^n, x \neq y. \quad (2.25)$$

If $y \in \partial\mathbb{R}_+^n$, we calculate (for $x \in \mathbb{R}_+^n$)

$$\frac{\partial G}{\partial \nu}(x, y) = -G_{y_n}(x, y) = \frac{2}{n\omega_n |x - y|^n}.$$

We might then expect

$$K(x, y) := \frac{2x_n}{n\alpha(n)} \frac{1}{|x - y|^n}, \quad x \in \mathbb{R}_+^n, y \in \partial\mathbb{R}_+^n$$

to play a role in our representation formula. Certainly, we call the above the *Poisson kernel for a half-space*. All of this will come together to yield the following result.

Theorem 2.26 (Poisson Formula for Half-Space). *Let $g \in C(\mathbb{R}^{n-1}) \cap L^\infty(\mathbb{R}^{n-1})$ be given and define*

$$u(x) := \frac{2x_n}{n\omega_n} \int_{\partial\mathbb{R}_+^n} \frac{g(y)}{|x - y|^n} dy$$

for $x \in \mathbb{R}_+^n$. Then,

1. $u \in C^\infty(\mathbb{R}_+^n) \cap L^\infty(\mathbb{R}_+^n)$,
2. $\Delta u \equiv 0$ in $\mathbb{R}_+^n$,
3. u can be continuously extended to $\partial\mathbb{R}_+^n$ by setting $u \equiv g$ on the boundary.

The proof is a standard estimating argument we have done multiple times. See [EVNS] for the proof. One can also proceed similarly to obtain an explicit Green's function for an open ball. We leave the details to the reader; as a reference, consult [EVNS]-[JOST].

2.2 The Heat Equation

We devote this section to the study of the heat equation in $\mathbb{R}^n$. Unless stated otherwise, Ω will denote an open and non-empty subset of $\mathbb{R}^n$. Also, t will always

be assumed to be non-negative, i.e. $t \in [0, \infty)$. A function u satisfying the heat equation will be assumed to be a map

$$u : \overline{\Omega} \times [0, \infty) \longrightarrow \mathbb{R}, \quad u = u(x, t).$$

The Laplacian will only be considered with respect to the *spatial variables*. More precisely, if $x = (x_1, \dots, x_n)$ then

$$\Delta u \equiv \Delta_x u := \sum_{j=1}^n \frac{\partial^2 u}{\partial x_j^2}.$$

With this out of the way, we can now state a classical problem in mathematics.

Definition 15 (Heat Equation). Assume $\Omega \subseteq \mathbb{R}^n$ is non-empty and open. We say a function u satisfies the *heat equation* if

$$u_t - \Delta u \equiv 0, \quad \text{in } \Omega \times (0, \infty). \quad (2.26)$$

Alternatively, such a function u is said to solve the *non-homogeneous heat equation* provided

$$u_t - \Delta u \equiv f, \quad \text{in } \Omega \times (0, \infty). \quad (2.27)$$

Of course, $f \in C(\Omega)$ is a given function.

Unlike the Laplacian, determining a fundamental solution to the homogeneous heat equation is a non-trivial process. One of the “simpler” ways to go about this involves the Fourier transform. Since we will not cover this transform in these notes, we will graciously omit the details of the derivation. The interested reader can simply consult [EVNS] or [JOST].

Definition 16 (Heat Kernel). The function $(x, t) \mapsto \Phi(x, t)$ defined by

$$\Phi(x, t) := \begin{cases} \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x|^2}{4t}}, & \text{if } x \in \mathbb{R}^n, t > 0, \\ 0, & \text{if } x \in \mathbb{R}^n, t \leq 0, \end{cases} \quad (2.28)$$

is called the *heat kernel* or *fundamental solution* to the heat equation. For historical reasons, we will prefer the term *heat kernel* for this book.

Direct calculation shows that the map $(x, t) \mapsto \Phi(x, t)$ satisfies the homogeneous heat equation away from $t = 0$, i.e.

$$\Phi_t - \Delta \Phi \equiv 0 \quad \text{for } t > 0.$$

The choice of the constant $(4\pi)^{-n/2}$ is a consequence of the following normalization property.

Lemma 2.27. *Let $t > 0$. Then,*

$$\int_{\mathbb{R}^n} \Phi(x, t) \, dx = 1.$$

Proof. The proof utilizes the Fubini-Tonelli theorem. Proceeding directly,

$$\begin{aligned} \int_{\mathbb{R}^n} \Phi(x, t) \, dx &= \frac{1}{(4\pi t)^{n/2}} \prod_{j=1}^n \int_{\mathbb{R}} e^{-\frac{x_j^2}{4t}} \, dx_j = \frac{1}{(4\pi t)^{n/2}} \prod_{j=1}^n \int_{\mathbb{R}} e^{-\frac{y^2}{4t}} \, dy \\ &\xrightarrow{z:=\frac{y}{\sqrt{4t}}} \frac{1}{\pi^{n/2}} \prod_{j=1}^n \int_{\mathbb{R}} e^{-z^2} \, dz \\ &= \frac{1}{\pi^{n/2}} \cdot \pi^{n/2} \\ &= 1. \end{aligned}$$

□

With this formality out of the way, we can turn towards results that actually involve solutions to the heat equation. As with the Laplacian, we will consider boundary value type problems.

2.2.1 Cauchy Problem

We now turn towards the following Cauchy (or initial value) problem for the heat equation:

$$\begin{cases} u_t - \Delta u \equiv 0, & \text{in } \mathbb{R}^n \times (0, \infty), \\ u \equiv g, & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases} \quad (2.29)$$

for a given function $g \in C(\mathbb{R}) \cap L^1(\mathbb{R}^n)$. If the our intuition from the treatment of the Laplacian is accurate, we should examine the convolution

$$u(x, t) := \int_{\mathbb{R}^n} g(y) \Phi(x - y, t) \, dy, \quad x \in \mathbb{R}^n, \, t > 0. \quad (2.30)$$

Notice that, for each $t > 0$, the function $x \mapsto \Phi(x, t)$ belongs to $L^\infty(\mathbb{R}^n)$. Since g is integrable on $\mathbb{R}^n$, it is a direct consequence of Hölder's inequality that $u(x, t)$ exists for all $x \in \mathbb{R}^n$ and $t > 0$.

Theorem 2.28. *Suppose that $g \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$ is bounded and let $u(x, t)$ be given by (2.30).*

1. u is smooth on $\mathbb{R}^n \times (0, \infty)$;
2. $u_t - \Delta u \equiv 0$ in $\mathbb{R}^n \times (0, \infty)$;
3. for each point $x^0 \in \mathbb{R}^n$ one has

$$\lim_{\substack{(x,t) \rightarrow (x^0,0) \\ x \in \mathbb{R}^n, t > 0}} u(x, t) = g(x^0).$$

Proof. First, let $\delta > 0$ be given. Since $\Phi(x, t)$ is smooth on $\mathbb{R}^n \times [\delta, \infty)$, it follows at once that $u(x, t)$ is smooth on $\mathbb{R}^n \times [\delta, \infty)$. Since $\delta > 0$ was arbitrary, we get that $u \in C^\infty(\mathbb{R}^n \times (0, \infty))$. Moreover,

$$u_t(x, t) - \Delta u(x, t) = \int_{\mathbb{R}^n} [\Phi_t(x - y, t) - \Delta \Phi(x - y, t)] g(y) dy = 0.$$

It therefore only remains to prove the third point. Fix a point $x^0 \in \mathbb{R}^n$ and let $\varepsilon > 0$ be given. Choose $\delta > 0$ with the property that

$$|g(y) - g(x^0)| < \varepsilon$$

whenever $|y - x^0| < \delta$, with $y \in \mathbb{R}^n$. Assume that $|x - x^0| < \frac{\delta}{2}$; for $t > 0$ the previous lemma yields

$$\begin{aligned} |u(x, t) - g(x^0)| &\leq \int_{\mathbb{R}^n} \Phi(x - y, t) |g(y) - g(x^0)| dy \\ &= I + J \end{aligned}$$

for

$$I := \int_{B(x^0, \delta)} \Phi(x - y, t) |g(y) - g(x^0)| dy$$

and

$$J := \int_{\mathbb{R}^n \setminus B(x^0, \delta)} \Phi(x - y, t) |g(y) - g(x^0)| dy.$$

Clearly, for each $t > 0$

$$I \leq \varepsilon \int_{B(x^0, \delta)} \Phi(x - y, t) dy \leq \varepsilon \int_{\mathbb{R}^n} \Phi(x - y, t) dy = \varepsilon$$

since $\int_{\mathbb{R}^n} \Phi(x, t) dx = 1$. If additionally $|y - x^0| \geq \delta$:

$$|y - x_0| \leq |y - x| + |x - x_0| \leq |y - x| + \frac{\delta}{2} \leq |y - x| + \frac{|y - x_0|}{2}.$$

This implies that

$$\frac{|y - x^0|}{2} \leq |y - x|$$

for all y with $|y - x^0| \geq \delta$. Hence, if $t > 0$, the inequality above gives

$$\begin{aligned} J &\leq 2M \int_{\mathbb{R}^n \setminus B(x^0, \delta)} \Phi(x - y, t) dy \leq \frac{C}{t^{n/2}} \int_{\mathbb{R}^n \setminus B(x^0, \delta)} e^{-\frac{|x-y|^2}{4t}} dy \\ &\leq \frac{C}{t^{n/2}} \int_{\mathbb{R}^n \setminus B(x^0, \delta)} e^{-\frac{|y-x^0|^2}{16t}} dy \\ &= C \int_{\mathbb{R}^n \setminus B(0, \delta/\sqrt{t})} e^{-\frac{|z|^2}{16}} dz \\ &\xrightarrow{t \rightarrow 0^+} 0. \end{aligned}$$

By taking $|x^0 - x| < \frac{\delta}{2}$ and allowing $0 < t \ll 1$, we can ensure that

$$|u(x, t) - g(x^0)| \leq I + J < 2\varepsilon,$$

as was required. $\square$

Remark 2.5. Assume the setting of the previous theorem and define u as in (2.30). The theorem tells us that, by extending u to be equal to g on $\mathbb{R}^n \times \{t = 0\}$, we obtain a continuous solution to (2.29).

Infinite Propagation Speed. In addition to the hypotheses of the theorem we have proven above, assume that $g \geq 0$ is *not* identically zero. In particular, there exists an open ball on which $g \neq 0$. The solution

$$u(x, t) := \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} g(y) e^{-\frac{|x-y|^2}{4t}} dy$$

is then positive at all points $x \in \mathbb{R}^n$ and all times $t > 0$. We interpret this phenomenon as follows:

Assume the temperature of a system is everywhere non-negative and the temperature is positive *somewhere* at a time $t_0 > 0$. Then, the temperature is positive *everywhere* at all times $t > t_0$.

2.2.2 Non-homogeneous Cauchy Problem

Let us now examine a non-homogeneous analogue to problem (2.29):

$$\begin{cases} u_t - \Delta u \equiv f, & \text{in } \mathbb{R}^n \times (0, \infty), \\ u \equiv 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \quad (2.31)$$

Here, we will obviously need to assume that f is sufficiently regular. To formalize this, we introduce a space of functions involving time. Let $\Omega \subseteq \mathbb{R}^n$ and $\Lambda \subseteq [0, \infty)$ both be non-empty sets. We will define $C_1^2(\Omega \times \Lambda)$ to be the set of all continuous functions $u = u(x, t) : \Omega \times \Lambda \rightarrow \mathbb{R}$ such that

1. All second order spatial partial derivatives of $u(x, t)$ exist and are continuous on $\Omega \times \Lambda$.
2. $u_t(x, t)$ exists and is continuous on $\Omega \times \Lambda$.

Using *Duhamel's principle*, we will strategically derive a solution to (2.31). Let $y \in \mathbb{R}^n$ and $s > 0$ be given. Note that the function

$$(x, t) \mapsto \Phi(x - y, t - s)$$

solves the homogeneous heat equation in $\mathbb{R}^n \times (s, \infty)$. For such a fixed s , the map

$$u \sim u(x, t; s) := \int_{\mathbb{R}^n} \Phi(x - y, t - s) f(y, s) dy$$

can be extended (as in Remark 2.5) to solve the problem

$$\begin{cases} u_t(\cdot; s) - \Delta u(\cdot; s) \equiv 0, & \text{in } \mathbb{R}^n \times (s, \infty), \\ u(\cdot; s) \equiv f(\cdot; s), & \text{on } \mathbb{R}^n \times \{t = s\}. \end{cases}$$

The process of this to obtain a solution to (2.31) is known as the *Duhamel principle*. This procedure involves defining a new function

$$u(x; t) := \int_0^t u(x, t; s) ds, \quad \text{for } x \in \mathbb{R}^n \text{ and } t \geq 0.$$

By Fubini-Tonelli, we see that this is equivalent to writing

$$u(x, t) = \int_0^t \int_{\mathbb{R}^n} \Phi(x - y, t - s) f(y, s) dy ds \quad (2.32)$$

$$= \int_0^t \frac{1}{(4\pi(t-s))^{n/2}} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{4(t-s)}} f(y, s) dy ds; \quad (2.33)$$

provided f is integrable over its domain. For simplicity, let us now assume that $f \in C_1^2(\mathbb{R}^n \times [0, \infty))$ has compact support.

Theorem 2.29. *Let $f \in C_1^2(\mathbb{R}^n \times [0, \infty))$ have compact support and define u by (2.32)-(2.33). The following hold true:*

1. $u \in C_1^2(\mathbb{R}^n \times (0, \infty))$;
2. $u_t(x, t) - \Delta u(x, t) = f(x, t)$ for $x \in \mathbb{R}^n$ and $t > 0$;
3. fix a point $x^0 \in \mathbb{R}^n$, then

$$\lim_{\substack{(x,t) \rightarrow (x^0,0) \\ x \in \mathbb{R}^n, t > 0}} u(x, t) = 0.$$

The messy proof can be found in [EVNS] and is very similar to that of the previous theorem. Our main goal here was to illustrate the intuition behind the Duhamel principle.

Remark 2.6. As in Remark 2.5, the theorem tells us how to construct a *continuous* solution to (2.31) in $\mathbb{R}^n \times [0, \infty)$. Certainly, this simply involves extending $u(x, t)$ to $\mathbb{R}^n \times \{t = 0\}$ via $u(x, 0) \equiv 0$.

Remark 2.7. We now know how to solve both (2.29) and (2.31). Combining these two strategies allows us to also solve the following problem:

$$\begin{cases} u_t - \Delta u = f, & \text{in } \mathbb{R}^n \times (0, \infty), \\ u \equiv g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \quad (2.34)$$

for a compactly supported function $f \in C_1^2(\mathbb{R}^n \times [0, \infty))$ and a bounded function $g \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$. Certainly, such a solution is given by

$$u(x, t) := \int_{\mathbb{R}^n} \Phi(x - y, t) g(y) dy + \int_0^t \int_{\mathbb{R}^n} \Phi(x - y, t - s) f(y, s) dy ds.$$

2.2.3 Qualitative Properties

We now turn towards the analysis of solutions to the heat equation, rather than solving the problem “explicitly”. As in the previous section, we will establish maximum principles and uniqueness results.

Throughout this subsection, assume that $\Omega \subset \mathbb{R}^n$ is non-empty, open, and bounded. Fix also a time $T > 0$. With these constants in mind, we introduce the following.

Definition 17 (Parabolic Domains). The parabolic cylinder (associated to Ω and T) is defined to be the set

$$\Omega_T := \Omega \times (0, T].$$

The *Parabolic boundary* of Ω_T is the set

$$\Gamma_T := \overline{\Omega_T} \setminus \Omega_T,$$

where Ω_T is viewed as a subset of the product space $\mathbb{R}^n \times [0, \infty)$.

We would like to formulate a mean value property specific to the heat equation. This is certainly possible (see [EVNS, §2.3]), but requires different conditions than for harmonic functions. For one, we are working in a region of space-time, which has a particular geometry. For this reason, we must give yet another definition.

Definition 18. Fix $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, and $r > 0$. We define

$$E(x, t; r) := \left\{ (y, s) \in \mathbb{R}^n \times \mathbb{R} : s \leq t, \Phi(x - y, t - s) \geq \frac{1}{r^n} \right\}.$$

We now state the following theorem from [EVNS], without proof.

Theorem 2.30. Let $u \in C_1^2(\Omega_T)$ satisfy

$$u_t - \Delta u \equiv 0 \quad \text{in } \Omega_T.$$

Then, for each $E(x, t; r) \subseteq \Omega_T$ one has

$$u(x, t) = \frac{1}{4r^n} \iint_{E(x, t; r)} u(y, s) \frac{|x - y|^2}{(t - s)^2} dy ds.$$

Using this, we will be able to prove the subsequent maximum principles.

Theorem 2.31 (Maximum Principles for the Heat Equation). Let $u \in C_1^2(\Omega_T)$ be continuous on $\overline{\Omega_T}$. Assume also that u satisfies $u_t - \Delta u \equiv 0$ in Ω_T .

1. Then,

$$\max_{\overline{\Omega_T}} u = \max_{\Gamma_T} u.$$

2. Suppose in addition that Ω is connected. If there exists a point $(x_0, t_0) \in \Omega_T$ such that

$$\max_{\overline{\Omega_T}} u = u(x_0, t_0),$$

then u is constant in $\overline{\Omega_{t_0}}$. Note we do not imply that u is constant on $\overline{\Omega_T}$.

Proof. 1. By assumption, the space $\overline{\Omega_T}$ is closed and bounded. Since furthermore u is continuous, it must achieve it's maximum in this space. In order to show that this maximum is always achieved on the boundary, we suppose that u achieves it's maximum in the interior. That is, suppose there exists $(x_0, t_0) \in \Omega_T$ such that

$$u(x_0, t_0) = \max_{\overline{\Omega_T}} u =: M$$

Now, we may pick $r > 0$ sufficiently small so that $E(x_0, t_0, r) \subseteq \Omega_T$. By the mean value property for the heat equation,

$$\begin{aligned} M = u(x_0, t_0) &= \frac{1}{4r^n} \iint_{E(x_0, t_0, r)} u(y, s) \frac{|x_0 - y|^2}{(t_0 - s)^2} dy ds \\ &\leq M \frac{1}{4r^n} \iint_{E(x_0, t_0, r)} \frac{|x_0 - y|^2}{(t_0 - s)^2} dy ds = M \end{aligned}$$

In the above, we have omitted the tedious computation of the integral. Now, we see that equality holds if and only if u is constant on $E(x_0, t_0, r)$ and everywhere equal to M .

We now pick a point (y_0, s_0) on the boundary of Ω_T with $s_0 < t_0$ such that the line segment L from (x_0, t_0) to (y_0, s_0) is contained in $\overline{\Omega_T}$.⁵ We wish to show that $u \equiv M$ on L .

To this end, define

$$r_0 = \min \{s \geq s_0 \mid u(x, t) = M \text{ for all points } (x, t) \in L, s \leq t \leq t_0\}$$

The minimum is indeed attained since u is continuous. Furthermore, it is clear that $s_0 \leq r_0$. If $s_0 < r_0$ then there exists a point $(z_0, r_0) \in L \cap \Omega_T$ such that $u(z_0, r_0) = M$. By our previous argument, $u \equiv M$ on $E(z_0, r_0, r)$ for all sufficiently small $r > 0$. But this is clearly a contradiction to the minimality of r_0 since

⁵This is a general fact for open bounded sets in $\mathbb{R}^n$ and can be shown with the help of a topological argument.

$E(z_0, r_0, r)$ contains $L \cap \{r_0 - \varepsilon \leq t \leq r_0\}$ for a sufficiently small $\varepsilon > 0$. Thus, $r_0 = s_0$ which shows that $u \equiv M$ on L .

We conclude that $u(y_0, s_0) = M$ and so u achieves its maximum on the boundary. That is,

$$\max_{\overline{\Omega_T}} u = \max_{\Gamma_T} u.$$

2. We pick an arbitrary point $(x, t) \in \Omega_{t_0}$ with $t < t_0$ and show that $u(x, t) = M$ where

$$M := \max_{\overline{\Omega_T}} u = u(x_0, t_0).$$

Since Ω is assumed to be connected in $\mathbb{R}^n$, we may pick a sequence of point $x_1, x_2, \dots, x_n = x \in \Omega$ such that the line segments from x_{i-1} to x_i are contained in Ω for $i = 1, \dots, n$. We also pick point $t_0 > t_1 > \dots > t_n = t$. Now, by the argument used in proving the first part of the theorem, we see that $u \equiv M$ on the line from (x_0, t_0) to (x_1, t_1) . Repeating this argument n times shows that $u \equiv M$ on each line segment from (x_{i-1}, t_{i-1}) to (x_i, t_i) for $i = 1, \dots, n$. In particular, we see that

$$u(x, t) = u(x_n, t_n) = M$$

Recalling that (x, t) was an arbitrary point in Ω_{t_0} with $t < t_0$, we see that $u \equiv M$ in $\Omega_{t_0} \cap \{t < t_0\}$. By continuity of u we see that $u \equiv M$ in all of Ω_{t_0} . $\square$

Infinite Propagation Speed. Suppose Ω is bounded domain and the function $u \in C_1^2(\Omega_T) \cap C(\overline{\Omega_T})$ satisfies

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \Omega_T \\ u = 0 & \text{on } \partial\Omega \times [0, T] \\ u = g & \text{on } \Omega \times \{t = 0\} \end{cases}$$

where $g \geq 0$. As a consequence of the above theorem, we see that if g is positive somewhere on Ω , then u is positive everywhere within Ω_T .

2.2.4 Properties of Solutions to the Heat equation

We now take a look at solutions to different variants of the Cauchy problem. We will provide sufficient conditions for uniqueness of the solution.

Theorem 2.32 (Uniqueness on bounded domains). *Suppose that the functions $g : \Gamma_T \rightarrow \mathbb{R}$ and $f : \Omega_T \rightarrow \mathbb{R}$ are continuous. Then there exists at most one continuous solution $u \in C_1^2(\Omega_T)$ of the initial value problem*

$$\begin{cases} u_t - \Delta u = f, & \text{in } \Omega_T, \\ u = g, & \text{on } \Gamma_T. \end{cases} \quad (2.35)$$

where Ω is an open and bounded subset of $\mathbb{R}^n$.

Proof. Suppose u and $\tilde{u}$ both solve the above problem. Then $w = u - \tilde{u}$ satisfies the requirements of Theorem 2.31. Ergo,

$$\max_{\overline{\Omega_T}} (u - \tilde{u}) = \max_{\overline{\Omega_T}} w = \max_{\Gamma_T} w = 0$$

Similarly, since $\tilde{w} = \tilde{u} - u$ satisfies the requirements of Theorem 2.31 it holds that

$$\max_{\overline{\Omega_T}} (\tilde{u} - u) = \max_{\overline{\Omega_T}} \tilde{w} = \max_{\Gamma_T} \tilde{w} = 0$$

We conclude that $u \equiv \tilde{u}$. □

Theorem 2.33 (Maximum Principle for the Cauchy problem). *Suppose the continuous function $u \in C_1^2(\mathbb{R}^n \times [0, T])$ solves*

$$\begin{cases} u_t - \Delta u = 0, & \text{in } \mathbb{R}^n \times (0, T), \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \quad (2.36)$$

and satisfies

$$|u(x, t)| \leq Ae^{a|x|^2} \quad (x, t) \in \mathbb{R}^n \times [0, T]$$

for constant $A, a > 0$. Then

$$\sup_{\mathbb{R}^n \times [0, T]} u = \sup_{\mathbb{R}^n} g$$

We shall omit the proof of this theorem since it is tedious, highly computational and not particularly instructive.

Theorem 2.34 (Uniqueness for Cauchy problem). *Suppose $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and $f : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}$ are continuous functions. Then there exists at most one continuous solution $u \in C_1^2(\mathbb{R}^n \times [0, T])$ of the initial value problem*

$$\begin{cases} u_t - \Delta u = f, & \text{in } \mathbb{R}^n \times (0, T), \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \quad (2.37)$$

satisfying the growth estimate

$$|u(x, t)| \leq Ae^{a|x|^2} \quad (x, t) \in \mathbb{R}^n \times [0, T]$$

for constant $A, a > 0$.

Proof. In a similar vain as the proof of Theorem 2.32, the result follows by applying the previous theorem to $w = \pm(u - \tilde{u})$. $\square$

Remark 2.8. In the above theorem, the growth estimate is a criteria which excludes the “wrong” solution. In fact, there are infinitely many solutions to the system

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, T) \\ u = 0 & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

Besides the obvious $u \equiv 0$, each solution to the above grows very rapidly as $|x| \rightarrow \infty$. These rapidly increasing solutions are called “nonphysical”.

2.2.5 Further Properties of the Heat Equation

An interesting fact is that solutions to the heat equation are automatically smooth. We shall also present some estimates for solutions of the heat equation and it’s derivatives. Finally, we provide an alternate argument for uniqueness based on energy methods and discuss the principle of backwards uniqueness.

Throughout this section, we assume that Ω is an open set. We begin with two theorems which shall be stated without proof.

Theorem 2.35 (Smoothness). *Suppose $u \in C_1^2(\Omega_T)$ solves $u_t - \Delta u = 0$ in Ω_T . Then u is smooth, i.e. $u \in C^\infty(\Omega_T)$. Note that we do not assume that u attains smooth boundary values on Γ_T .*

Theorem 2.36 (Estimates of derivatives). *Under the conditions of the above theorem, for each pair of non-negative integers k, ℓ there exists a constant $C_{k,\ell}$ such that*

$$\max_{C(x,t;r/2)} |D_x^k D_t^\ell u| \leq \frac{C_{k,\ell}}{r^{k+2\ell+n+2}} \|u\|_{L^1(C(x,t;r))}$$

for all cylinders $C(x, t; r/2) \subseteq C(x, t; r) \subseteq \Omega_T$ and all solutions u of the heat equation in Ω_T . Here, $C(x, t, r)$ denotes the cylinder of radius r and height r^2 with top centre (x, t) .

Recall the problem discussed in Theorem 2.32;

$$\begin{cases} u_t - \Delta u = f, & \text{in } \Omega_T, \\ u = g, & \text{on } \Gamma_T. \end{cases}$$

where Ω is an open and bounded set. We were able to prove uniqueness with the help of the Maximum principle. Integration by parts allows one to construct a different argument to prove uniqueness. Note that by Theorem 2.35 we can see that the assumption of u being continuous is not necessary.

An alternate proof to Theorem 2.32. Suppose u and $\tilde{u}$ both solve the above problem. Then $w = u - \tilde{u}$ solves

$$\begin{cases} w_t - \Delta w = 0 & \text{in } \Omega_T \\ w = 0 & \text{on } \Gamma_T \end{cases}$$

We now define the “energy” at time $0 \leq t \leq T$ to be

$$E(t) = \int_{\Omega} w^2(x, t) \, dx \geq 0$$

Since $w_t = \Delta w$ we see that

$$E'(t) = 2 \int_{\Omega} w \cdot w_t \, dx = 2 \int_{\Omega} w \Delta w \, dx = -2 \int_{\Omega} |Dw|^2 \, dx \leq 0$$

where the last equality follows from integrating by parts. The above shows that $E'(t)$ is always non-positive. In particular, for all $0 \leq t \leq T$ we see that

$$0 \leq E(t) \leq E(0) = 0$$

It then follows from our formula for $E(t)$ that $w(x, t) \equiv 0$ on Ω_T . That is, we have shown that $u \equiv \tilde{u}$ which concludes the proof. $\square$

Remark 2.9. Note that the proof relies upon the equality $E(0) = 0$. This follows from the fact that

$$w(x, 0) = u(x, 0) - \tilde{u}(x, 0) = 0 \quad \forall x \in \Omega$$

That is,

$$u(x, 0) = \tilde{u}(x, 0) \quad \forall x \in \Omega$$

We actually do not need w to be identically zero on all of Γ_T , but only at the time $t = 0$. Thus, we can view this result as a sort of “forward uniqueness”. That is, if u and $\tilde{u}$ both solve the heat equation and they “start off” having the same values, then $u \equiv \tilde{u}$. This observation leads us to the question of *backward uniqueness* which we present below.

Suppose that u and $\tilde{u}$ solve system

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \Omega_T \\ u = g & \text{on } \partial\Omega \times [0, T] \end{cases} \quad (2.38)$$

In this case, we cannot *yet* guarantee that u and $\tilde{u}$ are the same function. But what if we furthermore knew that

$$u(x, T) = \tilde{u}(x, T) \quad \forall x \in \Omega$$

Then $u \equiv \tilde{u}$ within all of Ω_T . This result is what we call *backward uniqueness*.

Theorem 2.37 (Backwards Uniqueness). *Suppose $u, \tilde{u} \in C^2(\Omega_T)$ both solve equation (2.38). If*

$$u(x, T) = \tilde{u}(x, T) \quad \forall x \in \Omega$$

then

$$u \equiv \tilde{u} \quad \text{within } \Omega_T$$

Proof. As in the alternate proof to Theorem 2.32 presented above, we write $w = u - \tilde{u}$ and define the “energy function” to be

$$E(t) = \int_{\Omega} w^2(x, t) \, dx \geq 0$$

If $E(t) \equiv 0$ then we are done. Thus, we will assume that $E(t)$ is not equivalently zero and show that this leads to a contradiction. Once again as in the alternate proof to Theorem 2.32, we see that

$$E'(t) = -2 \int_{\Omega} |Dw|^2 \, dx$$

We now use integration by parts to compute the second derivative;

$$E''(t) = -4 \int_{\Omega} Dw \cdot Dw_t \, dx = 4 \int_{\Omega} \Delta w w_t \, dx = 4 \int_{\Omega} (\Delta w)^2 \, dx$$

The last equality holds since w satisfies $w_t - \Delta w = 0$.

On the other hand, integrating by parts we also see that

$$E'(t) = -2 \int_{\Omega} |Dw|^2 dx = 2 \int_{\Omega} w \Delta w dx$$

Note that the boundary term vanishes since $w = 0$ on $\partial\Omega$. Combining the above result with the Cauchy-Schwartz inequality yields

$$(E'(t))^2 = 4 \left(\int_{\Omega} w \Delta w dx \right)^2 \leq \left(4 \int_{\Omega} (\Delta w)^2 dx \right) \left(\int_{\Omega} w^2 dx \right) = E''(t)E(t) \quad (2.39)$$

Recall our assumption that $E(t) \not\equiv 0$. Thus, there exists $0 \leq t_1 < t_2 \leq T$ such that $E(t)$ is strictly positive on $[t_1, t_2)$ and $E(t_2) = 0$. Note that we may suppose $E(t_2) = 0$ since $E(T) = 0$.

Consider now the function

$$f(t) = \log E(t) \quad t_1 \leq t < t_2$$

It follows from equation (2.39) that $f(t)$ is convex. Indeed,

$$f''(t) = \frac{E''(t)}{E(t)} - \frac{(E'(t))^2}{(E(t))^2} \geq \frac{E''(t)}{E(t)} - \frac{E''(t)E(t)}{(E(t))^2} = 0$$

So $f(t)$ is convex which means that for any $0 < \tau < 1$ and $t_1 < t < t_2$, we have

$$f((1-\tau)t_1 + \tau t) \leq (1-\tau)f(t_1) + \tau f(t)$$

Exponentiation either side yields

$$E((1-\tau)t_1 + \tau t) \leq E(t_1)^{1-\tau} E(t)^\tau$$

Thus,

$$E((1-\tau)t_1 + \tau t_2) \leq E(t_1)^{1-\tau} E(t_2)^\tau = 0$$

Since the above holds for any $0 < \tau < 1$, we conclude that

$$E(t) = 0 \quad \forall t_1 < t < t_2$$

which is a contradiction. □

2.3 Wave Equation

Chapter 3

Sobolev Spaces

We devote this chapter to the construction of Sobolev Spaces, and consequently the theory they induce. These spaces are in fact the *correct* setting for the analysis of partial differential equations. Despite the rather abstract nature of such spaces, the applications are far and wide. Even so, the reader with a significant background in real and functional analysis should not find the notion of a Sobolev space unmotivated.

In the long run, our goal is to “transform” a given partial differential equation into a functional analytic problem. That is, we will choose to view PDE as operators $X \rightarrow Y$, for the appropriate Banach spaces X and Y over $\mathbb{R}$. Surprisingly, it will be *very* convenient to take X and Y to be Sobolev spaces.

3.1 Constructing Sobolev Spaces

Much of the theory involving Sobolev spaces relies upon the notion of an L^p -space. For this reason, we assume the reader is familiar with abstract measure theory and functional analysis. Before diving into this construction, we must first introduce a weaker notion of differentiation. This is not too surprising since we are studying PDE and would like to search for a solution in as broad a space as possible (in order to apply results from functional analysis).

Throughout this section, or unless stated otherwise, we will denote by Ω a non-empty open subset of $\mathbb{R}^n$. We also assume that all functions are **real valued**. Note that real valued does *not* mean “extended real valued”. For the entirety of the chapter, functions are **not** allowed to take infinite values! From a physical point of view, such a concept is ill defined and it makes sense (intuitively) to omit

these functions. Mathematically speaking, it is only possible to study functions that are finite *almost everywhere*. At this point, we might as well redefine the function so that it is never infinity.

Given Ω , we define $C_c^\infty(\Omega)$ to be the real vector space of smooth functions $u : \Omega \rightarrow \mathbb{R}$ having compact support in Ω . We will sometimes choose to denote this by $\mathcal{D}(\Omega)$. The elements of $\mathcal{D}(\Omega)$ are frequently dubbed *test functions*.

3.1.1 Weak Derivatives

We would first like to motivate the abstract definition to come. For simplicity, suppose that $\partial\Omega$ is at least C^1 . Let $u \in C^1(\overline{\Omega})$ and fix a function $\phi \in \mathcal{D}(\Omega)$. Since $\text{supp}(\phi) \subsetneq \Omega$, integration by parts shows us that

$$\int_{\Omega} u(x) \phi_{x_i}(x) \, dx = - \int_{\Omega} u_{x_i}(x) \phi(x) \, dx, \quad \text{for } i = 1, \dots, n.$$

Now suppose that $u \in C^k(\overline{\Omega})$; if α is a multiindex then

$$\int_{\Omega} u(x) \partial^\alpha \phi(x) \, dx = (-1)^{|\alpha|} \int_{\Omega} \phi(x) \partial^\alpha u(x) \, dx.$$

This motivates the subsequent definition of a *weak derivative*. Henceforth, we relax the assumption that $\partial\Omega$ is of class C^1 .

Definition 19 (Weak Derivative). Let $u, v \in L_{\text{loc}}^1(\Omega)$ and suppose that α is a multiindex. We say that v is the α^{th} weak derivative of the function u provided

$$\int_{\Omega} u(x) \partial^\alpha \phi(x) \, dx = (-1)^{|\alpha|} \int_{\Omega} v(x) \phi(x) \, dx$$

for *all* test functions $\phi \in \mathcal{D}(\Omega)$. If this is the case, we will choose to write $\partial^\alpha u = v$.

The use of the symbol '=' is not yet justified. Indeed, we must check that a weak derivative is unique *whenever* it exists. However, since we are working in L^p -spaces, it suffice to show that weak derivatives are unique up to measure zero.

Proposition 3.1 (Uniqueness of Weak Derivative). *Let $u, v \in L_{\text{loc}}^1(\Omega)$ and suppose that v is the α^{th} weak derivative of u . If $w \in L_{\text{loc}}^1(\Omega)$ is any α^{th} weak derivative of u , then $v \equiv w$ almost everywhere on Ω .*

Proof. Fix a function $\phi \in \mathcal{D}(\Omega)$. Since v and w are weak derivatives of u , it follows by definition that

$$\int_{\Omega} v(x)\phi(x) \, dx = (-1)^{|\alpha|} \int_{\Omega} u(x)\partial^{\alpha}\phi(x) \, dx = \int_{\Omega} w(x)\phi(x) \, dx.$$

Thus, for each $\phi \in \mathcal{D}(\Omega)$,

$$\int_{\Omega} (v(x) - w(x)) \phi(x) \, dx = 0.$$

It follows that $v \equiv w$ almost everywhere on Ω . □

Let us now provide an easy example to solidify these ideas.

Example. Take $n = 1$ and $\Omega = (0, 2)$. Consider the functions

$$u(x) = \begin{cases} x, & \text{if } 0 < x \leq 1, \\ 1, & \text{if } 1 \leq x \leq 2 \end{cases} \quad \text{and} \quad v(x) = \begin{cases} 1, & \text{if } 0 < x \leq 1, \\ 0, & \text{if } 1 \leq x \leq 2. \end{cases}$$

Notice that both these functions belong to $L^1(\Omega)$. We claim that $u' = v$. To see that this is so, fix a test function $\phi \in \mathcal{D}(\Omega)$ and write

$$\int_{\Omega} u(x)\phi'(x) \, dx = \int_0^1 x\phi'(x) \, dx + \int_1^2 \phi'(x) \, dx.$$

Integration by parts then gives

$$\int_{\Omega} u(x)\phi'(x) \, dx = \phi(1) - \int_0^1 \phi(x) \, dx + \phi(2) - \phi(1).$$

Since ϕ has compact support in $(0, 2)$, we must have $\phi(2) = 0$ so that

$$\int_{\Omega} u(x)\phi'(x) \, dx = - \int_0^1 \phi(x) \, dx = - \int_0^2 \phi(x)v(x) \, dx.$$

3.1.2 The Sobolev Spaces

In this subsection, we provide the definition of a Sobolev space. To this end, we first fix $1 \leq p \leq \infty$ and $k \in \mathbb{N}_0$.

Definition 20. The Sobolev space $W^{k,p}(\Omega)$ is defined to be the set of all *real valued*¹ functions $u \in L^1_{\text{loc}}(\Omega)$ (considered up to equality almost everywhere) such that $\partial^\alpha u$ exists, and belongs to $L^p(\Omega)$, for all multiindices α with $|\alpha| \leq k$.

In the special case $p = 2$, we will often write $H^k(\Omega)$ instead of $W^{k,2}(\Omega)$. This is because, as we shall soon see, $H^k(\Omega)$ is a Hilbert space. As a subset of $L^1_{\text{loc}}(\cdot)$, notice that we always identify functions in $W^{k,p}(\Omega)$ up to sets of measure zero.

In order to conduct our analysis, we must obviously give $W^{k,p}(\Omega)$ additional structure. Intending to topologize the Sobolev spaces, we endow $W^{k,p}(\Omega)$ with the following norm:

Definition 21. The Sobolev norm of a function $u \in W^{k,p}(\Omega)$ is defined by the equation

$$\|u\|_{W^{k,p}(\Omega)} := \begin{cases} \left(\sum_{|\alpha| \leq k} \int_{\Omega} |\partial^\alpha u|^p \, dm \right)^{1/p}, & 1 \leq p < \infty, \\ \sum_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^\infty(\Omega)}, & p = \infty. \end{cases} \quad (3.1)$$

When no confusion can arise, we will often choose to instead denote this quantity by $\|u\|_{k,p}$.

The above certainly gives $W^{k,p}(\Omega)$ the structure of a normed space, provided it is indeed a norm on $W^{k,p}(\Omega)$. An obvious next step is to verify this fact. Unfortunately, this is easier said than done. Indeed, we have not even shown that $W^{k,p}(\Omega)$ is a vector space. Thus, we need to make some observations regarding these so-called weak derivatives. These are summarized within the proposition below.

Proposition 3.2. *Let $u, v \in W^{k,p}(\Omega)$ and suppose α is a multiindex with $|\alpha| \leq k$. Then, there hold the following:*

1. $W^{k,p}(\Omega)$ is closed under finite linear combinations over $\mathbb{R}$. Moreover,

$$\partial^\alpha (\lambda u + \mu v) = \lambda \partial^\alpha u + \mu \partial^\alpha v.$$

Hence, $W^{k,p}(\Omega)$ is a real vector space.

¹Again, a real valued function u cannot take infinite values anywhere.

2. For all open subsets U of Ω , we have $W^{k,p}(\Omega) \subseteq W^{k,p}(U)$.

3. If $|\alpha| + |\beta| \leq k$ then $\partial^\alpha u \in W^{k-|\alpha|,p}(\Omega)$ and

$$\partial^\beta (\partial^\alpha u) = \partial^\alpha (\partial^\beta u) = \partial^{\alpha+\beta} u.$$

Proof. First, observe that $W^{k,p}(\Omega)$ is a subset of $L^p(\Omega)$. Let now $\phi \in \mathcal{D}(\Omega)$ be given; we get

$$\begin{aligned} \int_{\Omega} [\lambda u(x) + \mu v(x)] \partial^\alpha \phi(x) \, dx &= \lambda \int_{\Omega} u(x) \partial^\alpha \phi(x) \, dx \\ &\quad + \mu \int_{\Omega} v(x) \partial^\alpha \phi(x) \, dx. \end{aligned}$$

Thus,

$$\begin{aligned} \int_{\Omega} [\lambda u(x) + \mu v(x)] \partial^\alpha \phi(x) \, dx &= (-1)^{|\alpha|} \int_{\Omega} \phi(x) \partial^\alpha u(x) \, dx \\ &\quad + (-1)^{|\alpha|} \int_{\Omega} \phi(x) \partial^\alpha v(x) \, dx. \end{aligned}$$

More compactly,

$$\int_{\Omega} [\lambda u(x) + \mu v(x)] \partial^\alpha \phi(x) \, dx = (-1)^{|\alpha|} \int_{\Omega} (\lambda \partial^\alpha u(x) + \mu \partial^\alpha v(x)) \phi(x) \, dx.$$

This yields the first point. The second is even easier to see. If $U \subseteq \Omega$ is open, then automatically $L^1_{\text{loc}}(\Omega) \subseteq L^1_{\text{loc}}(U)$. Thus, we have the desired inclusion by the simple fact that $\mathcal{D}(U) \subseteq \mathcal{D}(\Omega)$.

We must only demonstrate the final point. For this, the only non-trivial claim is the stated identity. By symmetry, we only need to show that the first equality holds. Fix a test function $\phi \in \mathcal{D}(\Omega)$ and note that $\partial^\alpha \phi$ is again a test function.

Therefore,

$$\begin{aligned}
 \int_{\Omega} \partial^{\beta} u(x) \partial^{\alpha} \phi(x) \, dx &= (-1)^{|\beta|} \int_{\Omega} u(x) \partial^{\beta} \partial^{\alpha} \phi(x) \, dx \\
 &= (-1)^{|\beta|} \int_{\Omega} u(x) \partial^{\beta+\alpha} \phi(x) \, dx \\
 &= (-1)^{|\beta|} \int_{\Omega} u(x) \partial^{\alpha+\beta} \phi(x) \, dx \\
 &= (-1)^{|\beta|} \cdot (-1)^{|\alpha+\beta|} \int_{\Omega} \partial^{\alpha+\beta} u(x) \phi(x) \, dx \\
 &= (-1)^{|\alpha|} \int_{\Omega} \partial^{\alpha+\beta} u(x) \phi(x) \, dx.
 \end{aligned}$$

Since ϕ was arbitrary, this concludes the proof. $\square$

Having established the linear structure of $W^{k,p}(\Omega)$, let us show that the function we defined earlier is indeed a norm. First, it is obvious that $\|u\|_{k,p} = 0$ if and only if u vanishes almost everywhere. In the case $p = \infty$, the fact that $\|\cdot\|_{k,p}$ is a norm follows directly from the previous proposition and the recollection that $\|\cdot\|_{\infty}$ is itself a norm.

Let now $p < \infty$. Clearly, for every $\lambda \in \mathbb{R}$ the previous result gives

$$\|\lambda u\|_{k,p} = |\lambda| \cdot \|u\|_{k,p}.$$

Assume that $u, v \in W^{k,p}(\Omega)$. We must show that the triangle inequality holds. This is done using the Minkowski inequality, which gives

$$\begin{aligned}
 \|u + v\|_{k,p} &= \left[\sum_{|\alpha| \leq k} \int_{\Omega} |\partial^{\alpha} (u + v)|^p \, dm \right]^{1/p} \\
 &= \left[\sum_{|\alpha| \leq k} \int_{\Omega} |\partial^{\alpha} u + \partial^{\alpha} v|^p \, dm \right]^{1/p} \\
 &= \left[\sum_{|\alpha| \leq k} \|\partial^{\alpha} u + \partial^{\alpha} v\|_{L^p(\Omega)}^p \right]^{1/p} \\
 &\leq \left[\sum_{|\alpha| \leq k} \left(\|\partial^{\alpha} u\|_{L^p(\Omega)} + \|\partial^{\alpha} v\|_{L^p(\Omega)} \right)^p \right]^{1/p}
 \end{aligned}$$

$$\leq \left[\sum_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^p(\Omega)}^p \right]^{1/p} + \left[\sum_{|\alpha| \leq k} \|\partial^\alpha v\|_{L^p(\Omega)}^p \right]^{1/p}.$$

In this last step, we applied the Minkowski inequality with the counting measure. Therefore, the triangle inequality holds true. We conclude that $W^{k,p}(\Omega)$ is a normed vector space over $\mathbb{R}$ when equipped with this norm. With these formalities out of the way, let us show that $W^{k,p}(\Omega)$ is in fact a *Banach space*.

Theorem 3.3. *For all $k \in \mathbb{N}$ and each $1 \leq p \leq \infty$, the Sobolev space $W^{k,p}(\Omega)$ is a real Banach space.*

Proof. We will “piggy back” off the completeness of L^p -spaces. Let $(u_m)_{m=1}^\infty$ be a Cauchy sequence in $W^{k,p}(\Omega)$. That is, a sequence such that differences

$$\|u_m - u_l\|_{k,p}$$

can be made arbitrarily small as $m, l \rightarrow \infty$. From the linearity of weak differentiation, we see that the corresponding sequences $(\partial^\alpha u_m)_{m=1}^\infty$ are Cauchy in $L^p(\Omega)$ for all multiindices $|\alpha| \leq k$. Since $L^p(\Omega)$ is a Banach space, we may choose functions $u_\alpha \in L^p(\Omega)$ such that

$$\lim_{m \rightarrow \infty} \partial^\alpha u_m = u_\alpha \quad \text{in } L^p(\Omega),$$

for each $|\alpha| \leq k$. In particular, there exists a function $u \in L^p(\Omega)$ such that

$$u = \lim_{m \rightarrow \infty} u_m \quad \text{in } L^p(\Omega).$$

We now argue that $u \in W^{k,p}(\Omega)$ and that $\partial^\alpha u = u_\alpha$ for each $|\alpha| \leq k$. To this end, let us fix a test function $\phi \in \mathcal{D}(\Omega)$. Suppose α is a multiindex with $|\alpha| \leq k$. For each $m \in \mathbb{N}$ we have

$$\begin{aligned} \left| \int_\Omega u \partial^\alpha \phi \, dm - \int_\Omega u_m \partial^\alpha \phi \, dm \right| &\leq \int_\Omega |u - u_m| |\partial^\alpha \phi| \, dm \\ &\leq \|u - u_m\|_{L^p(\Omega)} \|\partial^\alpha \phi\|_{L^q(\Omega)}. \end{aligned}$$

Hence, after using a similar argument for the sequence $(\partial^\alpha u_m)$,

$$\begin{aligned} \int_\Omega u \partial^\alpha \phi \, dm &= \lim_{m \rightarrow \infty} \int_\Omega u_m \partial^\alpha \phi \, dm = (-1)^{|\alpha|} \lim_{m \rightarrow \infty} \int_\Omega \partial^\alpha u_m \phi(x) \, dm \\ &= (-1)^{|\alpha|} \int_\Omega u_\alpha \phi(x) \, dm. \end{aligned}$$

Finally, $u \in L^1_{\text{loc}}(\Omega)$. Indeed, if $\Lambda \subset \Omega$ is compact then

$$\int_{\Lambda} |u| \, dm \leq \|u\|_{L^p(\Lambda)} m(\Lambda)^{1/q} < \infty.$$

Since $\partial^\alpha u_m$ converges to $\partial^\alpha u$ in $L^p(\Omega)$ for all multiindices α with $|\alpha| \leq k$, we conclude that u_m converges to u in $W^{k,p}(\Omega)$. $\square$

3.1.3 The Hilbert Space $H^k(\Omega)$

As mentioned before, the case $p = 2$ is of particular interest to us. Let $k \in \mathbb{N}_0$ and denote by $H^k(\Omega)$ the Sobolev space $W^{k,2}(\Omega)$. Let $\|\cdot\|_{H^k(\Omega)}$ denote the Sobolev norm on $H^k(\Omega)$. We have already shown that $H^k(\Omega)$ is a Banach space when equipped with this topological structure. We now show that $H^k(\Omega)$ is in fact a *Hilbert space*.

Let $u, v \in H^k(\Omega)$ be given; we define their inner product to be

$$\langle u, v \rangle := \sum_{|\alpha| \leq k} \int_{\Omega} \partial^\alpha u \partial^\alpha v \, dm.$$

By definition of this “inner product”, we have

$$\|u\|_{H^k(\Omega)} = \sqrt{\langle u, u \rangle}.$$

Thus, if we can show that $\langle \cdot, \cdot \rangle$ is an inner product on $H^k(\Omega)$, it will follow that $H^k(\Omega)$ is a real Hilbert space. First, it is clear from the definition that $\langle \cdot, \cdot \rangle$ is bilinear. In just the same way, it is immediate that $\langle u, u \rangle \geq 0$ for all $u \in W^{k,p}(\Omega)$ —with equality if and only if $u = 0$ almost everywhere.

We conclude the following.

Corollary 3.4. *For all $k \in \mathbb{N}$, the Sobolev space $H^k(\Omega) = W^{k,2}(\Omega)$ is a real Hilbert space. In the special case $k = 0$, we may identify $H^0(\Omega)$ with $L^2(\Omega)$.*

3.2 Approximation in $W^{k,p}(\Omega)$ by Smooth Functions

Much as with the theory of L^p -spaces, it is awkward to work directly with functions in $W^{k,p}(\Omega)$. To rectify this issue, it is convenient to work with density

results. Given the *differential* nature of the problems we hope to tackle, the obvious candidate for approximation are the smooth functions in Ω : $C^\infty(\Omega)$. Unfortunately, the process is somewhat arduous.

Unless otherwise stated, assume throughout this section that Ω is any non-empty open subset of $\mathbb{R}^n$. In some results, we will require it to be bounded.

3.2.1 Local Approximation

We first introduce some notation. Let $(u_m)_{m=1}^\infty$ be a sequence in $W^{k,p}(\Omega)$. We say that the sequence converges *locally* to a function $u \in W^{k,p}(\Omega)$, as $m \rightarrow \infty$, if for all open sets $U \Subset \Omega$ there holds

$$\lim_{m \rightarrow \infty} \|u_m - u\|_{W^{k,p}(U)} = 0.$$

In this case, we may write

$$u_m \xrightarrow[\text{loc}]{m \rightarrow \infty} u \quad \text{or} \quad u_m \rightarrow u \quad \text{in } W_{\text{loc}}^{k,p}(\Omega).$$

Evidently, $C_c^\infty(\Omega)$ forms a subset of $W^{k,p}(\Omega)$ for all $k \in \mathbb{N}_0$ and all $1 \leq p \leq \infty$. By linearity of differentiation, $C_c^\infty(\Omega)$ can be viewed as a vector subspace of the Banach space $W^{k,p}(\Omega)$. We then define

$$W_0^{k,p}(\Omega) \tag{3.2}$$

to be the *topological closure* of $C_c^\infty(\Omega)$ in $W^{k,p}(\Omega)$.

Mollifiers. Before proving some approximation theorems, we will need basic results about a class of functions allowing us to “smooth out” functions in Sobolev spaces. Such functions will be called *mollifiers*. This short discussion will only state the relevant results; a more accurate description of these functions can be found in Appendix C of [EVNS].

Let $\Omega \subseteq \mathbb{R}^n$ be a non-empty open set and fix $\varepsilon > 0$. As in [EVNS], we will use the notation

$$\Omega_\varepsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \varepsilon\}. \tag{3.3}$$

Notice that Ω_ε may be empty, as would be the case with $\Omega = (0, 1)$ and $\varepsilon = 2$. Let us also define a radially symmetric function $\eta \in C^\infty(\mathbb{R}^n)$ via the equation

$$\eta(x) := \begin{cases} C \exp\left(\frac{1}{|x| - 1}\right), & \text{if } |x| < 1, \\ 0, & \text{if } |x| \geq 1; \end{cases} \tag{3.4}$$

where C is chosen so that

$$\int_{\mathbb{R}^n} \eta(x) \, dx = 1.$$

This function $\eta \in C_c^\infty(\mathbb{R}^n)$ is often called the *standard mollifier*. For given $\varepsilon > 0$, we can construct a function $\eta_\varepsilon \in C_c^\infty(\Omega)$ by putting

$$\eta_\varepsilon(x) := \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right), \quad x \in \mathbb{R}^n. \quad (3.5)$$

It is not difficult to show that, for each $\varepsilon > 0$, one has both $\int_{\mathbb{R}^n} \eta_\varepsilon \, dm = 1$ and $\text{supp}(\eta_\varepsilon) \subset B(0, \varepsilon)$. These η_ε are also called mollifiers. Now, let $f \in L_{loc}^1(\Omega)$ be given. We define its mollification to be the function

$$f^\varepsilon(x) := (f * \eta_\varepsilon).$$

Since η_ε has compact support, the local integrability of f implies that f^ε is of class $C^\infty(\Omega_\varepsilon)$; this fact follows from Proposition 8.10 in [FLND].

We now summarize some properties of this mollification f^ε . The proof of the following results may be found in [EVNS, §C.5].

Proposition 3.5 (Mollification Properties). *Let $f \in L_{loc}^1(\Omega)$ and fix $\varepsilon > 0$.*

1. *The mollification f^ε is smooth on Ω_ε .*
2. *As $\varepsilon \searrow 0$, the family of functions $(f^\varepsilon)_{\varepsilon>0}$ converges to f pointwise-almost everywhere on $\mathbb{R}^n$.*
3. *If f is continuous on Ω , then f^ε converges uniformly to f on each compact subset of Ω .*
4. *Let $p \in [1, \infty)$ and $f \in L_{loc}^p(\Omega)$. Then, $f^\varepsilon \rightarrow f$ in $L_{loc}^p(\Omega)$.*

With this out of the way, we prove our first approximation theorem regarding smooth functions in $W^{k,p}(\Omega)$.

Theorem 3.6 (Local Approximation Theorem). *Let $\Omega \subseteq \mathbb{R}^n$ be a non-empty open set and suppose $u \in W^{k,p}(\Omega)$ for $p < \infty$. For each $\varepsilon > 0$ put*

$$u^\varepsilon(x) := \eta_\varepsilon * u, \quad \text{in } \Omega_\varepsilon.$$

Recall that each $u^\varepsilon \in C^\infty(\Omega_\varepsilon)$. Then, u^ε converges to u in $W_{loc}^{k,p}(\Omega)$ as $\varepsilon \searrow 0$.

Remark 3.1. Let $V \subset \Omega$ be compact, so that $V \neq \Omega$. We claim that there exists $\varepsilon > 0$ such that $V \subseteq \Omega_\varepsilon$. If this is indeed the case, then for all $0 < \delta \leq \varepsilon$ there would consequently hold

$$V \subseteq \Omega_\varepsilon \subseteq \Omega_\delta.$$

To see that this does hold, we argue by contradiction. Otherwise, for each $n \in \mathbb{N}$, there exists $x_n \in V$ such that $\text{dist}(x_n, \partial\Omega) \leq 1/n$. That is,

$$\inf_{y \in \partial\Omega} |x_n - y| \leq \frac{1}{n}.$$

Subsequently, one can choose $y_n \in \partial\Omega$ such that

$$\frac{1}{n} \leq |x_n - y_n| \leq \frac{2}{n}.$$

Since V is compact, after passing to a subsequence, we might as well assume that x_n converges to some point $x \in V$. By letting $n \rightarrow \infty$ in the above, we see that $x = \lim y_n$ as well. Using that $\partial\Omega$ is topologically closed, we obtain $x \in \partial\Omega$. Thus, $x \in \Omega \cap \partial\Omega$. This obviously contradicts the fact that Ω is open.

In light of the above, we see that the statement of our theorem “makes sense”. Indeed, each u^ε will belong to $C^\infty(\Omega_\varepsilon)$ thereby indicating that each partial derivative of u^ε will belong to $L^p(V)$, by compactness of V . Hence, each u^ε is in $W_{\text{loc}}^{k,p}(\Omega)$.

Proof of Theorem. We assume that $\text{supp}(u) \subseteq \Omega$. We also take $\varepsilon > 0$ to be sufficiently small in each step. By the previous proposition, each u^ε is smooth on Ω_ε . We now claim that for each multiindex $|\alpha| \leq k$,

$$D^\alpha u^\varepsilon \equiv \partial^\alpha u * \eta_\varepsilon, \quad \text{in } \Omega_\varepsilon.$$

Here, $D^\alpha u^\varepsilon$ denotes the ordinary α^{th} derivative of the smooth map u^ε while $\partial^\alpha u$ is the α^{th} weak derivative of u . Certainly, fix $x \in \Omega_\varepsilon$ and write

$$D^\alpha u^\varepsilon(x) = \int_{\Omega} D_x^\alpha \eta_\varepsilon(x - y) u(y) \, dy = (-1)^{|\alpha|} \int_{\Omega} D_y^\alpha \eta_\varepsilon(x - y) u(y) \, dy.$$

Now, since x is fixed, $\eta_\varepsilon(x - y)$ is a smooth function with compact support in Ω and, thus, we have

$$D^\alpha u^\varepsilon(x) = (-1)^{|\alpha|+|\alpha|} \int_{\Omega} \eta_\varepsilon(x - y) \partial^\alpha u(y) \, dy.$$

This proves our claim. Now, to prove our theorem. Let $V \Subset U$ be open; the first part of our proof shows that

$$(\partial^\alpha u^\varepsilon) = (\partial^\alpha u)^\varepsilon \quad \text{in } \Omega_\varepsilon.$$

Thus, by our remark and the previous proposition, we have

$$\lim_{\varepsilon \searrow 0} \|\partial^\alpha u^\varepsilon - \partial^\alpha u\|_{L^p(V)} = 0$$

for each multiindex $|\alpha| \leq k$. Hence, as $\varepsilon \searrow 0$,

$$u^\varepsilon \rightarrow u \quad \text{in } W_{\text{loc}}^{k,p}(\Omega).$$

This completes the proof. $\square$

3.2.2 Global Approximations

In the previous subsection, we established local approximation results allowing us to view functions in $W^{k,p}(\Omega)$ (locally) as limits of smooth functions. Obviously, this is not sufficient for our purposes and we now strengthen this result. Ultimately, we would like to globally approximate all functions in $W^{k,p}(\Omega)$ by smooth functions defined over the entirety of Ω . This is accomplished via the following theorem.

Theorem 3.7 (First Sobolev Approximation Theorem). *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and fix $p \in [1, \infty)$. For each $u \in W^{k,p}(\Omega)$, there exists a sequence $(u_m)_{m=1}^\infty$ in $W^{k,p}(\Omega) \cap C^\infty(\Omega)$ converging to u , as $m \rightarrow \infty$, in $W^{k,p}(\Omega)$.*

We would like to warn the reader that these functions u_m need not be smooth up to the boundary. To guarantee this, we will need to make smoothness assumptions regarding the boundary of the set Ω .

Before proving Theorem 3.7, we require a lemma that is honestly very overdue. For the lemma, we assume that $\Omega \subseteq \mathbb{R}^n$ is any non-empty open set.

Lemma 3.8. *Let $k \in \mathbb{N}_0$ and $1 \leq p \leq \infty$. Suppose that $u \in W^{k,p}(\Omega)$ and let $\zeta \in C_c^\infty(\Omega)$. Then, $\zeta u \in W^{k,p}(\Omega)$.*

The proof is messy and far from instructive, as it mostly involves algebraic manipulations and induction. The curious reader may find the mechanical proof in [EVNS]. Assuming the generalized Leibniz' formula for smooth functions of compact support, it is relatively easy to construct a proof, however.

Lemma 3.9 (Exhaustion Lemma). *Each open set in $\mathbb{R}^n$ admits an exhaustion.*

Proof. Let $\Omega \subseteq \mathbb{R}^n$ be open; if $\Omega = \emptyset, \mathbb{R}^n$, the claim is trivial. Hence, assume that $\Omega \neq \mathbb{R}^n$ is non-empty. For each $k \in \mathbb{N}$ define the set

$$\Lambda_k := \left\{ x \in \Omega : \text{dist}(x, \partial\Omega) \geq \frac{1}{k} \right\} \cap \overline{B(0, k)}.$$

It is not difficult to see that each Λ_k is closed, and hence compact in $\mathbb{R}^n$. Moreover, it is clear that $\Lambda_k \subseteq \Lambda_{k+1}$ for each index $k \in \mathbb{N}$. All that remains is to show that $\bigcup_{k=1}^{\infty} \Lambda_k = \Omega$. Let $x \in \Omega$ be fixed. Since $\partial\Omega$ is closed and $\Omega \cap \partial\Omega = \emptyset$,

$$\text{dist}(x, \partial\Omega) > 0.$$

Thus, x belongs to some V_k . □

We now begin the proof of our main theorem.

Proof of Theorem 3.7. We will break the proof down into several steps. Although short, piecing these together can seem confusing if they are not distinguished.

1. For each $i \in \mathbb{N}$ set

$$\Omega_i := \left\{ x \in \Omega : \text{dist}(x, \partial\Omega) > \frac{1}{i} \right\}$$

and subsequently put $V_i := \Omega_{i+3} \setminus \overline{\Omega_{i+1}}$. It is not too difficult to show that each Ω_i is open.² Hence, each V_i must also be open. Since Ω is bounded, there exists an open set $V_0 \Subset \Omega$ having the property that $\bigcup_{i=0}^{\infty} V_i = \Omega$.³

2. Choose now a family⁴ $\{\zeta_i\}_{i=0}^{\infty}$ of functions in $C_c^{\infty}(V_i)$ such that

$$\begin{cases} 0 \leq \zeta_i \leq 1, & i \in \mathbb{N}_0, \\ \sum_{j=0}^{\infty} \zeta_j \equiv 1, & \text{on } \Omega. \end{cases}$$

Let now $u \in W^{k,p}(\Omega)$ be given. By our lemma from [EVNS], each $\zeta_i u$ belongs to the Sobolev space $W^{k,p}(\Omega)$. Also, $\text{supp}(\zeta_i u) \subset V_i$ for each index $i = 0, 1, \dots$

² Let $x \in \Omega_i$ so that $\inf_{y \in \partial\Omega} |x - y| = \gamma > 1/i$. Let $\rho > 0$ be such that $B(x, \rho) \subseteq \Omega$. Without loss of generality, we may choose ρ so small that $\gamma - \rho > 1/i$. Then, for all $z \in B(x, \rho)$ and $y \in \partial\Omega$ one has

$$|y - z| \geq |y - x| - |z - x| > \gamma - \rho.$$

Thus, $\text{dist}(z, \partial\Omega) \geq \gamma - \rho > 1/i$. It follows that $B(x, \rho) \subseteq \Omega_i$. This proves the claim.

³ One could, for instance, take V_0 to be $\Omega \setminus \bigcup_{i=1}^{\infty} V_i$.

⁴ Such a family is called a smooth partition of unity subordinate to the family $\{V_i\}_{i=0}^{\infty}$.

3. Fix an index $i \geq 1$ and define for sufficiently small $\varepsilon_i > 0$ the quantity

$$u^i := (\zeta_i u) * \eta_{\varepsilon_i} \quad \text{in } \mathbb{R}^n.$$

so that, if $\varepsilon_i > 0$ is chosen sufficiently small,

$$\begin{aligned} \text{supp}(u^i) &\subseteq \text{supp}(\eta_{\varepsilon_i}) + \text{supp}(\zeta_i u) \subseteq B(0, \varepsilon_i) + \text{supp}(\zeta_i) \\ &\subseteq B(0, \varepsilon_i) + V_i \\ &\subseteq \Omega_{i+4} \setminus \overline{\Omega_i} \supseteq V_i. \end{aligned} \quad ^5$$

We will choose to denote the set $\Omega_{i+4} \setminus \overline{\Omega_i}$ by W_i , for $i \geq 1$. In the case $i = 0$, one can similarly show that u^i is supported on a compactly contained subset of Ω . Let $\delta > 0$ be given; we can thus use the previous theorem to choose $\varepsilon_i > 0$ so small that

$$\begin{cases} \|u^i - \zeta_i u\|_{W^{k,p}(\Omega)} \leq \frac{\delta}{2^{i+1}}, & i \geq 0, \\ \text{supp}(u^i) \subseteq W_i, & i \geq 1. \end{cases}$$

4. Define the function $v := \sum_{i \geq 0} u^i$. We claim that $v \in C^\infty(\Omega)$, and hence in $W_{\text{loc}}^{k,p}(\Omega)$. To see this, let $O \Subset \Omega$ be open and observe that all but finitely many u^i vanish in O . Indeed, this last point follows from the fact that $O \cap W_i$ will be empty for all i large (since $\overline{O}$ is compact in Ω and hence “far away” from $\partial\Omega$). Recall also that

$$u \equiv \sum_{i=0}^{\infty} \zeta_i u \quad \text{on } \Omega.$$

If $\Lambda \subset \Omega$ is compact, then by continuity of norms

$$\|v - u\|_{W^{k,p}(\Lambda)} \leq \sum_{i=0}^{\infty} \|u^i - \zeta_i u\|_{W^{k,p}(\Omega)} \leq \delta.$$

⁵Let $x \in \text{supp}(u^i)$. Then, $\int_{\mathbb{R}^n} \zeta_i(y) u(y) \eta_{\varepsilon_i}(x-y) dy \neq 0$. Assume by way of contradiction that x cannot be written in the form $y + z$, for any $y \in \text{supp}(\zeta_i u)$ and $z \in \text{supp}(\eta_{\varepsilon_i})$. Observe that

$$\int_{\mathbb{R}^n} \zeta_i(y) u(y) \eta_{\varepsilon_i}(x-y) dy = \int_{\text{supp}(\zeta_i u)} \zeta_i(y) u(y) \eta_{\varepsilon_i}(x-y) dy.$$

Therefore, for all $y \in \text{supp}(\zeta_i u)$, we will have $x - y \notin \text{supp}(\eta_{\varepsilon_i})$ for otherwise $x = (x - y) + y$ would belong to the sum of the supports. This is a clear contradiction.

5. We now show that the inequality derived in the last step extends to the entirety of Ω . Let $\{\Lambda_j\}_{j=1}^\infty$ be an *exhaustion* of the set Ω . For each $\ell \in \mathbb{N}$, we also define Γ_ℓ to be the compact union $\bigcup_{j=1}^\ell \Lambda_j$. By definition, we have $\Gamma_\ell \nearrow \Omega$. Hence, by the monotone convergence theorem,

$$\begin{aligned} \|u - v\|_{W^{k,p}(\Omega)}^p &= \sum_{|\alpha| \leq k} \int_{\Omega} |\partial^\alpha(u - v)|^p \, dm \\ &= \lim_{\ell \rightarrow \infty} \sum_{|\alpha| \leq k} \int_{\Gamma_\ell} |\partial^\alpha(u - v)|^p \, dm \\ &= \lim_{\ell \rightarrow \infty} \|u - v\|_{W^{k,p}(\Gamma_\ell)}^p \leq \delta^p. \end{aligned}$$

We therefore get that $u - v \in W^{k,p}(\Omega)$ —and $v \in W^{k,p}(\Omega)$. This completes the proof. □

As mentioned before, we can say slightly more when the boundary of Ω is “nice enough”. This is precisely what the subsequent theorem achieves.

Theorem 3.10 (First Sobolev Approximation Theorem). *Assume $\Omega \subseteq \mathbb{R}^n$ is open and bounded with C^1 boundary $\partial\Omega$. Let $u \in W^{k,p}(\Omega)$ for $1 \leq p < \infty$. There exists a sequence of functions $(u_m)_{m=1}^\infty$ in $C^\infty(\overline{\Omega}) \cap W^{k,p}(\Omega)$ such that*

$$\lim_{m \rightarrow \infty} \|u_m - u\|_{W^{k,p}(\Omega)} = 0.$$

We shall not prove this result, as it makes use of differential geometry beyond the scope of PDEs. As usual, the proof may be found in [EVNS], where is Theorem 3 in §5.3.3.

3.3 Sobolev Extensions

We now come to the question of whether one can extend a function $u \in W^{1,p}(\Omega)$ so that it becomes a function in $W^{1,p}(\mathbb{R}^n)$. Naturally, any such map u could have very bad discontinuities along at the boundary $\partial\Omega$. Thus, we cannot simply extend u to be zero on $\mathbb{R}^n \setminus U$ as this could affect the “weak differentiability” of u . Nonetheless, when $\partial\Omega$ is sufficiently nice, through some trickery, one can achieve the desired result.

Theorem 3.11 (Sobolev Extension Theorem). *Let $\Omega \Subset \mathbb{R}^n$ be non-empty and open with C^1 boundary $\partial\Omega$. Choose an open set $V \supset \Omega$. For each $1 \leq p \leq \infty$, there exists a bounded linear operator*

$$E : W^{1,p}(\Omega) \longrightarrow W^{1,p}(\mathbb{R}^n)$$

such that, for each $u \in W^{1,p}(\Omega)$, there holds

1. $Eu \equiv u$ almost everywhere in Ω ;
2. $\text{supp}(Eu) \subseteq V$;
3. *there exists a constant $C > 0$, independent of u , such that*

$$\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

We then call the function Eu a Sobolev extension of u to $\mathbb{R}^n$.

Proof of Theorem. We devote the remainder of this section to the proof of the extension theorem. We will try to focus mostly on the ideas behind *building up* the proof, and not so much on the techniques used within the proof. This is because, as with many things in PDE, it is too time consuming to give a detailed explanation of each technique. As before, we follow the proof from §5.4 in [EVNS]. Some of the techniques used are also described within the appendix of this *wonderful* book.

Step 1. Since $\Omega \neq \emptyset, \mathbb{R}^n$, we may choose a point x^0 from $\partial\Omega$. Let us also make the assumption that

$$\text{a neighbourhood of } x^0 \text{ lives within the plane } \{x_n = 0\}. \quad (3.6)$$

Thus, we may choose an open ball $B = B(x^0, r)$ in $\mathbb{R}^n$ such that

$$\begin{cases} B^+ := B \cap \{x_n \geq 0\} \subseteq \overline{\Omega}, \\ B^- := B \cap \{x_n \leq 0\} \subseteq \mathbb{R}^n \setminus \Omega. \end{cases}$$

Notice that $B^+ \cap B^-$ consists of $B \cap \{x_n = 0\}$.

Step 2. Assume for the moment that u is of class $C^1(\overline{\Omega})$. We define a higher-order reflection of u over $B^+ \vee B^-$ via the equation

$$\tilde{u}(x) := \begin{cases} u(x), & \text{if } x \in B^+, \\ -3u(x_1, \dots, x_{n-1}, -x_n) + 4u(x_1, \dots, x_{n-1}, -\frac{x_n}{2}), & \text{if } x \in B^-. \end{cases} \quad (3.7)$$

Notice that this yields a function $\tilde{u} : B \rightarrow \mathbb{R}$. In the remainder of this step, we show that $\tilde{u} \in C^1(B)$. To this end, put

$$u^+ := \tilde{u}|_{B^+} \quad \text{and} \quad u^- := \tilde{u}|_{B^-}.$$

To achieve this, we will require the identity

$$u_{x_n}^+ \equiv u_{x_n}^- \quad \text{on } \{x_n = 0\}. \quad (3.8)$$

Certainly, for $x \in B^-$, differentiation of (3.7) gives

$$u_{x_n}^-(x) = 3u_{x_n}(x_1, \dots, -x_n) - 2u_{x_n}\left(x_1, \dots, x_{n-1}, -\frac{x_n}{2}\right)$$

which is precisely $u_{x_n}^+$ if $x_n = 0$. Hence, we have established (3.8). Clearly, $u^+ \equiv u^-$ on $\{x_n = 0\}$. Also, similar arguments to that above shows that

$$u_{x_i}^+ \equiv u_{x_i}^- \quad \text{on } \{x_n = 0\}, \quad \text{for } i = 1, \dots, n-1.$$

Combining this with (3.8), we see that for all multiindices α with $|\alpha| \leq 1$ there holds

$$\partial^\alpha u^+ \equiv \partial^\alpha u^- \quad \text{on } \{x_n = 0\}.$$

Since the partial derivatives are continuous on B , it follows that $\tilde{u}$ is of class $C^1(B)$.

Step 3. We see from all of the above calculations that, for some constant $C > 0$ not depending on the function u , there holds the inequality

$$\|\tilde{u}\|_{W^{1,p}(B)} \leq C \|u\|_{W^{1,p}(B^+)} \quad (3.9)$$

for each $u \in W^{1,p}(\Omega)$. We now work to extend our arguments to the case where $\partial\Omega$ is not necessarily flat near x^0 . Choose a C^1 -straightening mapping Φ at x^0 , with inverse Ψ . Put $y = \Phi(x)$ and $x = \Psi(y)$. Define

$$u'(y) := u(\Psi(y)).$$

Choose an open ball B for $y = \Phi(x^0)$ as before; using the previous steps we extend u' from B^+ to a map $\bar{u}'$ defined on the entirety of B . Notice that we may take $\bar{u}'$ to be of class $C^1(B)$ with the bound

$$\|\bar{u}'\|_{W^{1,p}(B)} \leq C \|u'\|_{W^{1,p}(B^+)}.$$

Using Ψ to return to x -variables and editing the constant $C > 0$ as necessary, we obtain an extension $\tilde{u}$ of u to $W := \Psi(B)$ such that

$$\|\tilde{u}\|_{W^{1,p}(W)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

Step 4. Since $\partial\Omega$ is compact in $\mathbb{R}^n$, we may choose finitely many points $x_1^0, \dots, x_N^0$ in $\partial\Omega$ such that $\partial\Omega \subseteq \bigcup_{i=1}^N W_i$; where the W_i (as in the previous step) are associated to extensions $\tilde{u}_i$ of u to W_i . Let $W_0 \Subset \Omega$ be such that $\Omega \subseteq \bigcup_{i=0}^N W_i$.⁶ Let $\{\zeta_i\}_{i=0}^\infty$ be an associated partition of unity. Let $\tilde{u} := \sum_{i=0}^N \zeta_i \tilde{u}_i$, with $\tilde{u}_0 := u$. Using our previous estimates then yields

$$\|\tilde{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\Omega)}$$

for some constant $C > 0$ independent of u . By shrinking the W_i , we may also assume that the support of $\tilde{u}$ is contained within V .

Step 5. Define Eu to be $\tilde{u}$, which we constructed above. Notice that, by construction, this map is $\mathbb{R}$ -linear. We could have also made the assumption that $u \in C^\infty(\overline{\Omega})$. If $1 \leq p < \infty$, the set $C^\infty(\overline{\Omega})$ is dense in $W^{1,p}(\Omega)$ and one can choose a sequence u_m in $C^\infty(\overline{\Omega})$ converging to u in $W^{1,p}(\Omega)$. Since E is bounded by construction, the sequence Eu_m is Cauchy, and hence convergent, in $W^{1,p}(\mathbb{R}^n)$. It can also be shown that the limit is independent of approximating sequence. By taking limits, we see that the limit of this sequence satisfies the desired properties. This can be seen by extracting a subsequence converging to u almost everywhere. We omit the case $p = \infty$, as it requires additional methods not accessible to us.

3.4 Sobolev Inequalities

In this section, we focus on embeddings of Sobolev spaces into other function spaces. That is, we would like to find real Banach spaces $\mathcal{X}$ for which there exists an injective bounded linear map

$$W^{k,p}(\Omega) \hookrightarrow \mathcal{X}.$$

As should be expected, this is not an easy question in general. Throughout this section, we will handle the distinct cases

$$1 \leq p < n, \tag{3.10}$$

$$n < p \leq \infty. \tag{3.11}$$

⁶The existence of such a set follows from our “exhaustion lemma”.

We omit the borderline case $p = n$ and handle each of the above cases in different subsections. Before, however, we would like to extend Hölder's inequality to obtain a result we will make heavy use of through this section.

Lemma 3.12 (Extended Hölder Inequality). *Let $\Sigma \subseteq \mathbb{R}^n$ be Lebesgue measurable. Let $1 \leq p_1, \dots, p_k \leq \infty$ and assume*

$$\sum_{j=1}^k \frac{1}{p_j} = 1.$$

Let $u_1, \dots, u_k : \Sigma \rightarrow \mathbb{C}$ (or $\mathbb{R}^$) be measurable, then*

$$\left\| \prod_{j=1}^k u_j \right\|_{L^1(\Sigma)} \leq \prod_{j=1}^k \|u_j\|_{L^{p_j}(\Sigma)}.$$

Proof. We argue by induction on k . The case $k = 1$ is trivial and $k = 2$ is just Hölder's inequality. Assume the claim holds for $k - 1 \geq 2$, we must show it holds for k . Let $v := \prod_{j=1}^{k-1} u_j$ and apply Hölder's inequality to get

$$\left\| \prod_{j=1}^k u_j \right\|_{L^1(\Sigma)} \leq \|u_1\|_{L^{p_1}(\Sigma)} \|v\|_{L^r(\Sigma)}$$

where $r \in \mathbb{R}^*$ is such that

$$\frac{1}{r} = \sum_{j=2}^k \frac{1}{p_j}.$$

If $r = \infty$, this means that each $p_j = \infty$, $j > 1$, whence the claim follows without the use of the induction hypothesis. Otherwise, $r < \infty$. Thus,

$$\begin{aligned} \|v\|_{L^r(\Sigma)}^r &= \int_{\Sigma} |u_2 \cdots u_k|^r \, dm = \int_{\Sigma} |u_2|^r \cdots |u_k|^r \, dm \\ &= \|u_2^r \cdots u_k^r\|_{L^1(\Sigma)}. \end{aligned}$$

Now,

$$1 = \sum_{j=2}^k \frac{r}{p_j} = \sum_{j=2}^k \frac{1}{p_j/r}$$

so that, by our induction hypothesis,

$$\begin{aligned} \|v\|_{L^q(\Sigma)}^r &\leq \|u_2^r \cdots u_k^r\|_{L^1(\Sigma)} \leq \prod_{j=2}^k \|u_j^r\|_{L^{p_j/r}(\Sigma)} = \prod_{j=2}^k \|u_j^r\|_{L^{p_j/r}(\Sigma)} \\ &= \prod_{j=2}^k \left(\int_{\Sigma} |u_j|^{p_j} dm \right)^{r/p_j}. \end{aligned}$$

Thus, $\|v\|_{L^r(\Sigma)} \leq \prod_{j=2}^k \|u_j\|_{L^{p_j}(\Sigma)}$. This completes the proof. $\square$

Before turning towards inequalities related to Sobolev spaces, we give a final theorem.

Proposition 3.13 (Typewriter's Theorem). *Let $(X, \mathfrak{M}, \mu)$ be a complete measure space and assume $1 \leq p \leq \infty$. Fix $f \in L^p(X)$ and assume that $\{f_n\}_{n=1}^{\infty}$ is a sequence in $L^p(X)$, converging to f in $L^p(X)$. There exists a subsequence $\{f_{n_k}\}_{k=1}^{\infty}$ converging pointwise to f almost everywhere on X .*

Proof. The claim is immediate if $p = \infty$; assume henceforth that $p < \infty$. Since $\{f_n\}_{n=1}^{\infty}$ is convergent in $L^p(X)$, it is in particular Cauchy. Therefore, we may extract a subsequence $\{f_{n_k}\}_{k=1}^{\infty}$ such that

$$\|f_{n_k} - f_{n_{k-1}}\|_{L^p(X)} < \frac{1}{2^k}, \quad \text{for all } k > 2.$$

For $N \geq 1$ define

$$f_N(x) := f_{n_1}(x) + \sum_{k=2}^N (f_{n_k}(x) - f_{n_{k-1}}(x));$$

since each f_n belongs to $L^p(X)$, the above is well defined and finite for almost all $x \in X$. Moreover,

$$\sum_{k=2}^{\infty} \|f_{n_k} - f_{n_{k-1}}\|_{L^p(X)} \leq \sum_{k=2}^{\infty} \frac{1}{2^k} < \infty.$$

Since $L^p(X)$ is complete, $\sum_{k=2}^{\infty} (f_{n_k}(x) - f_{n_{k-1}}(x))$ exists in $L^p(X)$. In particular, this series is convergent for almost all $x \in X$. For all such x , put

$$g(x) := \lim_{N \rightarrow \infty} \left[f_{n_1}(x) + \sum_{k=2}^N (f_{n_k}(x) - f_{n_{k-1}}(x)) \right] = \lim_{k \rightarrow \infty} f_{n_k}(x)$$

and set $g(x) = 0$ otherwise. Since $(X, \mathfrak{M}, \mu)$ is complete, this map g is measurable as the almost everywhere limit of measurable functions. In fact, g will belong to $L^p(X)$ by the triangle inequality. On the other hand, by Fatou's lemma, we have

$$\begin{aligned} \|g - f\|_{L^p(X)}^p &= \int_X |g - f|^p d\mu \leq \liminf_{k \rightarrow \infty} \int_X |f_{n_k} - f|^p d\mu \\ &= \liminf_{k \rightarrow \infty} \|f_{n_k} - f\|_{L^p(X)}^p = 0. \end{aligned}$$

It follows that $f = g$ for almost all $x \in X$, and hence that $f_{n_k} \rightarrow f$ pointwise almost everywhere on X . This completes the proof. $\square$

3.4.1 The Gagliardo-Nirenberg-Sobolev Inequality

This subsection is devoted to the proof of a very famous inequality. Assume henceforth that

$$1 \leq p < n,$$

where n is the dimension of the space. We would like to determine whether there exists a constant $C > 0$ and $1 \leq q < \infty$ such that

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}$$

for all functions $u \in C_c^\infty(\mathbb{R}^n)$. The answer is a resounding YES, under the right conditions.

Definition 22 (Sobolev Conjugate). Let $1 \leq p < n$. We define the *Sobolev conjugate* of p as follows:

$$p^* := \frac{np}{n-p}. \quad (3.12)$$

Notice that, in particular, $p^* > p$. Moreover,

$$\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}.$$

Theorem 3.14 (Gagliardo-Nirenberg-Sobolev Inequality). *Let $1 \leq p < n$. There exists a constant $C > 0$ such that*

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)} \quad (3.13)$$

for all $u \in C_c^1(\mathbb{R}^n)$.

Remark 3.2. The assumption that u has compact support is critical. As a counter example, consider the *smooth* function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ given by $u(x) = 1$ on $\mathbb{R}^n$. Clearly, $u \notin L^q(\mathbb{R}^n)$ for any $1 \leq q < \infty$. However, ∇u is identically zero on $\mathbb{R}^n$, whence $\nabla u \in L^p(\mathbb{R}^n)$.

Proof of Theorem. Once again, this proof is best decomposed into steps.

Step 1. Since u has compact support, for each index $i = 1, \dots, n$ we have

$$u(x) = \int_{-\infty}^x u_{x_i}(x_1, \dots, x_{i-1}, y_i, \dots, x_n) dy_i, \quad \forall x \in \mathbb{R}^n.$$

A standard estimate then gives, for each $x \in \mathbb{R}^n$ and index i ,

$$|u(x)| \leq \int_{\mathbb{R}} |\nabla u(x_1, \dots, x_{i-1}, y_i, \dots, x_n)| dy_i.$$

Hence, for all $x \in \mathbb{R}^n$,

$$|u(x)|^{\frac{n}{n-1}} \leq \prod_{i=1}^n \left(\int_{\mathbb{R}} |\nabla u(x_1, \dots, x_{i-1}, y_i, \dots, x_n)| dy_i \right)^{\frac{1}{n-1}}. \quad (3.14)$$

Now, we integrate (3.14) with respect to the variable x_1 to obtain

$$\int_{\mathbb{R}} |u(x)|^{\frac{n}{n-1}} dx_1 \leq \int_{\mathbb{R}} \prod_{i=1}^n \left(\int_{\mathbb{R}} |\nabla u| dy_i \right)^{\frac{1}{n-1}} dx_1 \quad (3.15)$$

$$= \left(\int_{\mathbb{R}} |\nabla u| dy_1 \right)^{\frac{1}{n-1}} \int_{\mathbb{R}} \prod_{i=2}^n \left(\int_{\mathbb{R}} |\nabla u| dy_i \right)^{\frac{1}{n-1}} dx_1 \quad (3.16)$$

$$\leq \left(\int_{\mathbb{R}} |\nabla u| dy_1 \right)^{\frac{1}{n-1}} \prod_{i=2}^n \left(\int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| dx_1 dy_i \right)^{\frac{1}{n-1}}. \quad (3.17)$$

In this last step, we have used the extended Hölder inequality from the start of this section. Proceeding in a similar way, we integrate the above inequality with respect to x_2 which gives

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |u|^{\frac{n}{n-1}} dx_1 dx_2 \leq \left(\int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| dx_1 dy_2 \right)^{\frac{1}{n-1}} \int_{\mathbb{R}} \prod_{i \neq 2} I_i^{\frac{1}{n-1}} dx_2.$$

Here,

$$I_1 = \int_{\mathbb{R}} |\nabla u| \, dy_1, \quad I_i = \int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dx_1 \, dy_i, \quad i \geq 3.$$

As before, invoking the extended Hölder inequality yields

$$\begin{aligned} \int_{\mathbb{R}} \int_{\mathbb{R}} |u|^{\frac{n}{n-1}} \, dx_1 \, dx_2 &\leq \left(\int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dx_1 \, dy_2 \right)^{\frac{1}{n-1}} \left(\int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dy_1 \, dx_2 \right)^{\frac{1}{n-1}} \\ &\quad \cdot \prod_{i=3}^n \left(\int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dx_1 \, dx_2 \, dy_i \right). \end{aligned}$$

Continuing in this way, we eventually obtain

$$\int_{\mathbb{R}^n} |u|^{\frac{n}{n-1}} \, dm \leq \prod_{i=1}^n \left(\int_{\mathbb{R}} \cdots \int_{\mathbb{R}} |\nabla u| \, dx_1 \cdots dy_i \cdots dx_n \right)^{\frac{1}{n-1}} \quad (3.18)$$

$$= \left(\int_{\mathbb{R}^n} |\nabla u| \right)^{\frac{n}{n-1}} \, dm. \quad (3.19)$$

This yields the estimate in (3.13) for $p = 1$.

Step 2. We now handle the case $1 < p < n$. Fix now $\gamma > 1$, to be precisely chosen later. Using the estimates in (3.18)-(3.19) from the first step with $v := u^\gamma \in C_c^1(\mathbb{R}^n)$, we get

$$\begin{aligned} \left(\int_{\mathbb{R}^n} |u|^{\frac{\gamma n}{n-1}} \, dm \right)^{\frac{n-1}{n}} &\leq \int_{\mathbb{R}^n} |\nabla u^\gamma| \, dm \\ &= \gamma \int_{\mathbb{R}^n} |u^{\gamma-1}| |\nabla u| \, dm \\ &\leq \gamma \left(\int_{\mathbb{R}^n} \left| u^{\frac{(\gamma-1)p}{p-1}} \right| \, dm \right)^{\frac{p-1}{p}} \left(\int_{\mathbb{R}^n} |\nabla u|^p \, dm \right)^{1/p}. \end{aligned}$$

Choose now $\gamma > 1$ such that

$$\frac{\gamma n}{n-1} = \frac{(\gamma-1)p}{p-1},$$

i.e. define

$$\gamma := \frac{p(n-1)}{n-p} > 1.$$

Direct calculation confirms that

$$\frac{\gamma n}{n-1} = \frac{(\gamma-1)p}{p-1} = \frac{np}{n-p} = p^*.$$

Thus, this last inequality becomes

$$\left(\int_{\mathbb{R}^n} |u|^{p^*} dm \right)^{1/p^*} \leq \|\nabla u\|_{L^p(\mathbb{R}^n)},$$

as was required. This completes the proof. $\square$

Of course, (3.13) is only the first of many inequalities to come. However, it is still a *highly* non-trivial inequality that will serve as a bounds for functions in the Sobolev space $W^{1,p}(\Omega)$.

Theorem 3.15. *Let $\Omega \subseteq \mathbb{R}^n$ be bounded and open with C^1 boundary $\partial\Omega$. Let $1 \leq p < n$. There exists a constant $C > 0$ such that*

$$\|u\|_{L^{p^*}(\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)} \quad (3.20)$$

for all $u \in W^{1,p}(\Omega)$. In particular, $W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$.

Proof. By the Sobolev Extension Theorem, there exists a compactly supported function $\bar{u} \in W^{1,p}(\mathbb{R}^n)$ such that

1. $\bar{u} \equiv u$ almost everywhere in Ω ;
2. $\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\Omega)}$.

Recall that the convolution of two compactly supported functions is itself compactly supported, when it exists.⁷ In the notation from Theorem 3.6, for each $\varepsilon > 0$, we have $(\mathbb{R}^n)_\varepsilon = \mathbb{R}^n$. Thus, by this theorem, we may choose a sequence of functions $(u_m)_{m=1}^\infty$ in $C^\infty(\mathbb{R}^n)$ converging to $\bar{u}$ in $W_{\text{loc}}^{1,p}(\mathbb{R}^n)$. However, since $\bar{u}$

⁷Let $f \in L_{\text{loc}}^1(\mathbb{R}^n)$ be compactly supported and fix $\varphi \in C_c^\infty(\mathbb{R}^n)$. For each $x \in \mathbb{R}^n$, the convolution $(f * \varphi)(x) = \int_{\mathbb{R}^n} f(y)\varphi(x-y) dy$ exists and is finite. Indeed, this follows from the fact that φ has compact support. We have already seen that the support of $(f * \varphi)$ will be a subset of $\text{supp}(f) + \text{supp}(\varphi)$. Since both f and φ have compact support, both these sets are bounded. Hence, so is their “direct sum”. Taking closures shows that $(f * \varphi)$ has compact support.

is compactly supported, these functions u_m can be chosen in $C_c^\infty(\mathbb{R}^n)$ (see the footnote). Furthermore, for each $m \geq 1$ one has

$$\begin{aligned} \|\bar{u} - u_m\|_{W^{1,p}(\mathbb{R}^n)}^p &= \int_{\mathbb{R}^n} |\bar{u} - u_m|^p \, dm + \sum_{j=1}^n \int_{\mathbb{R}^n} |\bar{u}_{x_j} - (u_m)_{x_j}|^p \, dm \\ &= \int_{\Lambda_m} |\bar{u} - u_m|^p \, dm + \sum_{j=1}^n \int_{\Lambda_m} |\bar{u}_{x_j} - (u_m)_{x_j}|^p \, dm, \end{aligned}$$

where $\Lambda_m := \overline{\text{supp}(\bar{u}) \cup \text{supp}(u_m)}$ is a compact subset of $\mathbb{R}^n$. Let us briefly return to examine the proof of Theorem 3.6. An element of the sequence chosen will be of the form $u^\varepsilon := \bar{u} * \eta_\varepsilon$, for suitable $\varepsilon > 0$. Since our sequence is obtained by letting $\varepsilon \searrow 0$, we may assume that $\varepsilon \leq 1$. Then, η_ε is supported with the compact unit ball $\overline{B_1(0)}$ of $\mathbb{R}^n$. Thus,

$$\begin{aligned} \|\bar{u} - u_m\|_{W^{1,p}(\mathbb{R}^n)}^p &\leq \int_{\overline{\text{supp}(\bar{u}) \cup B_1(0)}} |\bar{u} - u_m|^p \, dm \\ &\quad + \sum_{j=1}^n \int_{\overline{\text{supp}(\bar{u}) \cup B_1(0)}} |\bar{u}_{x_j} - (u_m)_{x_j}|^p \, dm. \end{aligned}$$

Or, rather,

$$\|\bar{u} - u_m\|_{W^{1,p}(\mathbb{R}^n)} \leq \|\bar{u} - u_m\|_{W^{1,p}(\Sigma)}, \quad \forall m \geq 1,$$

where $\Sigma := \overline{\text{supp}(\bar{u}) \cup B_1(0)}$ is compact. Since $u_m \rightarrow \bar{u}$ in $W_{\text{loc}}^{1,p}(\mathbb{R}^n)$, the above implies that $u_m \rightarrow \bar{u}$ in $W^{1,p}(\mathbb{R}^n)$ as $m \rightarrow \infty$. In particular, $u_m \rightarrow \bar{u}$ in $L^p(\mathbb{R}^n)$. By the typewriter's theorem, after passing to a subsequence, we may assume that $u_m(x) \rightarrow \bar{u}(x)$ for almost all $x \in \mathbb{R}^n$.

Now, we invoke (3.13) to guarantee the existence of a constant $C > 0$ such that

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla f\|_{L^p(\mathbb{R}^n)},$$

for all $f \in C_c^1(\mathbb{R}^n) \supseteq C_c^\infty(\mathbb{R}^n)$. In particular, if $m, \ell \geq 1$ we have

$$\|u_m - u_\ell\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u_m - \nabla u_\ell\|_{L^p(\mathbb{R}^n)}. \quad (3.21)$$

On the other hand, if $f \in C_c^1(\mathbb{R}^n)$ then, by the Minkowski inequality,

$$\begin{aligned} \|\nabla f\|_{L^p(\mathbb{R}^n)} &= \left(\int_{\mathbb{R}^n} \left(\sqrt{f_{x_1}^2 + \cdots + f_{x_n}^2} \right)^p dm \right)^{1/p} \\ &\leq \left(\int_{\mathbb{R}^n} (|f_{x_1}| + \cdots + |f_{x_n}|)^p dm \right)^{1/p} \\ &= \| |f_{x_1}| + \cdots + |f_{x_n}| \|_{L^p(\mathbb{R}^n)} \\ &\leq \sum_{i=1}^n \|f_{x_i}\|_{L^p(\mathbb{R}^n)}. \end{aligned}$$

We then get that, $\|\nabla f\|_{L^p(\mathbb{R}^n)} \leq n \|f\|_{W^{1,p}(\mathbb{R}^n)}$. Thus, (3.21) shows that the sequence $(u_m)_{m=1}^\infty$ is Cauchy, and hence convergent, in $L^{p^*}(\mathbb{R}^n)$. Let then $g \in L^{p^*}(\mathbb{R}^n)$ be the limit of this sequence. Then, by Fatou's lemma,

$$\begin{aligned} \|g - \bar{u}\|_{L^p(\mathbb{R}^n)}^p &= \int_{\mathbb{R}^n} |g - \bar{u}|^p dm \leq \liminf_{m \rightarrow \infty} \int_{\mathbb{R}^n} |u_m - \bar{u}|^p dm \\ &= \liminf_{m \rightarrow \infty} \|u_m - \bar{u}\|_{L^p(\mathbb{R}^n)}^p. \end{aligned}$$

Therefore, $g = \bar{u}$ almost everywhere. This means that $u_m \rightarrow \bar{u}$ in $L^{p^*}(\mathbb{R}^n)$ as well. Then, by continuity of norms, we get

$$\begin{aligned} \|\bar{u}\|_{L^{p^*}(\mathbb{R}^n)} &= \lim_{m \rightarrow \infty} \|u_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C \lim_{m \rightarrow \infty} \|\nabla u_m\|_{L^p(\mathbb{R}^n)} \\ &= C \|\nabla \bar{u}\|_{L^p(\mathbb{R}^n)}. \end{aligned}$$

The existence of this last limit follows, as before, from the fact that $u_m \rightarrow \bar{u}$ in $W^{1,p}(\mathbb{R}^n)$. Finally, we use the above with the inequality in 2. to get that

$$\|u\|_{L^{p^*}(\Omega)} \leq \|\bar{u}\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla \bar{u}\|_{L^p(\mathbb{R}^n)} \leq C \|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C' \|u\|_{W^{1,p}(\Omega)}.$$

This completes the proof. $\square$

3.4.1.1 Estimates for $W_0^{1,p}$

Let $\Omega \subseteq \mathbb{R}^n$ be non-empty and open subset of $\mathbb{R}^n$. Recall that $C_c^\infty(\Omega)$ can be viewed as a vector subspace of $W^{k,p}(\Omega)$. Then, we defined $W_0^{k,p}(\Omega)$ to be the topological closure of $C_c^\infty(\Omega)$ in $W^{k,p}(\Omega)$. Thus, $W_0^{k,p}(\Omega)$ is itself a Banach space over the real numbers.

Theorem 3.16 (Poincaré Inequality). *Let $\Omega \subset \mathbb{R}^n$ be open and bounded. Let $u \in W_0^{1,p}(\Omega)$ for some $1 \leq p < n$. Then, for each $1 \leq q \leq p^*$,*

$$\|u\|_{L^q(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}$$

for a constant $C > 0$ depending only on p, q, n and Ω .

Proof. Let $u \in W_0^{1,p}(\Omega)$; by definition, there exists a sequence $(u_m)_{m=1}^\infty$ of functions in $C_c^\infty(\Omega)$ converging to u with respect to the $W^{1,p}(\Omega)$ -norm. In particular,

$$u_m \xrightarrow{m \rightarrow \infty} u \quad \text{in } L^p(\Omega).$$

After passing to a subsequence, we may assume by Proposition 3.13 that $u_m \rightarrow u$ almost everywhere on Ω . Let us extend these u_m to be identically zero on $\mathbb{R}^n \setminus \Omega$, thereby obtaining a sequence in $C_c^\infty(\mathbb{R}^n)$. By (3.13), there exists a constant $C > 0$ such that

$$\|v\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla v\|_{L^p(\mathbb{R}^n)}, \quad \text{for all } v \in C_c^\infty(\mathbb{R}^n). \quad (3.22)$$

Consequently, for all $m, \ell \geq 1$ we get, as in the previous theorem,

$$\begin{aligned} \|u_m - u_\ell\|_{L^{p^*}(\Omega)} &= \|u_m - u_\ell\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u_m - \nabla u_\ell\|_{L^p(\mathbb{R}^n)} \\ &= C \|\nabla u_m - \nabla u_\ell\|_{L^p(\Omega)} \\ &\leq C \|u_m - u_\ell\|_{W^{1,p}(\Omega)}. \end{aligned}$$

Thus, the sequence $(u_m)_{m=1}^\infty$ is Cauchy in $L^{p^*}(\Omega)$. It follows that u_m converges to some function $g \in L^{p^*}(\Omega)$. We now claim that $g = u$ for almost all $x \in \Omega$. To see this, write by Fatou's lemma

$$\|g - u\|_{L^{p^*}(\Omega)}^{p^*} = \int_\Omega |g - u|^{p^*} dm \leq \liminf_{m \rightarrow \infty} \int_\Omega |g - u_m|^{p^*} dm = 0.$$

Thus, $g = u$ for almost all $x \in \Omega$. In particular, $u \in L^{p^*}(\Omega)$ and $u_m \rightarrow u$ in $L^{p^*}(\Omega)$. This is precisely the convergence criteria we want. For each $m \geq 1$, (3.22) tells us that

$$\|u_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u_m\|_{L^p(\mathbb{R}^n)}.$$

Letting $m \rightarrow \infty$ in the above, we get that

$$\begin{aligned} \|u\|_{L^{p^*}(\Omega)} &= \lim_{m \rightarrow \infty} \|u_m\|_{L^{p^*}(\Omega)} = \lim_{m \rightarrow \infty} \|u_m\|_{L^{p^*}(\mathbb{R}^n)} \leq \lim_{m \rightarrow \infty} C \|\nabla u_m\|_{L^p(\mathbb{R}^n)} \\ &= \lim_{m \rightarrow \infty} C \|\nabla u_m\|_{L^p(\Omega)} \\ &= C \|\nabla u\|_{L^p(\Omega)}. \end{aligned}$$

To summarize, we have shown that $W_0^{1,p}(\Omega) \subseteq L^{p^*}(\Omega)$. Let now $1 \leq q \leq p^*$ and note that Ω has finite Lebesgue measure. Summoning Proposition 6.12 from [FLND] then gives $L^q(\Omega) \supseteq L^{p^*}(\Omega)$ whence

$$W_0^{1,p}(\Omega) \subseteq L^{p^*}(\Omega) \subseteq L^q(\Omega).$$

Let $m(\cdot)$ be the Lebesgue measure on $\mathbb{R}^n$, this same proposition implies that

$$\|f\|_{L^q(\Omega)} \leq m(\Omega)^{\frac{1}{q} - \frac{1}{p^*}} \|f\|_{L^{p^*}(\Omega)}, \quad \forall f \in L^{p^*}(\Omega).$$

Therefore, if $u \in W_0^{1,p}(\Omega)$ all these inequalities combine to yield

$$\|u\|_{L^q(\Omega)} \leq m(\Omega)^{\frac{1}{q} - \frac{1}{p^*}} \|u\|_{L^{p^*}(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}.$$

This completes the proof. $\square$

3.4.2 Morrey's Inequality

In this subsection, we pursue our previous endeavors. That is, we continue to search for potential embeddings $W^{1,p}(\Omega) \hookrightarrow \mathcal{X}$. In the last subsection, we investigated the case where $1 \leq p < n$. Let us instead consider the case of

$$1 \leq n < p \leq \infty.$$

Ultimately, we will show that any function $u \in W^{1,p}(\Omega)$ is equal (almost everywhere) to a Hölder continuous function $\Omega \rightarrow \mathbb{R}$. Thus, any function $u \in W^{1,p}(\Omega)$ can be made Hölder continuous after being redefined on a set of Lebesgue measure zero. This result will motivate the following definition.

Definition 23 (Version). Let $(X, \mathfrak{M}, \mu)$ be a measure space and consider two functions $f, f' : X \rightarrow \mathbb{R}^*$. We say that f is a *version* of f' if $f(x) = f'(x)$ for almost every $x \in X$.

Let us briefly develop the notion of Hölder continuity (closely related to the theory of Hölder spaces). Let $\Omega \subseteq \mathbb{R}^n$ be a non-empty open set and assume $0 < \gamma \leq 1$. A function $u : \Omega \rightarrow \mathbb{R}$ is said to be *Hölder continuous with exponent γ* if there exists $C > 0$ such that

$$|u(x) - u(y)| \leq C|x - y|^\gamma, \quad \forall x, y \in \Omega.$$

Definition 24 (Hölder Norm). Fix a non-empty open set $\Omega \subseteq \mathbb{R}^n$ and assume $u : \Omega \rightarrow \mathbb{R}$ is bounded and continuous. We then set

$$\|u\|_{C(\overline{\Omega})} := \sup_{x \in \overline{\Omega}} |u(x)| < \infty.$$

The γ^{th} -Hölder characteristic of u is subsequently defined as

$$[u]_{C^{0,\gamma}(\overline{\Omega})} := \sup_{\substack{x,y \in \overline{\Omega} \\ x \neq y}} \left\{ \frac{|u(x) - u(y)|}{|x - y|^\gamma} \right\}.$$

Then, the γ^{th} -Hölder norm is

$$\|u\|_{C^{0,\gamma}(\overline{\Omega})} := \|u\|_{C(\overline{\Omega})} + [u]_{C^{0,\gamma}(\overline{\Omega})}.$$

With these formalities out of the way, we move towards the aforementioned inequality.

Theorem 3.17 (Morrey's Inequality). *Let $n < p \leq \infty$, and define*

$$\gamma := 1 - \frac{n}{p} > 0.$$

There exists a constant $C > 0$ such that

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$$

for all functions $u \in C^1(\mathbb{R}^n)$.

The proof of this theorem is, unsurprisingly, quite the venture. Many sources (e.g. [EVNS]) give the proof in a single mess. We will instead break it into lemmas which, when combined, yield the proof of Morrey's inequality.

3.4.2.1 Morrey's Inequality – Helping Lemmas

Throughout this subsection we will assume $n < p < \infty$. An identical method of proof holds in the case $p = \infty$, but the notation is not consistent. For this reason, we leave the case $p = \infty$ as a nifty exercise to the reader.

Lemma 3.18. *Assume the hypothesis of Morrey's inequality. There exists a constant $C > 0$ such that*

$$\int_{B(x,r)} |u(x) - u(y)| \, dy \leq C \int_{B(x,r)} \frac{|\nabla u(y)|}{|y - x|^{n-1}} \, dy$$

for each $u \in C^1(\mathbb{R}^n)$ and every ball $B(x, r) \subset \mathbb{R}^n$.

Proof of Lemma 3.18. Fix any $u \in C^1(\mathbb{R}^n)$, a ball $B(x, r) \subset \mathbb{R}^n$, and a point $w \in \partial B(0, 1)$. For every $s \in (0, r)$, Cauchy-Schwarz gives

$$\begin{aligned} |u(x + sw) - u(x)| &= \left| \int_0^s \frac{d}{dt} u(x + tw) dt \right| \\ &= \left| \int_0^s w \cdot \nabla u(x + tw) dt \right| \\ &= \int_0^s |\nabla u(x + tw)| dt. \end{aligned}$$

Integrating both sides of this inequality, we discover that

$$\int_{\partial B(0,1)} |u(x + sw) - u(x)| dS(w) \leq \int_{\partial B(0,1)} \int_0^s |\nabla u(x + tw)| dt dS(w). \quad (3.23)$$

By Fubini-Tonelli, this translates into

$$\int_{\partial B(0,1)} |u(x + sw) - u(x)| dS(w) \leq \int_0^s \int_{\partial B(0,1)} |\nabla u(x + tw)| dS(w) dt.$$

Then, we make the substitution $y := x + tw$ to get

$$\begin{aligned} \int_{\partial B(0,1)} |u(x + sw) - u(x)| dS(w) &\leq \int_0^s \int_{\partial B(x,t)} \frac{|\nabla u(y)|}{t^{n-1}} dS(y) dt \\ &= \int_{B(x,s)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} dy \\ &\leq \int_{B(x,r)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} dy. \end{aligned}$$

Furthermore, making the substitution $z := x + sw$ in (3.23) gives

$$\int_{\partial B(0,1)} |u(x + sw) - u(x)| dS(w) = \frac{1}{s^{n-1}} \int_{\partial B(x,s)} |u(z) - u(x)| dS(z).$$

Putting this together with the previous estimates, we find that

$$\int_{\partial B(x,s)} |u(z) - u(x)| dS(z) \leq s^{n-1} \int_{B(x,r)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} dy.$$

Recall now that $s \in (0, r)$ was taken to be arbitrary. Let us now integrate through the above inequality with respect to s :

$$\begin{aligned} \int_{B(x,r)} |u(z) - u(x)| \, dS(z) &= \int_0^r \int_{\partial B(x,s)} |u(z) - u(x)| \, dS(z) \, ds \\ &= \int_0^r s^{n-1} \int_{B(x,r)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} \, dy \, ds \\ &= \frac{r^n}{n} \int_{B(x,r)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} \, dy, \end{aligned}$$

Or, rather,

$$\frac{1}{r^n \omega_n} \int_{B(x,r)} |u(z) - u(x)| \, dS(z) \leq \frac{1}{n \omega_n} \int_{B(x,r)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} \, dy.$$

With this, the lemma is proven. $\square$

There is much more to be said. Again, continue to assume the hypothesis of Morrey's inequality. Assume that $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is of class C^1 . For each $x \in \mathbb{R}^n$, an application of the lemma above gives

$$\oint_{B(x,1)} |u(x) - u(y)| \, dy \leq C \int_{B(x,1)} \frac{|\nabla u(y)|}{|y - x|^{n-1}} \, dy$$

for some constant $C > 0$ independent of u . In fact, successively applying Hölder's inequality allows us to say the following:

$$\begin{aligned} |u(x)| &= \oint_{B(x,1)} |u(x)| \, dy \leq \oint_{B(x,1)} |u(x) - u(y)| \, dy + \oint_{B(x,1)} |u(y)| \, dy \\ &\leq C \int_{B(x,1)} \frac{|\nabla u(y)|}{|y - x|^{n-1}} \, dy + C \|u\|_{L^p(B(x,1))} \\ &\leq C \|\nabla u\|_{L^p(B(x,1))} \cdot \left(\int_{B(x,1)} \frac{1}{|x - y|^{\frac{p(n-1)}{p-1}}} \, dy \right)^{\frac{p-1}{p}} \\ &\quad + C \|u\|_{L^p(B(x,1))} \\ &\leq C \|\nabla u\|_{L^p(\mathbb{R}^n)} + C \|u\|_{L^p(\mathbb{R}^n)} \\ &\leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}. \end{aligned}$$

This last inequality holds since

$$\frac{p(n-1)}{p-1} = \left(\frac{1}{p-1} + 1 \right) (n-1) = \underbrace{\frac{n-1}{p-1}}_{<1 \because p < n} + (n-1) < n;$$

whence

$$\int_{B(x,1)} \frac{1}{|x-y|^{\frac{p(n-1)}{p-1}}} dy < \infty.$$

Notice that the integral above is in fact *independent* of the point x , as can easily be seen after conversion to polar coordinates. Notice also that a very similar argument applies when $p = \infty$ to yield the “same” conclusion. Now, to summarize the discussion above:

Lemma 3.19. *Assume the conditions of Morrey’s inequality. There exists a constant $C > 0$ such that*

$$|u(x)| \leq C \|u\|_{W^{1,p}(\Omega)},$$

for all $u \in C^1(\mathbb{R}^n)$ and every $x \in \mathbb{R}^n$.

Corollary 3.20. *Under the conditions of Morrey’s inequality, there exists a constant $C > 0$ such that*

$$\sup_{x \in \mathbb{R}^n} |u(x)| = \|u\|_{L^\infty(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)},$$

for all functions $u \in C^1(\mathbb{R}^n)$.

3.4.2.2 Proof of Morrey’s Inequality

With these two lemmas, we can finally give a complete proof of Morrey’s inequality. As mentioned in §3.4.2.1, we continue to assume that $n < p < \infty$.

Proof of Theorem 3.17. First, let $x \in \mathbb{R}^n$ and assume $B = B(z, r)$ is an open ball containing the point x . Let $\bar{u}_B$ denote the average of u over the ball B :

$$\bar{u}_B := \oint_B u(\sigma) d\sigma.$$

Hence, a standard estimate yields

$$|u(x) - \bar{u}_B| = \left| u(x) - \oint_B u(\sigma) d\sigma \right| \leq \oint_B |u(x) - u(\sigma)| d\sigma.$$

Now, observe that $B = B(z, r) \subseteq B(x, 2r)$. Since $m(B(x, 2r)) = 2^n m(B)$ we then obtain

$$|u(x) - \bar{u}_B| \leq 2^n \int_{B(x, 2r)} |u(x) - u(\sigma)| d\sigma.$$

By Lemma 3.18, there exists a constant $C > 0$ independent of u and $r > 0$ having the property that

$$2^n \int_{B(x, 2r)} |u(x) - u(\sigma)| d\sigma \leq C \int_{B(x, 2r)} \frac{|\nabla u(\sigma)|}{|\sigma - x|^{n-1}} d\sigma.$$

Or, rather, such that

$$\begin{aligned} |u(x) - \bar{u}_B| &\leq C \int_{B(x, 2r)} \frac{|\nabla u(\sigma)|}{|\sigma - x|^{n-1}} d\sigma \\ &\leq C \|\nabla u\|_{L^p(B(x, 2r))} \left(\int_{B(x, 2r)} \frac{d\sigma}{|\sigma - x|^{\frac{(n-1)p}{p-1}}} \right)^{\frac{p-1}{p}}. \end{aligned}$$

Here, we have made use of Hölder's inequality in the last step. We are almost done. After a conversion to polar coordinates, we see that for some $c > 0$

$$\int_{B(x, 2r)} \frac{1}{|\sigma - x|^{\frac{(n-1)p}{p-1}}} d\sigma = cr^{\frac{p-n}{p-1}}.$$

Hence,

$$\begin{aligned} |u(x) - \bar{u}_B| &\leq C \|\nabla u\|_{L^p(B(x, 2r))} r^{\frac{p-n}{p}} = C \|\nabla u\|_{L^p(B(x, 2r))} r^{1-\frac{n}{p}} \\ &= C \|\nabla u\|_{L^p(B(x, 2r))} r^\gamma, \end{aligned}$$

where γ is as in the statement of Theorem 3.17. The proof is *almost complete*. Let $x \neq y \in \mathbb{R}^n$ be given and fix $\varepsilon > 0$. Let $r := |x - y| > 0$ and choose a ball $B = B(z, r)$ containing both x and y .⁸ After possibly editing the constant $C > 0$, the inequalities above give

$$\begin{aligned} |u(x) - u(y)| &\leq |u(x) - \bar{u}_B| + |u(y) - \bar{u}_B| \\ &\leq Cr^\gamma \left(\|\nabla u\|_{L^p(B(x, 2r))} + \|\nabla u\|_{L^p(B(y, 2r))} \right) \\ &\leq C \|\nabla u\|_{L^p(\mathbb{R}^n)} |x - y|^\gamma. \end{aligned}$$

⁸Take z to be the midpoint along the straight line connecting x and y .

Therefore,

$$\sup_{\substack{x, y \in \mathbb{R}^n \\ x \neq y}} \frac{|u(x) - u(y)|}{|x - y|^\gamma} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}.$$

This last inequality together with Corollary (3.20) complete the proof. $\square$

3.4.2.3 Almost Hölder Continuous Functions

As mentioned at the start of the subsection, our ultimate goal is to prove that functions in $W^{1,p}(\Omega)$ are “almost Hölder continuous”, when $p > n$. Now that we are equipped with Morrey’s inequality, this result is now within reach.

Theorem 3.21. *Let $\Omega \subset \mathbb{R}^n$ be non-empty, open, and bounded with C^1 boundary. Suppose $n < p \leq \infty$ and let $u \in W^{1,p}(\Omega)$. There exists a version $u^* \in C^{0,\gamma}(\overline{\Omega})$ of u and a constant $C > 0$ such that*

$$\|u^*\|_{C^{0,\gamma}(\overline{\Omega})} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

This constant $C > 0$ does not depend on u or u^ .*

Proof. By the Sobolev extension theorem, there exists a compactly supported function $\bar{u} \in W^{1,p}(\mathbb{R}^n)$, equal to u almost everywhere in Ω , with the property that

$$\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\Omega)} < \infty.$$

By this same theorem, we may choose $C > 0$ independent of u and $\bar{u}$. Since $\bar{u}$ has compact support, the argument used in Theorem 3.15 allows us to construct a sequence $(u_m)_{m=1}^\infty$ in $C_c^\infty(\mathbb{R}^n)$ converging to $\bar{u}$ in $W^{1,p}(\mathbb{R}^n)$, as $m \rightarrow \infty$. In particular, u_m converges to $\bar{u}$ in $L^p(\Omega)$. By Morrey’s inequality, for all $m, \ell \geq 1$ one has the estimate

$$\|u_m - u_\ell\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u_m - u_\ell\|_{W^{1,p}(\mathbb{R}^n)}.$$

Here, $\gamma \in (0, 1)$ is taken to be as in the statement of Morrey’s inequality. Similarly,

$$\|u_m\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u_m\|_{W^{1,p}(\mathbb{R}^n)}$$

for each index $m \in \mathbb{N}$. Therefore, $(u_m)_{m=1}^\infty$ is a Cauchy sequence in $C^{0,\gamma}(\mathbb{R}^n)$. Since Hölder spaces are complete, there exists a function $u^* \in C^{0,\gamma}(\mathbb{R}^n)$ with

$$\lim_{m \rightarrow \infty} \|u_m - u^*\|_{C^{0,\gamma}(\mathbb{R}^n)} = 0.$$

This function u^* is Hölder continuous on the entirety of $\mathbb{R}^n$. In particular,

$$\lim_{m \rightarrow \infty} \sup_{x \in \mathbb{R}^n} |u_m(x) - u^*(x)| = 0.$$

Now, Fatou's lemma yields

$$\begin{aligned} \|u^* - u\|_{L^p(\Omega)}^p &= \int_{\Omega} |u^* - u|^p \, dm = \int_{\Omega} |u^* - \bar{u}|^p \, dm \\ &\leq \liminf_{m \rightarrow \infty} \int_{\Omega} |u_m - \bar{u}|^p \, dm \\ &\leq \liminf_{m \rightarrow \infty} \|u_m - \bar{u}\|_{W^{1,p}(\mathbb{R}^n)}^p = 0. \end{aligned}$$

Hence, $u = u^*$ almost everywhere on Ω . $\square$

We can draw the same conclusion in the case $p = \infty$, although we must adopt an entirely different argument.

Theorem 3.22. *Let $\Omega \subset \mathbb{R}^n$ be non-empty, open, and bounded. Assume also that $\partial\Omega$ is C^1 . Then, any function $u \in W^{1,\infty}(\Omega)$ is Hölder continuous with exponent 1. That is, $W^{1,\infty}(\Omega) \hookrightarrow \text{Lip}(\Omega)$.*

Proof. Let $u \in W^{1,\infty}(\Omega)$ and invoke the Sobolev extension theorem to guarantee the existence of a compactly supported function $\bar{u} \in W^{1,\infty}(\mathbb{R}^n)$, equal to u almost everywhere on Ω . Let $\eta \in C_c^\infty(\mathbb{R}^n)$ be the standard mollifier and set for $\varepsilon > 0$:

$$u^\varepsilon(x) := \int_{\mathbb{R}^n} u(y) \eta_\varepsilon(x - y) \, dy, \quad x \in \mathbb{R}^n.$$

Since we are interested these as $\varepsilon \searrow 0$, we might as well restrict ourselves to $0 < \varepsilon < 1$. In this case, the η_ε will all have compact support within the unit ball $B(0, 1)$. Since u has compact support, it follows that

$$u^\varepsilon \longrightarrow \bar{u} \quad \text{uniformly in } \mathbb{R}^n \text{ as } \varepsilon \rightarrow 0.$$

Fix now any index $i = 1, \dots, n$ and a point $x \in \mathbb{R}^n$. It is known from the “properties of mollifiers” that u^ε is smooth on $\mathbb{R}^n$ for each $\varepsilon > 0$. Since $u \in L^1(\mathbb{R}^n)$ we consequently have

$$\frac{\partial}{\partial x_i} u^\varepsilon(x) = \int_{\mathbb{R}^n} \bar{u}(y) \frac{\partial}{\partial x_i} \eta_\varepsilon(x - y) \, dy = (-1) \int_{\mathbb{R}^n} \partial_{x_i} \bar{u}(y) \eta_\varepsilon(x - y) \, dy.$$

Thus, an application of Hölder's inequality yields

$$\left| \frac{\partial}{\partial x_i} u^\varepsilon(x) \right| \leq \int_{\mathbb{R}^n} |\partial_{x_i} \bar{u}(y) \eta_\varepsilon(x-y)| dy \leq \|\partial_{x_i} \bar{u}\|_{L^\infty(\mathbb{R}^n)} \leq \|\bar{u}\|_{L^\infty(\mathbb{R}^n)}.$$

Hence, for each $\varepsilon \in (0, 1)$,

$$\|\nabla u^\varepsilon\|_{L^\infty(\mathbb{R}^n)} \leq \sqrt{n} \|\bar{u}\|_{L^\infty(\mathbb{R}^n)}.$$

Let now $x \neq y$ in Ω . For $\varepsilon \in (0, 1)$, we have

$$\begin{aligned} u^\varepsilon(x) - u^\varepsilon(y) &= \int_0^1 \frac{d}{dt} u^\varepsilon(tx + (1-t)y) dt \\ &= \int_0^1 \nabla u^\varepsilon(tx + (1-t)y) \cdot (x-y) dt \end{aligned}$$

whence

$$|u^\varepsilon(x) - u^\varepsilon(y)| \leq \|u^\varepsilon\|_{L^\infty(\mathbb{R}^n)} |x-y| \leq C \|\bar{u}\|_{L^\infty(\mathbb{R}^n)} |x-y|.$$

Letting $\varepsilon \searrow 0$, it follows that

$$|u(x) - u(y)| \leq L|x-y|$$

for a constant $L > 0$ that does not depend on x and y . □

3.4.3 A General Sobolev Inequality

This subsection consists of a single “general inequality” for Sobolev spaces. Making use of the inequalities developed in this section, we will deduce the following theorem that is difficult to memorize.

Theorem 3.23. *Let $\Omega \subset \mathbb{R}^n$ be non-empty, open, and bounded. Suppose further that $\partial\Omega$ is C^1 . Let $u \in W^{k,p}(\Omega)$.*

1. If

$$k < \frac{n}{p},$$

then $u \in L^p(\Omega)$, with

$$\frac{1}{q} = \frac{1}{p} - \frac{k}{n} \quad \text{and} \quad \|u\|_{L^q(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)}.$$

Here, $C > 0$ is a constant independent of the function u .

2. If $k \geq 1$ and

$$k > \frac{n}{p},$$

then $u \in C^{k-\lfloor \frac{n}{p} \rfloor - 1, \gamma}(\overline{\Omega})$ for

$$\gamma := \begin{cases} \lfloor \frac{n}{p} \rfloor + 1 - \frac{n}{p}, & \text{if } \frac{n}{p} \notin \mathbb{Z}, \\ \text{any positive number} < 1, & \text{if } \frac{n}{p} \in \mathbb{Z}. \end{cases}$$

Additionally, there exists a constant $C > 0$, independent of u , such that

$$\|u\|_{C^{k-\lfloor \frac{n}{p} \rfloor - 1, \gamma}(\overline{\Omega})} \leq C \|u\|_{W^{k,p}(\Omega)}.$$

Here, we identify u with a version that is continuous.

Proof. We shall only prove the first claim. The proof of the second is not difficult, but it is filled with trivialities and case checking. Essentially, the proof of the second part boils down to L^p -interpolation and successive application of Morrey's inequality. The *heretic* reader may find the proof in [EVNS, §5.6.3]. I firmly believe that the proof does not convey any enlightening information, and it is therefore safe to omit it.

Let us now establish the first claim. Suppose that $k < \frac{n}{p}$. In particular, $p < \infty$. Claim 1. is trivial if $k = 0$. Thus, we assume $k \geq 1$. By assumption, $\partial^\alpha u$ belongs to $L^p(\Omega)$ for all multiindices $|\alpha| \leq k$. If $|\beta| \leq k - 1$, then Theorem 3.15 yields a constant $C > 0$ (independent of u) such that

$$\|\partial^\beta u\|_{L^{p^*}(\Omega)} \leq C \|\partial^\beta u\|_{W^{1,p}(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)}.$$

Indeed, the condition $k < \frac{n}{p}$ implies, especially, that $p < n$. Therefore, we have $u \in W^{k-1,p^*}(\Omega)$. Now, let $|\beta| \leq k - 2$ (if this makes sense). Then, $\partial^\beta u \in W^{1,p^*}(\Omega)$ so that, by the same argument, $\partial^\beta u \in L^{p^{**}}(\Omega)$. Hence, $u \in W^{k-2,p^{**}}(\Omega)$. Here,

$$p^{**} = \frac{1}{p^*} - \frac{1}{n} = \frac{1}{p} - \frac{2}{n}.$$

Proceeding in this fashion, we get $u \in W^{0,q}(\Omega) \cong L^q(\Omega)$ where

$$\frac{1}{q} = \frac{1}{p} - \frac{1}{k},$$

as was asserted. This procedure also ensures that the stated estimate holds true. $\square$

3.5 Compact Embeddings

We now finalize our treatment of Sobolev spaces and, fittingly, our consideration of their embeddings. Let $1 \leq p < n$ and assume $\Omega \subset \mathbb{R}^n$ is non-empty, open, and bounded with C^1 -boundary. If p^* denotes the Sobolev conjugate to p , we have already shown (see Theorem 3.15) that $W^{1,p}(\Omega)$ can be embedded into $L^{p^*}(\Omega)$. In this section, we show that this embedding into $L^q(\Omega)$ —for $q \in [1, p^*)$ —is actually *compact*. To clarify what we mean by a “compact embedding”, let us first recall a definition from functional analysis.

Definition 25 (Compact Embedding). Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces with ground field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. We say that $\mathcal{X}$ is *compactly embedded* in $\mathcal{Y}$, denoted $\mathcal{X} \Subset \mathcal{Y}$, if the following hold true:

1. $\mathcal{X} \subseteq \mathcal{Y}$;
2. there exists a constant $C > 0$ such that

$$\|u\|_{\mathcal{Y}} \leq C \|u\|_{\mathcal{X}}$$

for all $u \in \mathcal{X}$;

3. if $(u_m)_{m=1}^\infty$ is a bounded sequence in $\mathcal{X}$, there exists a subsequence converging to some $u \in \mathcal{Y}$.

The first two criteria of this definition simply require that $\mathcal{X}$ be *embedded* in $\mathcal{Y}$. In fact, points 1-2 of the definition boil down to saying that the inclusion map $\mathcal{X} \hookrightarrow \mathcal{Y}$ is well defined and continuous. The compactness itself comes from the final requirement. For the sake of clarification, a sequence $(u_m)_{m=1}^\infty$ is called *bounded* in $\mathcal{X}$ if

$$\sup_{m \in \mathbb{N}} \|u_m\|_{\mathcal{X}} < \infty.$$

Our cornerstone theorem (for the section) is the following.

Theorem 3.24 (Rellich-Kondrachov). *Let $\Omega \subset \mathbb{R}^n$ be non-empty, bounded, and open with C^1 -boundary. Suppose further that $1 \leq p < n$. Then, $W^{1,p}(\Omega) \Subset L^q(\Omega)$ for each $1 \leq q < p^*$.*

Proof. We will decompose this long proof into several steps.

Step 1. Let $q \in [1, p^*)$ be given. Invoking Theorem 3.15 yields

$$W^{1,p}(\Omega) \subseteq L^{p^*}(\Omega) \subseteq L^q(\Omega).$$

In fact, we obtain the existence of a constant $C > 0$ such that

$$\|u\|_{L^q(\Omega)} \leq C \|u\|_{L^{p^*}(\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}$$

for every $u \in W^{1,p}(\Omega)$. In short, $W^{1,p}(\Omega)$ is continuously embedded in $L^q(\Omega)$ for every $1 \leq q < p^*$. It hence only remains to establish the compactness of this embedding.

Step 2. Let $(v_m)_{m=1}^\infty$ be a bounded sequence in $W^{1,p}(\Omega)$. The Sobolev Extension theorem guarantees the existence of a bounded linear operator

$$E : W^{1,p}(\Omega) \longrightarrow W^{1,p}(\mathbb{R}^n)$$

such that $Ev \equiv v$ almost everywhere in Ω , for each $v \in W^{1,p}(\Omega)$. Evidently, we may also assume that each Ev has compact support within some precompact set $V_0 \subset \mathbb{R}^n$. This bounded linear operator E allows us to then associate a sequence $(u_m)_{m=1}^\infty$ in $W^{1,p}(\mathbb{R}^n)$ by setting $u_m := E(v_m)$, for $m \geq 1$. Since E is bounded, there exists a constant $C > 0$ with the property that

$$\sup_{m \geq 1} \|u_m\|_{W^{1,p}(V_0)} = \sup_{m \geq 1} \|u_m\|_{W^{1,p}(\mathbb{R}^n)} \leq C \sup_{m \geq 1} \|v_m\|_{W^{1,p}(\Omega)} < \infty.$$

Notice how we may choose to view $(u_m)_{m=1}^\infty$ as a sequence in $W^{1,p}(V_0)$. Denote by η the standard mollifier. For $0 < \varepsilon < 1$ and $m \in \mathbb{N}$ let us define

$$u_m^\varepsilon := u_m * \eta_\varepsilon, \quad \text{in } \mathbb{R}^n.$$

This definition makes sense since $u_m \in L^1(\mathbb{R}^n)$ for each index m .⁹ Now, the u_m and the η_ε both have uniform compact support (in m and ε , respectively). Thus, there exists a precompact set $V \supset V_0$ such that the u_m^ε and u_m *uniformly* have compact support in V (for $m \geq 1$ and $0 < \varepsilon < 1$). By taking V to be a sufficiently large ball, we can ensure that ∂V be of class C^∞ .

Step 3. We claim that $\{u_m^\varepsilon\}_{\varepsilon \in (0,1)}$ converges to u_m in $L^q(V)$, uniformly in m , as $\varepsilon \searrow 0$. First, observe that each $u_m^\varepsilon \in L^q(V)$ since these are smooth functions

⁹Since $u_m \in W^{1,p}(\mathbb{R}^n)$, it holds in particular that $u_m \in L^1_{\text{loc}}(\mathbb{R}^n)$. Recalling that u_m has compact support, the claim follows.

having compact support. Also, $u_m \in L^q(V)$ for every index m by the first step (clearly, $u_m \in W^{1,p}(V)$).

Suppose for the moment that $u_m \in C^\infty(V)$. Then, for each $x \in V$ we easily calculate

$$\begin{aligned} u_m^\varepsilon(x) - u_m(x) &= \int_{\mathbb{R}} \eta_\varepsilon(x-z) (u_m(z) - u_m(x)) \, dz \\ &= \frac{1}{\varepsilon^n} \int_{B(x,\varepsilon)} \eta\left(\frac{x-z}{\varepsilon}\right) (u_m(z) - u_m(x)) \, dz \\ &= \frac{1}{\varepsilon^n} \int_{B(x,\varepsilon)} \eta\left(\frac{z-x}{\varepsilon}\right) (u_m(z) - u_m(x)) \, dz \\ &= \int_{B(0,1)} \eta(y) (u_m(x+y\varepsilon) - u_m(x)) \, dy. \end{aligned}$$

Of course, we may choose to rewrite this as

$$\begin{aligned} \int_{B(0,1)} \eta(y) \int_0^1 \frac{d}{dt} u_m(x+ty\varepsilon) \, dt \, dy \\ = \varepsilon \int_{B(0,1)} \eta(y) \int_0^1 \nabla u_m(x+ty\varepsilon) \cdot y \, dt \, dy. \end{aligned}$$

Therefore, integrating the above inequality yields

$$\begin{aligned} \|u_m^\varepsilon - u_m\|_{L^1(V)} &\leq \varepsilon \int_{B(0,1)} \eta(y) \int_0^1 \int_V |\nabla u_m(x+t\varepsilon y)| \, dx \, dt \, dy \\ &\leq \varepsilon \int_V |\nabla u_m(z)| \, dz. \end{aligned}$$

Here, we have used that $\int_{\mathbb{R}^n} \eta(y) \, dy = 1$ and the fact that ∇u_m is compactly supported in V . We would now like to show that this estimate holds for functions $u_m \in W^{1,p}(V)$. Fix an index $m \geq 1$ and choose a sequence $(u_{m,k})_{k \geq 1}$ in

$$W^{1,p}(V) \cap C^\infty(\overline{V})$$

converging to u_m in $W^{1,p}(V)$. In particular, this sequence converges to u_m in $L^p(V)$. Since $p \geq 1$, we can actually say (c.f. [FLND]) that these $u_{m,k}$ converge to u_m in $L^1(V)$. Indeed, because V has finite measure, there exists a constant $C > 0$ such that

$$\|f\|_{L^1(V)} \leq C \|f\|_{L^p(V)}, \quad \forall f \in L^p(V).$$

Similarly, one can show that $\nabla u_{m,k}$ converges to ∇u_m in $L^1(V)$. Certainly,

$$\begin{aligned} \int_V |\nabla u_{m,k} - \nabla u_m| \, dm &= \int_V \left[\sum_{l=1}^n (\partial_{x_l} u_{m,k} - \partial_{x_l} u_m)^2 \right]^{1/2} \, dm \\ &\leq \sum_{j=1}^n \int_V |\partial_{x_l} u_{m,k} - \partial_{x_l} u_m| \, dm \\ &\leq C \sum_{j=1}^n \|\partial_{x_l} u_{m,k} - \partial_{x_l} u_m\|_{L^p(V)} \\ &\leq C \sum_{j=1}^n \|u_{m,k} - u_m\|_{W^{1,p}(V)} . \end{aligned}$$

Next, we claim that $u_{m,k}^\varepsilon$ converges to u_m^ε in $L^1(V)$, as $k \rightarrow \infty$. If we set each $u_{m,k}$ to be identically zero outside of $\bar{V}$, we obtain a function in $L^1(\mathbb{R}^n)$. Thus, one has by Young's inequality:

$$\begin{aligned} \|u_{m,k}^\varepsilon - u_m^\varepsilon\|_{L^p(V)} &\leq \|u_{m,k}^\varepsilon - u_m^\varepsilon\|_{L^p(\mathbb{R}^n)} = \|(u_{m,k} - u_m) * \eta_\varepsilon\|_{L^p(\mathbb{R}^n)} \\ &\leq \|\eta_\varepsilon\|_{L^1(\mathbb{R}^n)} \|u_{m,k} - u_m\|_{L^p(\mathbb{R}^n)} \\ &= \|u_{m,k} - u_m\|_{L^p(V)} . \end{aligned}$$

As before, this implies that

$$\|u_{m,k}^\varepsilon - u_m^\varepsilon\|_{L^1(V)} \leq C \|u_{m,k}^\varepsilon - u_m^\varepsilon\|_{L^p(V)} \xrightarrow{k \rightarrow \infty} 0$$

for each $m \geq 1$ and $0 < \varepsilon < 1$. Hence, we conclude by approximation that

$$\|u_m^\varepsilon - u_m\|_{L^1(V)} \leq \varepsilon \int_V |\nabla u_m(z)| \, dz \leq \varepsilon C \|u_m\|_{W^{1,p}(V)} .$$

Since the sequence u_m is uniformly bounded in $\|\cdot\|_{W^{1,p}(V)}$, we see that u_m^ε converges in $L^1(V)$ to u_m *uniformly* with respect to m , as $\varepsilon \rightarrow 0$.

Standard L^p -interpolation results (see [FLND]) give the existence of a constant $\theta \in (0, 1)$ such that

$$\|u_m^\varepsilon - u_m\|_{L^p(V)} \leq \|u_m^\varepsilon - u_m\|_{L^1(V)}^\theta \|u_m^\varepsilon - u_m\|_{L^{p^*}(V)}^{1-\theta} ,$$

for all $\varepsilon \in (0, 1)$ and $m \geq 1$. Our previous embedding theorem then implies that, for suitable $C > 0$,

$$\|u_m^\varepsilon - u_m\|_{L^p(V)} \leq C \|u_m^\varepsilon - u_m\|_{L^1(V)}^\theta \|u_m^\varepsilon - u_m\|_{W^{1,p}(V)}^{1-\theta} ,$$

Now, the u_m are uniformly bounded in $W^{1,p}(V)$. Applying Young's inequality also gives

$$\|u_m^\varepsilon\|_{L^p(V)} \leq \|\eta_\varepsilon\|_{L^1(V)} \|u_m\|_{L^p(V)}$$

and, for each index $l = 1, \dots, n$,

$$\|\partial_{x_l} u_m^\varepsilon\|_{L^p(V)} \leq \|\eta_\varepsilon\|_{L^1(V)} \|\partial_{x_l} u_m\|_{L^p(V)} \leq \|\eta_\varepsilon\|_{L^1(V)} \|u_m\|_{W^{1,p}(V)}$$

whence the family $\{u_m^\varepsilon\}_{\substack{m \geq 1, \\ \varepsilon \in (0,1)}}$ is also uniformly bounded in $W^{1,p}(V)$. Consequently,

$$\|u_m^\varepsilon - u_m\|_{L^p(V)} \leq C \|u_m^\varepsilon - u_m\|_{L^1(V)}^\theta$$

for a suitable constant $C > 0$ independent of m and $\varepsilon \in (0, 1)$. In light of this, we conclude that u_m^ε converges uniformly in m , to u_m in $L^q(V)$, as $\varepsilon \searrow 0$. This proves the claim in this step.

Step 4. Fix $\varepsilon \in (0, 1)$; we claim that the sequence $(u_m^\varepsilon)_{m=1}^\infty$ is uniformly bounded and equicontinuous on V . Let us first prove uniform boundedness. Fix $x \in \mathbb{R}^n$ and observe that, for each $m \geq 1$, there holds

$$\begin{aligned} |u(x)| &\leq \int_{B(x, \varepsilon)} \eta_\varepsilon(x - y) |u_m(y)| \, dy \leq \|\eta_\varepsilon\|_{L^\infty(\mathbb{R}^n)} \|u_m\|_{L^1(V)} \\ &\leq \frac{C}{\varepsilon^n} \|u_m\|_{L^p(V)} \\ &\leq \frac{C}{\varepsilon^n}. \end{aligned}$$

Since u_m^ε is smooth on $\mathbb{R}^n$ and compactly supported on V , it suffices to show that ∇u_m^ε is uniformly bounded on V . Certainly, the mean value theorem would then imply the equicontinuity of the sequence on V . To this end, observe that for each $x \in V$ one has

$$\begin{aligned} |\nabla u_m^\varepsilon(x)| &\leq \int_{\mathbb{R}^n} |\nabla \eta_\varepsilon(x - y)| |u_m(y)| \, dy \leq \|\nabla \eta_\varepsilon\|_{L^\infty(\mathbb{R}^n)} \|u_m\|_{L^1(V)} \\ &\leq \frac{C}{\varepsilon^{n+1}}. \end{aligned}$$

Here, we have used the fact that $L^p(V) \subseteq L^1(V)$ as well as the corresponding bound on the L^1 -norm of a function in $L^p(V)$, i.e.

$$\|f\|_{L^1(V)} \leq C \|f\|_{L^p(V)} \leq C \|f\|_{W^{1,p}(V)}.$$

This completes the step.

Step 5. Let $\delta > 0$ be given. We show there exists a subsequence $(u_{m_j})_{j=1}^\infty$ such that

$$\limsup_{j,k \rightarrow \infty} \|u_{m_j} - u_{m_k}\|_{L^q(V)} \leq \delta.$$

To see that this is so, we first choose $\varepsilon > 0$ such that

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \leq \frac{\delta}{2},$$

for all $m \geq 1$. The existence of such an ε is guaranteed by Step 3. By construction, the u_m^ε are all smooth on $\mathbb{R}^n$ and compactly supported within the precompact set V . Hence, by the Arzela-Ascoli theorem, we may extract a subsequence $(u_{m_j}^\varepsilon)_{j \geq 1}$ converging uniformly on $\bar{V}$, and hence V . In particular,

$$\limsup_{j,k \rightarrow \infty} \|u_{m_j}^\varepsilon - u_{m_k}^\varepsilon\|_{L^q(V)} = 0.$$

But then, for each $j, k \geq 1$ we have

$$\begin{aligned} \|u_{m_j} - u_{m_k}\|_{L^q(V)} &\leq \|u_{m_j} - u_{m_j}^\varepsilon\|_{L^q(V)} + \|u_{m_j}^\varepsilon - u_{m_k}^\varepsilon\|_{L^q(V)} \\ &\quad + \|u_{m_k}^\varepsilon - u_{m_k}\|_{L^q(V)}. \end{aligned}$$

Hence, letting $j, k \rightarrow \infty$ yields

$$\limsup_{j,k \rightarrow \infty} \|u_{m_j} - u_{m_k}\|_{L^q(V)} \leq \delta,$$

as was asserted. $\square$

Now, we use a standard diagonalization argument (with $\delta = 1, \frac{1}{2}, \dots$) to obtain a subsequence $(u_{m_j})_{j=1}^\infty$ with the property that

$$\limsup_{j,k \rightarrow \infty} \|u_{m_j} - u_{m_k}\|_{L^q(V)} = 0.$$

Since $L^q(V)$ is complete, this sequence will converge to a function $u \in L^q(V)$. That is,

$$\lim_{\ell \rightarrow \infty} \|u_{m_\ell} - u\|_{L^q(V)} = 0.$$

Now, let v denote the restriction of u to $\Omega \subseteq V$. Clearly, $u \in L^q(\Omega)$. Also,

$$\limsup_{\ell \rightarrow \infty} \|v_{m_\ell} - v\|_{L^q(\Omega)} = \limsup_{\ell \rightarrow \infty} \|u_{m_\ell} - u\|_{L^q(\Omega)} \leq \lim_{\ell \rightarrow \infty} \|u_{m_\ell} - u\|_{L^q(V)} = 0.$$

This completes the proof.

3.6 Distributions

For most of the chapter we focused on Sobolev spaces and therefore made heavy use of *weak derivatives*. This was of course intended to serve as a generalization of the standard derivative from calculus. However, not all locally integrable functions will have a weak derivative as the following example shows.

Example 3.1. Define a function $f : \mathbb{R} \rightarrow \mathbb{R}$ via

$$x \mapsto \begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Clearly, $f \in L^1_{\text{loc}}(\mathbb{R}) \cap L^\infty(\mathbb{R})$ and is in fact smooth on $\mathbb{R} \setminus \{0\}$. We show that f does not have a weak derivative on $\mathbb{R}$. If f does indeed have a weak derivative on $\mathbb{R}$, there exists a function $g \in L^1_{\text{loc}}(\mathbb{R})$ such that

$$\int_{\mathbb{R}} f \phi' \, dm = - \int_{\mathbb{R}} g \phi \, dm$$

for all compactly supported smooth functions $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Since ϕ has compact support, it is easily seen that

$$\int_{\mathbb{R}} f(x) \phi'(x) \, dx = \int_0^\infty \phi'(x) \, dx = -\phi(0).$$

Hence, g must satisfy

$$\int_{\mathbb{R}} g(x) \phi(x) \, dx = \phi(0)$$

for all functions $\phi \in C_c^\infty(\mathbb{R})$. We now claim that no such function $g \in L^1_{\text{loc}}(\mathbb{R})$ can exist. Let $(\varphi_k)_{k=1}^\infty$ be a sequence in $C_c^\infty(\mathbb{R})$ such that for all $k \geq 1$:

$$\varphi_k(0) = 1, \quad \|\varphi_k\|_{L^\infty(\mathbb{R})} = 1, \quad \text{supp}(\varphi_k) \subseteq \left[-\frac{1}{k}, \frac{1}{k}\right].$$

Note that $\varphi_k(x) \rightarrow 0$ as $k \rightarrow \infty$ for almost all $x \in \mathbb{R}^n$. Through dominated convergence we obtain

$$\lim_{k \rightarrow \infty} \int_{\mathbb{R}} g(x) \varphi_k(x) \, dx = 0.$$

On the other hand,

$$\lim_{k \rightarrow \infty} \int_{\mathbb{R}} g(x) \varphi_k(x) \, dx = \lim_{k \rightarrow \infty} \varphi_k(0) = \lim_{k \rightarrow \infty} 1 = 1.$$

This is an obvious contradiction.

The example above puts us in an awkward position. We generalized the standard definition of the derivative in order to “differentiate” a larger space of functions. However, there still exist functions that have no such derivative. To rectify this issue once and for all, we introduce *distributions*.

If $\Omega \subseteq \mathbb{R}^n$ is open, we defined $\mathcal{D}(\Omega)$ to be the space of all compactly supported smooth functions $\phi : \Omega \rightarrow \mathbb{R}$. Of course, $\mathcal{D}(\Omega)$ can be made into a real vector space when endowed with the obvious notions of addition and scalar multiplication. To allow for complex valued “test functions”, we will introduce the following notation.

Definition 26. Let $\Omega \subseteq \mathbb{R}^n$ be non-empty and open. We define $\mathfrak{D}(\Omega)$ to be the collection of all smooth functions

$$\phi : \Omega \rightarrow \mathbb{C}$$

such that $\overline{\text{supp}(\phi)} \subset \Omega$. That is, $\mathfrak{D}(\Omega)$ consists of all *complex valued* smooth functions having compact support in Ω . Note that $\mathcal{D}(\Omega) \subset \mathfrak{D}(\Omega)$.

Clearly, $\mathfrak{D}(\Omega)$ is a complex vector space with the usual pointwise operations. Let us now “topologize” $\mathfrak{D}(\Omega)$ by introducing a notion of convergence for sequences in $\mathfrak{D}(\Omega)$.

Definition 27. Let $(\phi_m)_{m=1}^{\infty}$ be a sequence in $\mathfrak{D}(\Omega)$. We say that the sequence $(\phi_m)_{m=1}^{\infty}$ converges to a function $\phi \in \mathfrak{D}(\Omega)$ provided

1. There exists a compact set $\Lambda \subset \Omega$ such that

$$\text{supp}(\phi_m) \subseteq \Lambda$$

for all $m \in \mathbb{N}$.

2. For all multiindices $\alpha = (\alpha_1, \dots, \alpha_n)$ we have

$$\lim_{m \rightarrow \infty} \|\partial^\alpha \phi_m - \partial^\alpha \phi\|_{\infty} = \lim_{m \rightarrow \infty} \sup_{x \in \Omega} |\partial^\alpha \phi_m(x) - \partial^\alpha \phi(x)| = 0.$$

In this case, we will simply write “ $\phi_m \rightarrow \phi$ ” when no confusion is possible. Observe also that $\text{supp}(\phi) \subseteq \Lambda$ as well. In fact, $\phi_m(x) \rightarrow \phi(x)$ pointwise on Ω .

We are now in a position to introduce distributions, which will serve as generalization of classical functions $\mathbb{R}^n \rightarrow \mathbb{C}$.

Definition 28 (Distribution). Let $\Omega \subseteq \mathbb{R}^n$ be open and non-empty. A *distribution* on Ω is a linear functional

$$F[\cdot] : \mathcal{D}(\Omega) \longrightarrow \mathbb{C}$$

such that

$$\lim_{m \rightarrow \infty} F[\phi_m] = F[\phi]$$

whenever $(\phi_m)_{m=1}^\infty$ is a sequence in $\mathcal{D}(\Omega)$ converging to the function $\phi \in \mathcal{D}(\Omega)$.

In a sense, this last requirement merely says that $F[\cdot]$ is a continuous linear functional. This quite abstract mapping will not serve as a good generalization if they do not “include” classical functions. To this end, we seek to identify functions in $L^1_{\text{loc}}(\Omega)$ with these newly introduced distributions.

Functions as distributions. Let $\Omega \subseteq \mathbb{R}^n$ be open and non-empty. Fix a function $f \in L^1_{\text{loc}}(\Omega)$. We can associate to f a distribution $\mathcal{D}(\Omega) \rightarrow \mathbb{C}$ according to the following map:

$$F_f[\cdot] : \mathcal{D}(\Omega) \longrightarrow \mathbb{C}, \quad \phi \mapsto \int_{\Omega} f(x)\phi(x) \, dm.$$

Since ϕ has compact support, this is always well defined. Let us now show that this map $F_f[\cdot]$ is continuous. Let $(\phi_m)_{m=1}^\infty$ be a sequence in $\mathcal{D}(\Omega)$ converging to $\phi \in \mathcal{D}(\Omega)$. Thus, each ϕ_m and ϕ have support within some compact set $\Lambda \subset \Omega$. Since $(\phi_m)_{m=1}^\infty$ is convergent within the sup-norm, the sequence must also be uniformly bounded on Ω . Therefore, an application of the dominated convergence theorem yields

$$\lim_{m \rightarrow \infty} F_f[\phi_m] = \lim_{m \rightarrow \infty} \int_{\Omega} f(x)\phi_m(x) \, dm = \int_{\Omega} f(x)\phi(x) \, dm. = F_f[\phi]$$

Hence, $F_f[\cdot]$ is indeed a distribution. We then choose to identify f with this distribution.

Now, we turn to the differentiation of distributions, and hence all locally integrable functions defined on Ω . Let $F[\cdot]$ be a distribution on Ω and assume α is a multiindex. We define a new distribution on Ω , which we shall call the α^{th} (distributional) derivative of $F[\cdot]$:

$$\partial^\alpha F[\cdot] : \mathfrak{D}(\Omega) \longrightarrow \mathbb{C}, \quad \phi \mapsto (-1)^{|\alpha|} F[\partial^\alpha \phi].$$

In particular, if $F_f[\cdot]$ is a distribution representing a function f :

$$\partial^\alpha F[\phi] = (-1)^{|\alpha|} \int_{\Omega} f(x) \partial^\alpha \phi(x) \, dm.$$

It is customary to then denote $\partial^\alpha F[\cdot]$ by $\partial^\alpha f$. This would be called the α^{th} distributional derivative of f .

It is clear from the definitions that this notion of differentiation generalizes the weak derivative; since the latter generalizes the standard derivative, this will also prove that distributional derivatives are weaker than standard derivatives. As a testament to the power of this concept, let us rectify our motivating problem.

Example 3.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be as in Example 3.1. It was shown there that f does not admit a weak derivative on $\mathbb{R}$. Here, we shall prove that f has a distributional derivative.

Let $F_f[\cdot]$ be the distribution associated to f and let $\phi \in \mathfrak{D}(\Omega)$ be given. We directly evaluate

$$\int_{\mathbb{R}} f(x) \phi'(x) \, dx = \int_0^\infty \phi'(x) \, dx = -\phi(0).$$

Hence,

$$\partial_x F_f[\phi] = -F_f[\phi'] = - \int_{\Omega} f(x) \phi'(x) \, dx = \phi(0).$$

It follows that f' is the Dirac delta distribution

$$\delta_0 : \mathfrak{D}(\Omega) \longrightarrow \mathbb{C}, \quad \phi \mapsto \phi(0).$$

Finally, it is easy to check that every locally integrable function on Ω admits a distributional derivative (since all distributions are differentiable).

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Bibliography

- [EVNS] Evans, Lawrence. *Partial differential equations*. Second edition. Graduate Studies in Mathematics, Volume 19. 2010.
- [FLND] Folland, Gerald. *Real Analysis: Modern Techniques and Their Applications*. Second edition. Wiley.
- [JOST] Jost, Jürgen. *Partial differential equations*. Third edition. Graduate texts in mathematics. Springer.